Algebraic integration stress test: Trager and ParallelMixedTower on *Diabolically Difficult Integrals*, Vol. 1

2026-10-30 (Mathilda v0.6087)

Summary

Two of Mathilda's algebraic integrators were run over every integral of S. Blake, *Diabolically Difficult Integrals*, Volume 1, with FriCAS as a third opinion:

- `Integrate[f, x, Method -> "Trager"]`: Trager's algorithm for a rational function of x and one radical over Q.
- `Integrate[f, x, Method -> "ParallelMixedTower"]` (PMT): the parallel Risch method over mixed towers.

There were 803 integrands. 207 have symbolic parameters, and each of those also appears with its parameters set to small integers.

The main results

1. **Trager is sound throughout.** It gave no wrong answers and no false non-elementarity certificates. It finds 338 antiderivatives and 15 proofs of non-elementarity. In 113 cases it gets an answer where FriCAS does not.
   - Its median time is 0.64 s.
   - Its limits come from its scope. It declines nested radicals (207), symbolic parameters (109 at the entry gate), several radicals (51) and transcendental integrands (52).
2. **PMT covers more classes but had soundness defects.**
   - It returned a leaked internal symbol as the "antiderivative" in 56 cases. This is fixed.
   - It issued 2 false certificates, through a precision bug in `Series`. A fix is in hand.
   - It returned 13 answers that are right only on part of the domain.
   - One case crashed the kernel through a FLINT exception.
   - 201 cases time out at 120 s.
3. **Together the two methods decide 476 integrands.** FriCAS decides 528 at a 120 s timeout.
   - 130 integrands are decided by Mathilda but not FriCAS.
   - 182 are decided by FriCAS but by neither Mathilda method.
   - 145 are decided by no system.
4. **On your parameter question.** Trager decides 86 of the 207 parametric integrals once the parameters are instantiated: 83 antiderivatives and 3 proofs. Extending Trager to Q(a, b, …) should win most of those 86. PMT already accepts parameters but decides only 29 symbolic cases. FriCAS solves 123.

Method

Corpus

- File: `tests/data/ddi_vol1_corpus.m`, one record per line: id, chapter, integrand, expected status, book reference.
- Transcribed from the LaTeX chapter files (`*_problems.tex`, with `*_hints.tex` and `*_solutions.tex` used for the expected status).
- Families over n are expanded at the listed n.
- Skipped: definite integrals, non-integration exercises, ArcSin/ArcCsc integrands, and templates without a concrete integrand.
- **Audit:** every record was checked against the LaTeX, symbol by symbol. No integrand errors were found. 16 expected labels were corrected to say only what the book states.

Harness

- Tools: `tools/ddi_stress.py` runs the cases and `tools/ddi_report.py` writes this report.
- Each (integrand, method) pair runs in a fresh process under `TimeConstrained[…, 120]`.

- With an explicit `Method -> "ParallelMixedTower"`, PMT runs with no internal time budget (changed in v0.6087).
- A Mathilda antiderivative is checked two ways:
  - `PossibleZeroQ[D[F, x] - f]`;
  - a 40-digit numeric check at three real points where the integrand is real, with the parameters instantiated.
- FriCAS answers are checked numerically.
- A FriCAS answer in special functions (erf, dilog, …) counts as FriCAS deciding the integral is not elementary.

Status labels

- WRONG: the derivative check fails.
- BRANCH: D[F] = f on part of the real line and −f elsewhere.
- DISAGREE: a non-elementarity certificate for an integral that FriCAS or the book shows to be elementary.
- BOOK-CONFLICT: a verified antiderivative for an integral the book calls non-elementary.

Caveat about verification. These baseline verdicts accept an answer that passes the numeric check even when `PossibleZeroQ` says False. That happened for 24 Trager and 54 PMT answers. The bar is now higher on two counts:

- An antiderivative must be valid for all complex x, not just on the real line.
- `PossibleZeroQ` alone decides. A `PossibleZeroQ` false negative on a correct answer is a `PossibleZeroQ` bug to fix, not something to patch over with numerics.

Under that bar some of these 78 answers will turn out to be branch errors, and the rest are `PossibleZeroQ` false negatives. Both are in the fix plan below.

Defects found

#	defect	component	status
1	PMT returned the unassigned local `ParallelMixed`Private`r$NNN` as an antiderivative (56 cases, also through the default `Integrate`). `Return[False]` inside a `With` in `eulerStep` left only the `With`; a self-referential `Dwp` then overflowed `$RecursionLimit`.	`ParallelMixed.m` (BuildTower)	**fixed** (v0.6087), regression test
2	False certificates: `Sqrt[1-x] Sqrt[1+x] Sqrt[1-x^4]` and multiple_radicals_4 (iii) "proved" non-elementary. `Series` came back short of the requested order at poles of order ≥ 7 (fixed padding), so PMT read a residue of −5/4 as 0.	`src/calculus/series.c`	fix in hand (Phase 1a)
3	The verify gate let a recursion-overflow result through. After `$RecursionLimit` the evaluator runs degraded, and the gate's decline branches were skipped. `fac` is a global variable.	`ParallelMixed.m`, `integrate.c`	planned (Phase 1b)
4	A kernel abort (FLINT `qqbar_pow_fmpq: division by zero`) on `((2x^3-10) Sqrt[2x^3+3x^2+5])/(2x^3+5)^2`	`src/poly/flint_qqbar.c`	planned (Phase 1c)
5	13 answers right on one real interval and of the wrong sign on another: separate radicals are rebuilt as one principal radical, and a square factor's sign is fixed at one sample point	`ParallelMixed.m` (`rootNormalize`, `signFix`, `branchUnit`)	planned (Phase 2b)
6	78 verified answers where `PossibleZeroQ` says False	PossibleZeroQ / both methods	planned (Phase 2d)
7	PMT: 201 timeouts and 151 "tower construction failed" declines (one radical only, no u = x^k substitution, conics with no rational point)	`ParallelMixed.m`	planned (Phase 3)
8	Trager: rational functions (11) and algebraic constants (8) declined; no parameters (207)	`src/internal/trager/`	planned (Phase 4)
9	Trager: nested radicals (207) and several radicals (51)	`src/internal/trager/`	separate plan (general function fields)

Harness defects found and fixed along the way:

- An unverifiable parametric answer was counted as WRONG. Parameters are now instantiated before differentiating.
- FriCAS answers in special functions were counted as elementary.
- The PMT decline reasons were parsed wrongly.

The parameter question

- AXIOM and FriCAS integrate over constant fields Q(a, b, …). For the torsion and reduction steps they specialise the parameters.
- Mathilda's Trager works over Q only. That is not just the entry gate: the constant fields in `Field.m`, the residue fields in `Residues.m`, and the reductions mod p in `TorsionModP.m`, `Genus.m` and `Jacobian.m` all assume Q.
- On this corpus, 86 of the 207 parametric integrals are decided by Trager once instantiated. So extending Trager to Q(params) has a measured payoff, and it is Phase 4b of the plan.
- One soundness rule carries over. A non-elementarity verdict must be proved generically; specialisation may only supply bounds and orders.

Fix plan (in order)

1. **Soundness.**
   - The `Series` padding bug.
   - The PMT gate accepts only on positive evidence.
   - FLINT exceptions never abort the kernel.
2. **Answers valid for all complex x.**
   - Complex sample points in the harness.
   - PMT back-substitutes in the integrand's own radicals.
   - Power-free radicands.
   - `PossibleZeroQ` becomes the sole verifier, with its false negatives fixed.
3. **PMT coverage and speed.**
   - A u = x^k substitution pre-pass.
   - Euler steps through algebraic points.
   - Partially ramified places.
   - Profile and fix the timeouts by algorithm, with no time caps.
4. **Trager.**
   - Route rational integrands and algebraic constants.
   - Work over Q(params).
   - Fix the 4 timeouts.
5. **Separate plan:** Trager for general algebraic function fields (primitive element and integral basis), for nested and multiple radicals.

Findings about the book

- $\int$ArcTan[x²] dx is elementary: `x ArcTan[x^2] - Integrate[2 x^2/(1 + x^4), x]`. The book (inverse_function_3) says it is not. Mathilda's verified antiderivative is listed as a book conflict.
- Possible typos:
  - realis_2 (v): the sign pairing contradicts special case (i);
  - realis_2 (iv);
  - multiple_radicals_0 (vi) has a stray constant `+(a+b)`;
  - timofeev_3 (iv): a stale LaTeX comment says 1/3 where the text says 3/2;
  - multiple_methods problem 1: the LaTeX comment differs from the displayed integral.
- Some expected labels rest on the chapter's framing rather than an explicit statement: generalised_gunther and multiple_radicals have empty hint and solution files. The book's own "elementary" counts above use these labels.

Results

Run of 2026-10-30 23:10:42: **803 integrands** (207 with symbolic parameters, 596 over Q), per-method timeout 120 s, FriCAS timeout 120 s, wall time 76 min; ParallelMixedTower re-run on 803 cases (33 min); ParallelMixedTower re-run on 55 cases (2 min); ParallelMixedTower re-run on 803 cases (38 min); ParallelMixedTower, Trager re-run on 803 cases (44 min); ParallelMixedTower re-run on 2 cases (0 min); ParallelMixedTower re-run on 2 cases (0 min); Trager, ParallelMixedTower re-run on 803 cases (44 min); Trager, ParallelMixedTower re-run on 18 cases (2 min); ParallelMixedTower re-run on 803 cases (39 min); ParallelMixedTower re-run on 2 cases (0 min).

Headline

	Trager	ParallelMixedTower	FriCAS
verified antiderivative	351	260	482
proved non-elementary	15	21	46
declined	433	323	0
timeout	4	199	253
error / crash	0	0	21
wrong answer	0	0	1
right on part of the real line only (branch)	0	0	0
correct, but PossibleZeroQ cannot confirm it (PZQ-FN)	0	0	0
answer not verifiable in budget	0	0	0

Of the **646** integrands the book states are elementary: Trager solves **294**, ParallelMixedTower solves **223**, FriCAS **419**, either Mathilda method **369**.

- over Q (no parameters) (596): Trager solved 351, decided 366; ParallelMixedTower solved 239, decided 260; FriCAS solved 359.
- with symbolic parameters (207): Trager solved 0, decided 0; ParallelMixedTower solved 21, decided 21; FriCAS solved 123.

Classification against FriCAS and the book

class	Trager	ParallelMixedTower
agree elem	238	223
Mathilda only	113	36
agree non-elem	9	15
non-elem unconf.	6	6
FriCAS only	244	258
neither	190	152
timeout	3	112
book conflict	0	1

Per chapter

Cells are *verified antiderivatives / proved non-elementary / declined / timeout+error+wrong*.

chapter	n	book elem	Trager	ParallelMixedTower	FriCAS (elem/non)
warmup	73	73	46 / 0 / 26 / 1	51 / 0 / 11 / 11	73 / 0
laurent_poly_subst	60	60	43 / 0 / 17 / 0	16 / 0 / 13 / 31	42 / 0
rational_subs	32	32	27 / 0 / 5 / 0	13 / 0 / 5 / 14	16 / 0
gunther	33	29	21 / 4 / 8 / 0	7 / 4 / 11 / 11	26 / 0
power_expand	6	6	5 / 0 / 1 / 0	4 / 0 / 1 / 1	6 / 0
generalised_gunther	88	80	66 / 2 / 20 / 0	14 / 0 / 16 / 58	19 / 2
chebychev	87	51	3 / 7 / 77 / 0	18 / 8 / 48 / 13	32 / 32
inverse_function	18	10	0 / 0 / 18 / 0	2 / 5 / 10 / 1	6 / 5
product_rule	21	21	19 / 0 / 2 / 0	20 / 0 / 1 / 0	21 / 0
log_term	16	16	16 / 0 / 0 / 0	15 / 0 / 1 / 0	16 / 0
realis	22	7	17 / 2 / 3 / 0	9 / 2 / 10 / 1	18 / 0
nested_radicals	161	159	0 / 0 / 161 / 0	27 / 1 / 113 / 20	86 / 2
multiple_radicals	55	34	0 / 0 / 55 / 0	12 / 0 / 36 / 7	40 / 2
euler	22	22	11 / 0 / 10 / 1	6 / 0 / 9 / 7	13 / 0
timofeev	23	7	19 / 0 / 4 / 0	13 / 0 / 5 / 5	23 / 0
potpourri	43	7	31 / 0 / 12 / 0	20 / 0 / 17 / 6	27 / 0
mult_piecewise	1	0	0 / 0 / 1 / 0	0 / 0 / 1 / 0	0 / 0
danger_zone	20	18	17 / 0 / 1 / 2	1 / 0 / 9 / 10	4 / 0

chapter	n	book elem	Trager	ParallelMixedTower	FriCAS (elem/non)
multiple_methods	4	0	2 / 0 / 2 / 0	1 / 0 / 1 / 2	0 / 0
introduction	18	14	8 / 0 / 10 / 0	11 / 1 / 5 / 1	14 / 3

Timing (seconds, cases each system answered — antiderivative or proof)

system	answered	median	mean	max
Trager	366	0.645	1.778	34.87
ParallelMixedTower	281	0.278	4.598	119.14
FriCAS	528	0.200	2.452	90.90

On the 169 integrands both methods solve, median LeafCount of the antiderivative: Trager 118, ParallelMixedTower 97.

Why each method declines

Trager (433 declines)

reason	count
nested radicals	207
radicand is not a rational function over Q	109
transcendental function in the integrand	52
more than one radicand (a radical compositum)	51
integrand is not a rational function of x and the radical over Q	6
algebraic constant in the integrand (coefficients must be rational)	6
integrand is not algebraic over Q(x)	2

ParallelMixedTower (323 declines)

reason	count
tower construction failed	151
verification failed	60
no solution within bounds	47
needs torsion realisation (milestone iii)	30
partially ramified branch prime	15
residue class lost	9
certificate withheld	5
internal	4
deep residue sheet undecided	2

Wrong answers (failed derivative check) (0)

None.

Right on part of the real line only (D[F] = f on one interval, -f on another) (0)

None.

Correct answers PossibleZeroQ cannot confirm (PossibleZeroQ bugs) (0)

None.

Antiderivatives not verifiable within the budget (0)

None.

Disagreements (proved non-elementary, but FriCAS or the book has an antiderivative) (0)

None.

Book conflicts (verified antiderivative where the book says non-elementary) (1)

id	method	integrand	book ref	detail
inverse_function-inverse_function_3-iii	PMT	ArcTan[x^2]	Problem inverse_function_3 (iii) BOOK CL…	(-ArcTan[x^2] + Sqrt[2] x ArcTan[x^2])/Sqrt[2] - Sqrt[2] ArcTan[1 + Sqrt[2] x] - Log[1 - Sqrt[2] x + x^2]/(2 S…

Crashes / errors (0)

None.

Timeouts (203)

id	method	integrand	book ref	detail
warmup-prob_warmup_4-iii	PMT	Sqrt[a*x^2+b*x+c]/x^2	Problem 0.4 (iii) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-iv	PMT	x^2*Sqrt[a*x^2+b*x+c]	Problem 0.4 (iv) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-v	PMT	(a*x^2+b*x+c)^(3/2)/x^2	Problem 0.4 (v) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-vi	PMT	x*(a*x^2+b*x+c)^(3/2)	Problem 0.4 (vi) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-vii	PMT	(a*x^2+b*x+c)^(5/2)/x	Problem 0.4 (vii) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-viii	PMT	x*(a*x^2+b*x+c)^(5/2)	Problem 0.4 (viii) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-ix	PMT	1/((x+b)^2*(a*x^2+b*x+c)^(3/2))	Problem 0.4 (ix) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-x	PMT	1/((x+b)^2*(a*x^2+b*x+c)^(5/2))	Problem 0.4 (x) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_4-xi	PMT	x/((a*x^2+b*x+c)^(3/2)-(a*x^2+b*x+c)^(5/2))	Problem 0.4 (xi) Euler substitution	FriCAS ELEM; book elem
warmup-prob_warmup_5-inst	Trager	((2*x+3)^(2/3)+1)/((2*x+3)^(1/2)+(2*x+3)^(1/3))	Problem 0.5 multiple radicals of ax+b in…	FriCAS ELEM; book elem
warmup-prob_warmup_6	PMT	(((x-1)/(2*x+1))^(1/4)-3*((x-1)/(2*x+1))^(3/4))/((x-1)*(x+1)^2*(2*x-1))	Problem 0.6 Moebius radicand	FriCAS ELEM; book elem
warmup-prob_warmup_13-iii	PMT	1/(x*Sqrt[a*x^2+b*x+c])	Problem 1.13 (iii)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_1	PMT	(a*x^2-b)/((a*x^2+b)*Sqrt[a^2*x^4+b^2])	Problem 1	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_3	PMT	(x^12+1)/((x^12-1)*(x^8+1)^(1/4))	Problem 3	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_5-i-…	PMT	((x^5+1)*(4*x^5-1))/(x*(x^5-2*x+1)*Sqrt[x^6+x])	Problem 5 (i) instantiated a=2	FriCAS ELEM; book elem

id	method	integrand	book ref	detail
laurent_poly_subst-prob_laurent_sub_5-ii…	PMT	((x^5-1)*(4*x^5+1))/(x*(x^5-2*x-1)*Sqrt[x^6-x])	Problem 5 (ii) instantiated a=2	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_7-i	PMT	(c*x^3-d)/(a*x^3-b)^(1/3)	Problem 7 (i)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_7-i-…	PMT	(5*x^3-7)/(2*x^3-3)^(1/3)	Problem 7 (i) instantiated a=2,b=3,c=5,d…	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_7-ii	PMT	(c*x^3+d)/(a*x^3-b)^(1/3)	Problem 7 (ii)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_7-ii…	PMT	(5*x^3+7)/(2*x^3-3)^(1/3)	Problem 7 (ii) instantiated a=2,b=3,c=5,…	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_8-i-…	PMT	(2*x^4+3*x^3)^(1/4)	Problem 8 (i) instantiated a=2,b=3	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_8-ii…	PMT	(2*x^8+3*x^7)^(1/4)	Problem 8 (ii) instantiated a=2,b=3	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_8-ii…	PMT	(2*x^12+3*x^11)^(1/4)	Problem 8 (iii) instantiated a=2,b=3	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_9	PMT	(a*x^4-b)/(x^4*(2*a*x^4-b)^(1/4))	Problem 9 (same as Problem 14b (i))	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_10	PMT	((x^6-2)*(x^6+1)*(x^6-x^4+1)^(1/4))/(x^6*(x^6-2*x^4+1))	Problem 10	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_12-i…	PMT	((x^3-1)*(x^6-x^3+1)*(x^6+1)^(2/3))/(x^6*(x^3+1))	Problem 12 (ii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_13-i	PMT	(a*x^2+b)/((a*x^2-b)*Sqrt[a*x^3+b*x^2-b*x])	Problem 13 (i)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_13-i…	PMT	((2*x^3-2*5)*Sqrt[2*x^3+3*x^2+5])/(2*x^3+5)^2	Problem 13 (ii) instantiated a=2,b=3,c=5	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_13-v…	PMT	((2*x^5-3)*Sqrt[x^6+2*x^4+x])/((x^5+1)*(x^5+x^3+1))	Problem 13 (vi)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_13-v…	PMT	((2*c-a*x^3)*Sqrt[a*x^3+b*x^2+c])/((a*x^3+(b-3)*x^2+c)*(a*x^3+(b-2)*x^2+c))	Problem 13 (vii)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_14-i	PMT	(x^2+4)/((x^2-4)*(x^3-x^2)^(1/3))	Problem 14 (i)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_14-i…	PMT	((x^2-4)*(x^3+x)^(1/3))/(x^4*(x^2+2))	Problem 14 (ii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_14-i…	PMT	((x^3-1)*(x^6+1)^(1/3))/(x^2*(x^3+1))	Problem 14 (iii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_14-i…	PMT	((x^2-2)*(x^3+x)^(1/3))/(x^2*(x^4-2*x^2+4))	Problem 14 (iv)	FriCAS ERROR; book elem
laurent_poly_subst-prob_laurent_sub_14-v	PMT	(x^3*(x^2+3))/((x^2+1)*(x^3+x^2+1)*(x^3-x^2-1)^(1/3))	Problem 14 (v)	FriCAS ELEM; book elem

id	method	integrand	book ref	detail
laurent_poly_subst-prob_laurent_sub_14b-…	PMT	(x^3+4*b)/((x^3+b)*(a*x^4-x^3-b)^(1/4))	Problem 14b (ii)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_14b-…	PMT	(x^3+4*3)/((x^3+3)*(2*x^4-x^3-3)^(1/4))	Problem 14b (ii) instantiated a=2,b=3	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_14b-…	PMT	((x^2-2)*(x^2-1)*(x^4+x^2-1)^(1/4))/(x^6*(2*x^4+x^2-1))	Problem 14b (iii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_16-i…	PMT	((a*x^2-b)/(x*(a*x^2+b)))*Sqrt[(a*x^2-c*x+b)/(a*x^2+c*x+b)]	Problem 16 (ii)	FriCAS ELEM; book elem
laurent_poly_subst-prob_laurent_sub_17-i…	PMT	((2*a*x^3+b)*(a^2*x^6-2*a*b*x^3-a*b*x^2+b^2))/(x*(a*x^3-b)*(a^2*x^6-2*a*b*x^3+x^2+b^2))*Sq…	Problem 17 (ii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_18-i	PMT	((2*x^3+1)/(x*(x^3+x-1)))*((x^3-x-1)/(x^3+x-1))^(1/3)	Problem 18 (i)	FriCAS ERROR; book elem
laurent_poly_subst-prob_laurent_sub_18-i…	PMT	((2*a*x^3+b)/(x*(a*x^3+x-b)))*((a*x^3-a*x-b)/(a*x^3+b*x-b))^(1/3)	Problem 18 (ii)	FriCAS TIMEOUT; book elem
laurent_poly_subst-prob_laurent_sub_18-i…	PMT	((2*2*x^3+3)/(x*(2*x^3+x-3)))*((2*x^3-2*x-3)/(2*x^3+3*x-3))^(1/3)	Problem 18 (ii) instantiated a=2,b=3	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-i	PMT	1/((x+1)*(3*x^2+1)^(1/3))	Problem 1 (i)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-ii	PMT	1/((x-1)*(3*x^2+1)^(1/3))	Problem 1 (ii)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-iii	PMT	1/((x+1)*(x^3+6*x^2+3*x+2)^(1/3))	Problem 1 (iii)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-iv	PMT	1/((x-1)*(x^3-6*x^2+3*x-2)^(1/3))	Problem 1 (iv)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-v	PMT	1/((x+1)*((a+1)*x^3+3*(1-a)*x^2+3*(a+1)*x-a+1)^(1/3))	Problem 1 (v)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-v-inst	PMT	1/((x+1)*((2+1)*x^3+3*(1-2)*x^2+3*(2+1)*x-2+1)^(1/3))	Problem 1 (v) instantiated a=2	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-vi	PMT	1/((x-1)*((a-1)*x^3+3*(a+1)*x^2+3*(a-1)*x+a+1)^(1/3))	Problem 1 (vi)	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_1-vi-ins…	PMT	1/((x-1)*((2-1)*x^3+3*(2+1)*x^2+3*(2-1)*x+2+1)^(1/3))	Problem 1 (vi) instantiated a=2	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_2-iii	PMT	(b*x^2-2*a*x-a*b)/((x^2+a)*Sqrt[2*x^4+2*b*x^3+(2*a+b^2)*x^2+a^2])	Problem 2 (iii)	FriCAS ELEM; book elem
rational_subs-prob_rational_sub_2-iv	PMT	(b*x^2-2*a*x-a*b)/(x*(x+b)*Sqrt[2*x^4+2*b*x^3+(2*a+b^2)*x^2+a^2])	Problem 2 (iv)	FriCAS ELEM; book elem
rational_subs-prob_rational_sub_6-i-plus	PMT	1/((x+1)*(x^3+x)^(1/4))	Problem 6 (i) x+1	FriCAS TIMEOUT; book elem

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rational_subs-prob_rational_sub_6-ii-plu…	PMT	(x+2)/((x^2-1)*(x^3+x)^(1/4))	Problem 6 (ii) x+2	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_6-i-minu…	PMT	1/((x-1)*(x^3+x)^(1/4))	Problem 6 (i) x-1	FriCAS TIMEOUT; book elem
rational_subs-prob_rational_sub_6-ii-min…	PMT	(x-2)/((x^2-1)*(x^3+x)^(1/4))	Problem 6 (ii) x-2	FriCAS TIMEOUT; book elem
gunther-prob_gunther_2-ii	PMT	(a*x-b)/((a*x+b)*Sqrt[a^2*x^3+b^2*x])	Problem 2 (ii)	FriCAS ELEM; book elem
gunther-prob_gunther_3-iv	PMT	(a*x^2+b)/((a*x^2-b)*Sqrt[a*x^3+c*x^2-b*x])	Problem 3 (iv)	FriCAS ELEM; book elem
gunther-prob_gunther_3-vi	PMT	(x^2-1)/((x^2+1)*Sqrt[a*x^4+b*x^3+c*x^2+b*x+a])	Problem 3 (vi)	FriCAS ELEM; book elem
gunther-prob_gunther_4	PMT	(2*a*x^4+4*a*x^3-b*x+b-2*c)/((a*x^4+x^2+(b+2)*x+c+1)*Sqrt[a*x^4+b*x+c])	Problem 4	FriCAS TIMEOUT; book elem
gunther-prob_gunther_4-inst	PMT	(2*2*x^4+4*2*x^3-3*x+3-2*5)/((2*x^4+x^2+(3+2)*x+5+1)*Sqrt[2*x^4+3*x+5])	Problem 4 instantiated a=2,b=3,c=5	FriCAS ELEM; book elem
gunther-prob_gunther_5-i	PMT	x/((x^3+8)*Sqrt[x^3-1])	Problem 5 (i)	FriCAS ELEM; book elem
gunther-prob_gunther_5-ii-pp	PMT	x/((x^3+4)*Sqrt[x^3+1])	Problem 5 (ii) upper signs	FriCAS ELEM; book elem
gunther-prob_gunther_5-ii-mm	PMT	x/((x^3-4)*Sqrt[x^3-1])	Problem 5 (ii) lower signs	FriCAS ELEM; book elem
gunther-prob_gunther_5-iii-pp	PMT	x^3/((x^6+4)*Sqrt[x^6+1])	Problem 5 (iii) upper signs, from (ii) b…	FriCAS ELEM; book elem
gunther-prob_gunther_5-iii-mm	PMT	x^3/((x^6-4)*Sqrt[x^6-1])	Problem 5 (iii) lower signs, from (ii) b…	FriCAS ELEM; book elem
gunther-prob_gunther_5-iv-pm	PMT	x^3/((x^6+8)*Sqrt[x^6-1])	Problem 5 (iv) x^6+8 with x^6-1, from (i…	FriCAS ELEM; book elem
power_expand-power_expand_2-ii	PMT	1/(x^20-3*x^16+2*x^12+2*x^8-3*x^4+1)^(1/4)	Problem 2 (ii)	FriCAS ELEM; book elem
generalised_gunther-intro-1	PMT	(2*x^2-1)/((x^2-1)*(3*x^2-1)*(2*x^3-x)^(1/3))	Chapter intro worked example (solved)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_1-main	PMT	(2*x^2+1)/((x^2+1)*(3*x^2+1)*(2*x^3-x)^(1/3))	Problem gg1 (unlabelled, 1st of chapter)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_2-i	PMT	1/(x*(x^2-3*x+2)^(1/3))	Problem gg2 (i)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_2-ii	PMT	(x-2*a+b)/((x^2+(a*b^2-2*b)*x+b^2-a^2*b^2)*((x-a)*(x-b))^(1/3))	Problem gg2 (ii)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_2-ii-inst	PMT	(x-2*2+3)/((x^2+(2*3^2-2*3)*x+3^2-2^2*3^2)*((x-2)*(x-3))^(1/3))	Problem gg2 (ii) instantiated a=2,b=3	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_4-main	PMT	(x+1)/(x*(x^4-4*x^3+7*x^2-4*x+1)^(1/4))	Problem gg4	FriCAS ELEM; book elem
generalised_gunther-prob_gg_5-i	PMT	(3*x^2+1)/((x^2-1)*(x^4+6*x^2+1)^(3/4))	Problem gg5 (i)	FriCAS TIMEOUT; book elem

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generalised_gunther-prob_gg_5-ii	PMT	(x-1)^2/((x+1)*(x^4+6*x^2+1)^(3/4))	Problem gg5 (ii)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_5-iii	PMT	(x+1)^2/((x-1)*(x^4+6*x^2+1)^(3/4))	Problem gg5 (iii)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_6-i	PMT	(x+a)^2/((x+b)*(2*x^4+4*(a+b)*x^3+6*(a^2+b^2)*x^2+4*(a^3+b^3)*x+2*a^4+2*b^4)^(3/4))	Problem gg6 (i)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_6-i-inst	PMT	(x+2)^2/((x+3)*(2*x^4+4*(2+3)*x^3+6*(2^2+3^2)*x^2+4*(2^3+3^3)*x+2^4+2*3^4)^(3/4))	Problem gg6 (i) instantiated a=2,b=3	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_6-ii	PMT	(x-a)^2/((x-b)*(2*x^4-4*(a+b)*x^3+6*(a^2+b^2)*x^2-4*(a^3+b^3)*x+a^4+b^4)^(3/4))	Problem gg6 (ii)	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_6-ii-inst	PMT	(x-2)^2/((x-3)*(2*x^4-4*(2+3)*x^3+6*(2^2+3^2)*x^2-4*(2^3+3^3)*x+2^4+3^4)^(3/4))	Problem gg6 (ii) instantiated a=2,b=3	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_7-ix	PMT	(x^2+x+2)*(x^3-1)^(1/3)/(x*(x^2-1)^2*(x^3+x^2-2*x-3))	Problem gg7 (ix) cube roots of cubics	FriCAS ERROR; book elem
generalised_gunther-prob_gg_8-ii	PMT	(x-3)/((x+2)*(x^2+1)*(x^2-3)^(1/4))	Problem gg8 (ii) fourth roots of quadrat…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_8-iv	PMT	(x+3)*(x^2-3)^(1/4)/((x-2)*(x-1)^2*(x^2+1))	Problem gg8 (iv) fourth roots of quadrat…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_8-v	PMT	(x-3)*(x^2-3)^(3/4)/((x+1)^4*(x+2)*(x^2+1))	Problem gg8 (v) fourth roots of quadrati…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_8-vii	PMT	x^2*(x+6)/((x^3+x+4)*(x^2+4*x)^(3/4))	Problem gg8 (vii) fourth roots of quadra…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_8-viii	PMT	(x+3)*(2*x+9)/(x*(x^3+x+3)*(x^2+3*x)^(1/4))	Problem gg8 (viii) fourth roots of quadr…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_8-ix	PMT	(x+2)*(x+3)/(x^3*(x^3+x+2)*(x^2+2*x)^(1/4))	Problem gg8 (ix) fourth roots of quadrat…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_9-i-inst	PMT	x/((2*2*x^3+3)*(2*x^3+3)^(1/3))	Problem gg9 (i) instantiated a=2,b=3	FriCAS ELEM; book elem
generalised_gunther-prob_gg_9-ii-inst	PMT	x^3/((2*2*x^3+3)*(2*x^3+3)^(1/3))	Problem gg9 (ii) instantiated a=2,b=3	FriCAS ELEM; book elem
generalised_gunther-prob_gg_9-iii_n1	PMT	(2*a*x^3+b)*x/((a^2*m*x^6+a*b*m*x^3+b^2*(m-3))*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=1	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_9-iii_n1-ins…	PMT	(2*2*x^3+3)*x/((2^2*5*x^6+2*3*5*x^3-3^2*(m-3))*(2*x^3+3)^(1/3))	Problem gg9 (iii) general form with n=1 …	FriCAS ELEM; book elem
generalised_gunther-prob_gg_9-iii_n2	PMT	(3*a*x^3+2*b)*x^3/((a^2*m*x^9+a*b*m*x^6+b^2*(m-4))*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=2	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_9-iii_n2-ins…	PMT	(3*2*x^3+2*3)*x^3/((2^2*5*x^9+2*3*5*x^6-3^2*(m-4))*(2*x^3+3)^(1/3))	Problem gg9 (iii) general form with n=2 …	FriCAS ELEM; book elem
generalised_gunther-prob_gg_9-iii_n3	PMT	(4*a*x^3+3*b)*x^5/((a^2*m*x^12+a*b*m*x^9+b^2*(m-5))*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=3	FriCAS ELEM; book elem

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generalised_gunther-prob_gg_9-iii_n3-ins…	PMT	(4*2*x^3+3*3)*x^5/((2^2*5*x^12+2*3*5*x^9+3^2)^(1/2)*x^3+3^2)^(1/2))	Problem gg9 (iii) general form with n=3 …	FriCAS ELEM; book elem
generalised_gunther-prob_gg_10-n2	PMT	1/((2*b*c/a*x^2+c)*Sqrt[a+b*x^2])	Problem gg10 Albert Rich form x^(n-2)/((…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_10-n3-inst	PMT	x/((2*3*5/2*x^3+5)*(2+3*x^3)^(1/3))	Problem gg10 Albert Rich form x^(n-2)/((…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_10-n4	PMT	x^2/((2*b*c/a*x^4+c)*(a+b*x^4)^(1/4))	Problem gg10 Albert Rich form x^(n-2)/((…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_10-n4-inst	PMT	x^2/((2*3*5/2*x^4+5)*(2+3*x^4)^(1/4))	Problem gg10 Albert Rich form x^(n-2)/((…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_11-i_plus	PMT	1/((x+1)*(x^3+x)^(1/4))	Problem gg11 (i) sign +	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-i_minus	PMT	1/((x-1)*(x^3+x)^(1/4))	Problem gg11 (i) sign -	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-ii_minus	PMT	1/((x-1)*(x^2-2*x^3)^(1/4))	Problem gg11 (ii) signs - -	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-iii_plus	PMT	1/((a*x+b)*(a^2*x^3+b^2*x)^(1/4))	Problem gg11 (iii) sign +	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-iii_plus-…	PMT	1/((2*x+3)*(2^2*x^3+3^2*x)^(1/4))	Problem gg11 (iii) sign + instantiated a…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-iii_minus	PMT	1/((a*x-b)*(a^2*x^3+b^2*x)^(1/4))	Problem gg11 (iii) sign -	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_11-iii_minus…	PMT	1/((2*x-3)*(2^2*x^3+3^2*x)^(1/4))	Problem gg11 (iii) sign - instantiated a…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_12-i_plus	PMT	4*x^2/((x^4+1)*(x^8+1)^(1/4))	Problem gg12 (i) sign + (variable u rena…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_12-i_minus	PMT	4*x^2/((x^4-1)*(x^8+1)^(1/4))	Problem gg12 (i) sign - (variable u rena…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_12-ii_plus	PMT	2/((x^2+1)*(1+2*x^2)^(1/4))	Problem gg12 (ii) signs + + (u renamed x…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_12-ii_minus	PMT	2/((x^2-1)*(1-2*x^2)^(1/4))	Problem gg12 (ii) signs - - (u renamed x…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_12-main	PMT	(x^3+x)^(1/4)/((x-1)^2*(x+1))	Problem gg12 unnumbered target integral	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_12-iii	PMT	4*x^4*(x^8+1)^(1/4)/((x^4-1)^2*(x^4+1))	Problem gg12 (iii) (u renamed x)	FriCAS ELEM; book elem
generalised_gunther-prob_gg_13-i-inst	PMT	(2*5*x^4+2*3*5*x^2+3*7)*Sqrt[(2*x^2+3)/(5*x^2+7)]/(5*x^2+7)^2/(2*2^2)*x^4-2*2*3*x…	Problem gg13 (i) rational radicand insta…	FriCAS ELEM; book elem
generalised_gunther-prob_gg_13-ii	PMT	(2*a*c*x^4+(a*d+3*b*c)*x^2+2*b*d)*Sqrt[(b+a*x^2)/(c+d*x^2)]/(c+d*x^2)^2/(2*a^2*c^2-a*x^…	Problem gg13 (ii) rational radicand	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_13-ii-inst	PMT	(2*2*5*x^4+(2*7+3*3*5)*x^2+2*3*7)*Sqrt[(2+3*x^2)/(5*x^2+7)]/(5*x^2+7)^2/(2*2^2*x^…	Problem gg13 (ii) rational radicand inst…	FriCAS ELEM; book elem

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generalised_gunther-prob_gg_13-iii	PMT	(2*a*c*x^3+a*d*x^2+4*b*c*x+3*b*d)/(a*x^2-b)*((a*x^2+b)/(c*x+d))^(1/3)…	Problem gg13 (iii) rational radicand	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_13-iii-inst	PMT	(2*2*5*x^3+2*7*x^2+4*3*5*x+3*3*7)/(2*x^2-3)*((2*x^2+3)/(5*x+7))^(1/3)…	Problem gg13 (iii) rational radicand ins…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_13-iv	PMT	(a*c*x^6+2*b*c*x^3+b*d)*((a*x^3+b)/(c*x^3+d))^(1/3)/(a*x^3-b)^2/(a*x^3-b))	Problem gg13 (iv) rational radicand	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_13-iv-inst	PMT	(2*5*x^6+2*3*5*x^3+3*7)*((2*x^3+3)/(5*x^3+7))^(1/3)/(2*x^3-3)^2/(2*x^3-3))	Problem gg13 (iv) rational radicand inst…	FriCAS TIMEOUT; book elem
generalised_gunther-prob_gg_14-i_plus	PMT	1/((2*x+3)*(1-4*x^2)^(1/3))	Problem gg14 (i) sign + (no statement)	FriCAS TIMEOUT; book ?
generalised_gunther-prob_gg_14-i_minus	PMT	1/((2*x-3)*(1-4*x^2)^(1/3))	Problem gg14 (i) sign - (no statement)	FriCAS TIMEOUT; book ?
generalised_gunther-prob_gg_14-ii_plus	PMT	1/((x+1)*(1-9*x^2)^(1/3))	Problem gg14 (ii) sign + (no statement)	FriCAS TIMEOUT; book ?
generalised_gunther-prob_gg_14-ii_minus	PMT	1/((x-1)*(1-9*x^2)^(1/3))	Problem gg14 (ii) sign - (no statement)	FriCAS TIMEOUT; book ?
generalised_gunther-prob_gg_15-ii_plus	PMT	1/((2*x+1)*(1-8*x^3)^(1/3))	Problem gg15 (ii) 2x+1 with 1-8x^3 (no s…	FriCAS TIMEOUT; book ?
generalised_gunther-prob_gg_15-ii_minus	PMT	1/((2*x-1)*(1+8*x^3)^(1/3))	Problem gg15 (ii) 2x-1 with 1+8x^3 (no s…	FriCAS TIMEOUT; book ?
chebychev-chebychev_1-iii	PMT	x^(4/3)/(a*Sqrt[x]+b)^(2/3)	Problem chebychev_1 (iii)	FriCAS TIMEOUT; book elem
chebychev-chebychev_1-iii-inst	PMT	x^(4/3)/(2*Sqrt[x]+3)^(2/3)	Problem chebychev_1 (iii) instantiated a…	FriCAS TIMEOUT; book elem
chebychev-chebychev_1-iv	PMT	(a*x^(3/4)+b)^(2/3)/x^3	Problem chebychev_1 (iv)	FriCAS TIMEOUT; book elem
chebychev-chebychev_3-iii-inst	PMT	x^(1/3)*(2*x^2+3)^(1/3)	Problem chebychev_3 (iii) instantiated a…	FriCAS TIMEOUT; book elem
chebychev-chebychev_3-v-inst	PMT	1/(2*x^3+3)^(1/3)	Problem chebychev_3 (v) instantiated a=2…	FriCAS ELEM; book elem
chebychev-chebychev_3-viii	PMT	x^(4/5)*(a/x^(4/5)+b)^(5/4)	Problem chebychev_3 (viii) (expect by bo…	FriCAS TIMEOUT; book elem
chebychev-chebychev_3-x	PMT	x^(3/2)/Sqrt[a/Sqrt[x]+b]	Problem chebychev_3 (x) (expect by book …	FriCAS ELEM; book elem
chebychev-chebychev_3c-i_n3	PMT	1/(Sin[x])^(1/3)	Problem chebychev_3c (i) 1/sin^(1/n) n=3…	FriCAS NONELEM; book nonelem
chebychev-chebychev_3c-iv_n3	PMT	(Sin[x])^(1/3)	Problem chebychev_3c (iv) 1/csc^(1/n)=si…	FriCAS NONELEM; book nonelem

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chebychev-chebychev_5-v	PMT	(a*Sqrt[x]+b)^(2/3)/(c*Sqrt[x]+d)^(2/3)	Problem chebychev_5 (v) type IV conjectu…	FriCAS TIMEOUT; book elem
chebychev-chebychev_5-v-inst	PMT	(2*Sqrt[x]+3)^(2/3)/(5*Sqrt[x]+7)^(2/3)	Problem chebychev_5 (v) type IV conjectu…	FriCAS TIMEOUT; book elem
chebychev-chebychev_5-viii	PMT	(a*x^(1/3)+b)^(5/3)/(c*x^(1/3)+d)^(2/3)	Problem chebychev_5 (viii) type V conjec…	FriCAS TIMEOUT; book elem
chebychev-chebychev_5-viii-inst	PMT	(2*x^(1/3)+3)^(5/3)/(5*x^(1/3)+7)^(2/3)	Problem chebychev_5 (viii) type V conjec…	FriCAS TIMEOUT; book elem
inverse_function-inverse_function_2-v	PMT	(2^(1/3)*(Sqrt[x^2+4]+x)^(2/3)-2)^(1/3)/(Sqrt[x^2+4]+x)^(1/9)	Problem inverse_function_2 (v)	FriCAS TIMEOUT; book elem
realis-prob_realis_4-ii	PMT	(a-2*x^2)/((2*x^4-2*a*x^2+a)*Sqrt[(a-x^2)*(x^4-a*x^2+a)])	Problem prob_realis_4 (ii) (a=2 gives (i…	FriCAS ELEM; book elem
nested-nested_3-i-inst	PMT	(2/x^3+3*x*Sqrt[x])^(1/3)	Problem nested_3 (i), n=-3; inst a=2, b=…	FriCAS TIMEOUT; book elem
nested-nested_3-iii	PMT	(a*x+b*x*Sqrt[x])^(1/3)	Problem nested_3 (iii), n=1	FriCAS TIMEOUT; book elem
nested-nested_3-iii-inst	PMT	(2*x+3*x*Sqrt[x])^(1/3)	Problem nested_3 (iii), n=1; inst a=2, b…	FriCAS TIMEOUT; book elem
nested-nested_4-ex1-inst	PMT	(2/x^4+3*x^(4/3))^(1/4)	Problem nested_4, solution example 1 (n=…	FriCAS TIMEOUT; book elem
nested-nested_6-i	PMT	Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]/Sqrt[a/x+b]	Problem nested_6 (i)	FriCAS ELEM; book elem
nested-nested_6-ii	PMT	Sqrt[a/x+b]/Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]	Problem nested_6 (ii)	FriCAS ELEM; book elem
nested-nested_6-iii	PMT	Sqrt[a/x+b]*Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]	Problem nested_6 (iii)	FriCAS ELEM; book elem
nested-nested_6-iv	PMT	1/(Sqrt[a/x+b]*Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]])	Problem nested_6 (iv)	FriCAS ELEM; book elem
nested-nested_9-i-inst	PMT	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/4)	Problem nested_9 (i); inst a=2, b=3, c=5	FriCAS ELEM; book elem
nested-nested_9-ii-inst	PMT	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(3/4)	Problem nested_9 (ii); inst a=2, b=3, c=…	FriCAS ELEM; book elem
nested-nested_9-iii-inst	PMT	(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/4)/(3*x+5)	Problem nested_9 (iii); inst a=2, b=3, c…	FriCAS ELEM; book elem
nested-nested_9-iv-inst	PMT	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(5/6)	Problem nested_9 (iv); inst a=2, b=3, c=…	FriCAS ELEM; book elem
nested-nested_9-v-inst	PMT	x/(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/8)	Problem nested_9 (v); inst a=2, b=3, c=5	FriCAS ELEM; book elem
nested-nested_12-ii-inst	PMT	(2*x-Sqrt[2^2*x^2+3])^(1/3)/(x*Sqrt[2^2*x^2+3])	Problem nested_12 (ii); inst a=2, b=3	FriCAS ELEM; book elem

id	method	integrand	book ref	detail
nested-nested_12-iii-inst	PMT	(2*x-Sqrt[2^2*x^2+3])^(1/3)/((2^2*x^2+3)^2*Sqrt[2^2*x^2+3])	Problem nested_12 (iii); inst a=2, b=3	FriCAS ELEM; book elem
nested-nested_14-iv-inst	PMT	Sqrt[Sqrt[2*x^2+3^2]+3]/((2*x^2-3)*Sqrt[2*x^2+3^2])	Problem nested_14 (iv); inst a=2, b=3	FriCAS TIMEOUT; book elem
nested-nested_14-v-inst	PMT	(2*x^2+3)^2*Sqrt[2*x^2+3^2]/((2*x^2-3)^2*Sqrt[Sqrt[2*x^2+3^2]+3])	Problem nested_14 (v); inst a=2, b=3	FriCAS TIMEOUT; book elem
nested-nested_17-i	PMT	((x^2+1)^(3/2)+x^3)^(1/3)/(x^2*(x^2+1))	Problem nested_17 (i)	FriCAS TIMEOUT; book elem
nested-nested_17-iii	PMT	(2*x^2+1)*(x^3*(x^2+1)^(3/2)+1)^(1/3)/(x^6*(x^2+1))	Problem nested_17 (iii)	FriCAS ERROR; book elem
nested-nested_19-xi	PMT	(2+1/x^2-Sqrt[8*x^2+1]/x^2)^(1/4)	Problem nested_19 (xi)	FriCAS TIMEOUT; book elem
multrad-multiple_radicals_0-v	PMT	x^2*(x^4-2*b*x^2+a*b)*Sqrt[x^2-a]/((d*x^8-2*a*d*x^6+(a^2*d+c)*x^4-2*b*c*x^2+b^2*c)*Sqrt[x^…	Problem multiple_radicals_0 (v)	FriCAS TIMEOUT; book elem
multrad-multiple_radicals_0-vi	PMT	(x^4+2*a*x^2+a^2)*(x^6+3*a*x^4-a*(2*a+3*b)*x^2+a^2*b)/((c*x^12-2*c*x^10+(a^2*c+4*a*b*c+b^2…	Problem multiple_radicals_0 (vi)	FriCAS TIMEOUT; book elem
multrad-multiple_radicals_0-vi-inst	PMT	(x^4+2*2*x^2+2^2)*(x^6+3*2*x^4-2*(2*2+3*3)*x^2+2^2*3^2)/((5*x^12-2*5*x^10+(2^2*5+4*2*3*5+3^2…	Problem multiple_radicals_0 (vi); inst a…	FriCAS TIMEOUT; book elem
multrad-multiple_radicals_2-iii	PMT	(x^4+2*a*x^2-a*b)*Sqrt[x^2+a]/((x^4-(b+1)*x^2-a)*(x^4+(1-b)*x^2+a)*Sqrt[x^2-b])	Problem multiple_radicals_2 (iii)	FriCAS ELEM; book ?
multrad-multiple_radicals_3-i	PMT	(x^2-2*a*x+a*b+a*c-b*c)*Sqrt[x-b]*Sqrt[x-c]/((x^2-(b+c-1)*x+b*c-a)*(x-a)^(3/2))	Problem multiple_radicals_3 (i)	FriCAS TIMEOUT; book ?
multrad-multiple_radicals_3-iii	PMT	(x^2-2*b*x+a*b-a*c+b*c)*Sqrt[x-b]/((x^2-(a+c-1)*x-b+a*c)*(x^2-(a+c+1)*x+b+a*c)*Sqrt[x-a]*S…	Problem multiple_radicals_3 (iii)	FriCAS ELEM; book ?
multrad-multiple_radicals_7-iv	PMT	ArcTan[d*Sqrt[a*x+b]-d*Sqrt[a*x+c]]	Problem multiple_radicals_7 (iv)	FriCAS ELEM; book elem
euler-prob_euler_outrageous_1-ii	PMT	(a^2*x^2-2*a*b*x+b^2)*(a^2*x^2+2*a*b*x+b^2)/((a^2*x^2+b^2)*(a^4*x^4+6*a^2*b^2*x^2+b^4)^(3/…	Problem prob_euler_outrageous_1 (ii)	FriCAS TIMEOUT; book elem
euler-prob_euler_outrageous_1-ii-inst	PMT	(2^2*x^2-2*2*3*x+3^2)*(2^2*x^2+2*2*3*x+3^2)/((2^2*x^2+3^2)*(2^4*x^4+6*2^2*3^2*x^2+3^4)^(3/…	Problem prob_euler_outrageous_1 (ii); in…	FriCAS TIMEOUT; book elem
euler-prob_euler_outrageous_2-m	PMT	1/((3-x^2)*(1-3*x^2)^(1/3))	Problem prob_euler_outrageous_2 (- sign)…	FriCAS TIMEOUT; book elem
euler-prob_euler_outrageous_3-ii-inst	Trager	(2*x^2-x-2*3)/((x^4-2*x^2+3)*(2*x^2-3)^(1/4))	Problem prob_euler_outrageous_3 (ii); in…	FriCAS TIMEOUT; book elem
euler-prob_euler_outrageous_4-iii	PMT	Sqrt[a*x^4+c*x^2+b]/(b-a*x^4)	Problem prob_euler_outrageous_4 (iii)	FriCAS ELEM; book elem
euler-prob_euler_outrageous_4-iv	PMT	x^2/((b-a*x^4)*Sqrt[a*x^4+b])	Problem prob_euler_outrageous_4 (iv)	FriCAS ELEM; book elem

id	method	integrand	book ref	detail
euler-prob_euler_outrageous_4-iv-inst	PMT	x^2/((3-2*x^4)*Sqrt[2*x^4+3])	Problem prob_euler_outrageous_4 (iv); in…	FriCAS ELEM; book elem
euler-prob_euler_outrageous_5	PMT	(a*x^2-b)/((a*x^2+b)*Sqrt[a^2*x^4+b^2])	Problem prob_euler_outrageous_5	FriCAS ELEM; book elem
timofeev-timofeev_2-i	PMT	1/((x^3-1)*(x^3+2)^(1/3))	Problem timofeev_2 (i)	FriCAS ELEM; book elem
timofeev-timofeev_2-v	PMT	(x^5-2)^2/(x^5+3)^(16/5)	Problem timofeev_2 (v)	FriCAS ELEM; book elem
timofeev-timofeev_2-vi	PMT	(x^5-2)^2/((x^5+3)^3*(x^5+3)^(1/5))	Problem timofeev_2 (vi)	FriCAS ELEM; book elem
timofeev-timofeev_2-vii	PMT	1/((x^3+3*x^2+3*x)*(x^3+3*x^2+3*x+3)^(1/3))	Problem timofeev_2 (vii)	FriCAS ELEM; book elem
timofeev-timofeev_3-v	PMT	((x^2+2)/x^2)^(7/9)/(x^2+2)^(3/2)	Problem timofeev_3 (v)	FriCAS ELEM; book ?
potpourri-p6-v	PMT	1/((x-1)*(x^3+x)^(1/4))	Potpourri problem 6 (v)	FriCAS TIMEOUT; book ?
potpourri-p6-xi	PMT	(x^4+4)*(x^4-1)^(3/4)/(x^8*(x^4-4))	Potpourri problem 6 (xi)	FriCAS ELEM; book ?
potpourri-p6-xiii	PMT	(x-2)/((x^2+2)*(2*x^2+x-1)^(1/3))	Potpourri problem 6 (xiii)	FriCAS TIMEOUT; book ?
potpourri-p6-xiv	PMT	(x^3+2)*(2*x^3+1)^(2/3)/(x^6*(x^3+1))	Potpourri problem 6 (xiv)	FriCAS TIMEOUT; book ?
potpourri-p6-xv	PMT	(x^6+x^3+1)/((1-x^6)*(x^5+x^3)^(1/4))	Potpourri problem 6 (xv)	FriCAS ERROR; book ?
potpourri-p6-xvi	PMT	x^3/((x^6-1)*(x^4+x^2)^(1/3))	Potpourri problem 6 (xvi)	FriCAS TIMEOUT; book ?
danger-d1-i-p	PMT	1/((3*x^4-4)*(x^2+1)^(1/4))	Danger zone problem 1 (i), x^2+1; book t…	FriCAS TIMEOUT; book elem
danger-d1-i-m	PMT	1/((3*x^4-4)*(x^2-1)^(1/4))	Danger zone problem 1 (i), x^2-1; book t…	FriCAS TIMEOUT; book elem
danger-d1-ii-p	PMT	x^2/((3*x^4-4)*(x^2+1)^(1/4))	Danger zone problem 1 (ii), x^2+1; book …	FriCAS TIMEOUT; book elem
danger-d1-ii-m	PMT	x^2/((3*x^4-4)*(x^2-1)^(1/4))	Danger zone problem 1 (ii), x^2-1; book …	FriCAS TIMEOUT; book elem
danger-d1-iii-p	PMT	1/((x^4+4)*(x^2+1)^(1/4))	Danger zone problem 1 (iii), x^2+1; book…	FriCAS TIMEOUT; book elem
danger-d1-iii-m	PMT	1/((x^4+4)*(x^2-1)^(1/4))	Danger zone problem 1 (iii), x^2-1; book…	FriCAS TIMEOUT; book elem
danger-d1-iv-p	PMT	x^2/((x^4+4)*(x^2+1)^(1/4))	Danger zone problem 1 (iv), x^2+1; book …	FriCAS TIMEOUT; book elem
danger-d1-iv-m	PMT	x^2/((x^4+4)*(x^2-1)^(1/4))	Danger zone problem 1 (iv), x^2-1; book …	FriCAS TIMEOUT; book elem
danger-d1-v-p	Trager	1/((x+1)*(1-5*x^2)^(1/5))	Danger zone problem 1 (v), x+1	FriCAS ERROR; book elem
danger-d1-v-p	PMT	1/((x+1)*(1-5*x^2)^(1/5))	Danger zone problem 1 (v), x+1	FriCAS ERROR; book elem

id	method	integrand	book ref	detail
danger-d1-v-m	Trager	1/((x-1)*(1-5*x^2)^(1/5))	Danger zone problem 1 (v), x-1	FriCAS ERROR; book elem
danger-d1-v-m	PMT	1/((x-1)*(1-5*x^2)^(1/5))	Danger zone problem 1 (v), x-1	FriCAS ERROR; book elem
multmeth-m1	PMT	(a^3*x^3-b^3)^(1/3)/(a*x+b)	Multiple methods problem 1 [4]	FriCAS ERROR; book ?
multmeth-m1-inst	PMT	(2^3*x^3-3^3)^(1/3)/(2*x+3)	Multiple methods problem 1 [4]; inst a=2…	FriCAS TIMEOUT; book ?
intro-multiple_approaches-1	PMT	1/((x-1)*Sqrt[x^2+b*x+c])	Introduction, Multiple Approaches to Int…	FriCAS ELEM; book elem

Elementary per book or FriCAS, but declined (646)

id	method	integrand	book ref	detail
warmup-prob_warmup_2b-minus-n1	PMT	((2*1+1)*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^1-1])	Problem 0.2b Realis lower sign n=1	{“failed”, “verification failed”}
warmup-prob_warmup_2b-plus-n2	PMT	((2*2+1)*x-1)/((x^2-1)*Sqrt[(x+1)*(x^2-1)^2-1])	Problem 0.2b Realis upper sign n=2	{“needs torsion realisation (milestone iii)”, {{-1 + Sqrt[x], {-2*I, 2*I}}, {1 + Sqrt[x], {2*I, -2*I}}, {1 + x…
warmup-prob_warmup_2b-minus-n2	PMT	((2*2+1)*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^2-1])	Problem 0.2b Realis lower sign n=2	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-3*I, 3*I}}, {1 + x, {-2*I, 2*I}}}}
warmup-prob_warmup_2b-plus-n3	PMT	((2*3+1)*x-1)/((x^2-1)*Sqrt[(x+1)*(x^2-1)^3-1])	Problem 0.2b Realis upper sign n=3	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-3*I, 3*I}}, {1 + x, {-4*I, 4*I}}}}
warmup-prob_warmup_2b-minus-n3	PMT	((2*3+1)*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^3-1])	Problem 0.2b Realis lower sign n=3	{“needs torsion realisation (milestone iii)”, {{-1 + Sqrt[x], {-4*I, 4*I}}, {1 + Sqrt[x], {4*I, -4*I}}, {1 + x…
warmup-prob_warmup_2c-iii	Trager	x^4/(x^6-b^6)	Problem 0.2c (iii) rational	{“failed”, “integrand is not algebraic over Q(x)”}
warmup-prob_warmup_2c-iv	Trager	(x^4+a)/(x^4+b)^2	Problem 0.2c (iv) rational	{“failed”, “integrand is not algebraic over Q(x)”}
warmup-prob_warmup_2cc-i	PMT	x/(x^8+1)	Problem 1.2cc (i) rational	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, 1 + x^8, (-1/8…
warmup-prob_warmup_2e	PMT	(189*x^5+945*x^4+1131*x^3-147*x^2-110*x-28)/(9*x^6+54*x^5+74*x^4-14*x^3+25*x^2-8*x+4)	Problem 2.2e (and 2.2f) Lazard-Rioboo-Tr…	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, 4 - 8 x + 25 x…
warmup-prob_warmup_3-i	Trager	(x^2-a)/(Sqrt[x+1]+b)	Problem 0.3 (i) Q(x,(ax+b)^(m/n))	{“failed”, “integrand is not a rational function of x and the radical over Q”}

id	method	integrand	book ref	detail
warmup-prob_warmup_3-ii	Trager	((a*x+b)^(1/3)+c)/((a*x+b)^(1/3)+d)	Problem 0.3 (ii) Q(x,(ax+b)^(m/n))	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_3-iii	Trager	(Sqrt[a*x+b]+a)/((a*x+b)^(1/4)+b)	Problem 0.3 (iii) Q(x,(ax+b)^(m/n))	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-i	Trager	1/Sqrt[a*x^2+b*x+c]	Problem 0.4 (i) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-i	PMT	1/Sqrt[a*x^2+b*x+c]	Problem 0.4 (i) Euler substitution	{“failed”, “no solution within bounds”, {0}}
warmup-prob_warmup_4-ii	Trager	Sqrt[a*x^2+b*x+c]	Problem 0.4 (ii) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-ii	PMT	Sqrt[a*x^2+b*x+c]	Problem 0.4 (ii) Euler substitution	{“failed”, “no solution within bounds”, {2}}
warmup-prob_warmup_4-iii	Trager	Sqrt[a*x^2+b*x+c]/x^2	Problem 0.4 (iii) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-iv	Trager	x^2*Sqrt[a*x^2+b*x+c]	Problem 0.4 (iv) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-v	Trager	(a*x^2+b*x+c)^(3/2)/x^2	Problem 0.4 (v) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-vi	Trager	x*(a*x^2+b*x+c)^(3/2)	Problem 0.4 (vi) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-vii	Trager	(a*x^2+b*x+c)^(5/2)/x	Problem 0.4 (vii) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-viii	Trager	x*(a*x^2+b*x+c)^(5/2)	Problem 0.4 (viii) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-ix	Trager	1/((x+b)^2*(a*x^2+b*x+c)^(3/2))	Problem 0.4 (ix) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-x	Trager	1/((x+b)^2*(a*x^2+b*x+c)^(5/2))	Problem 0.4 (x) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_4-xi	Trager	x/((a*x^2+b*x+c)^(3/2)-(a*x^2+b*x+c)^(5/2))	Problem 0.4 (xi) Euler substitution	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_5	Trager	((a*x+b)^(2/3)+1)/((a*x+b)^(1/2)+(a*x+b)^(1/4))	Problem 0.5 multiple radicals of ax+b	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_7	Trager	((x-5)*(3*x-1)^(5/2))/((x+2)*(8-7*x)^(3/2))	Problem 0.7 two linear square roots	{“failed”, “more than one radicand (a radical compositum)”}
warmup-prob_warmup_7	PMT	((x-5)*(3*x-1)^(5/2))/((x+2)*(8-7*x)^(3/2))	Problem 0.7 two linear square roots	{“failed”, “no solution within bounds”, {5}}
warmup-prob_warmup_10-i	Trager	1/(1-Sin[x])	Problem 0.10 (i) tangent half-angle	{“failed”, “transcendental function in the integrand”}
warmup-prob_warmup_10-ii	Trager	(Sin[x]-1)/(Sin[x]+Cos[x])	Problem 0.10 (ii) tangent half-angle	{“failed”, “transcendental function in the integrand”}
warmup-prob_warmup_10-iii	Trager	(1/Cos[x]-Cos[x]/Sin[x])/(Cos[x]-Sin[x]+1)	Problem 0.10 (iii) tangent half-angle, s…	{“failed”, “transcendental function in the integrand”}
warmup-prob_warmup_11-ii	PMT	1/(x*(x/(1-x))^(3/2))	Problem 0.11 (ii)	{“failed”, “verification failed”}
warmup-prob_warmup_13-iii	Trager	1/(x*Sqrt[a*x^2+b*x+c])	Problem 1.13 (iii)	{“failed”, “radicand is not a rational function over Q”}
warmup-prob_warmup_14-ii	Trager	((Exp[x]-Exp[-x])/2)/(1+(Exp[x]-Exp[-x])/2)	Problem 1.14 (ii) sinh(x)/(1+sinh(x)), s…	{“failed”, “transcendental function in the integrand”}
warmup-prob_warmup_14-iii	Trager	1/(((Exp[x]+Exp[-x])/2)^3-((Exp[x]+Exp[-x])/2))	Problem 1.14 (iii) 1/(cosh(x)^3-cosh(x))…	{“failed”, “transcendental function in the integrand”}
warmup-prob_warmup_15-i	Trager	1/((x-c)*Sqrt[a*x+b])	Problem 1.15 (i)	{“failed”, “radicand is not a rational function over Q”}

id	method	integrand	book ref	detail
warmup-prob_warmup_15-ii	Trager	Sqrt[a*x+b]/(x-c)	Problem 1.15 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_1	Trager	(a*x^2-b)/((a*x^2+b)*Sqrt[a^2*x^4+b^2])	Problem 1	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_5-i	Trager	((x^5+1)*(4*x^5-1))/(x*(x^5-a*x+1)*Sqrt[x^6+x])	Problem 5 (i)	{“failed”, “integrand is not a rational function of x and the radical over Q”}
laurent_poly_subst-prob_laurent_sub_5-i	PMT	((x^5+1)*(4*x^5-1))/(x*(x^5-a*x+1)*Sqrt[x^6+x])	Problem 5 (i)	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_5-ii	Trager	((x^5-1)*(4*x^5+1))/(x*(x^5-a*x-1)*Sqrt[x^6-x])	Problem 5 (ii)	{“failed”, “integrand is not a rational function of x and the radical over Q”}
laurent_poly_subst-prob_laurent_sub_5-ii	PMT	((x^5-1)*(4*x^5+1))/(x*(x^5-a*x-1)*Sqrt[x^6-x])	Problem 5 (ii)	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_7-i	Trager	(c*x^3-d)/(a*x^3-b)^(1/3)	Problem 7 (i)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_7-ii	Trager	(c*x^3+d)/(a*x^3-b)^(1/3)	Problem 7 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_8-i	Trager	(a*x^4+b*x^3)^(1/4)	Problem 8 (i)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_8-i	PMT	(a*x^4+b*x^3)^(1/4)	Problem 8 (i)	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_8-ii	Trager	(a*x^8+b*x^7)^(1/4)	Problem 8 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_8-ii	PMT	(a*x^8+b*x^7)^(1/4)	Problem 8 (ii)	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_8-ii…	Trager	(a*x^12+b*x^11)^(1/4)	Problem 8 (iii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_8-ii…	PMT	(a*x^12+b*x^11)^(1/4)	Problem 8 (iii)	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_9	Trager	(a*x^4-b)/(x^4*(2*a*x^4-b)^(1/4))	Problem 9 (same as Problem 14b (i))	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_11	Trager	(b*x^4+c)^4/(x^4+a*x^2)^(1/4)	Problem 11	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_11	PMT	(b*x^4+c)^4/(x^4+a*x^2)^(1/4)	Problem 11	{“failed”, “partially ramified branch prime”, x}
laurent_poly_subst-prob_laurent_sub_11-i…	PMT	(3*x^4+5)^4/(x^4+2*x^2)^(1/4)	Problem 11 instantiated a=2,b=3,c=5	{“failed”, “partially ramified branch prime”, x}
laurent_poly_subst-prob_laurent_sub_13-i	Trager	(a*x^2+b)/((a*x^2-b)*Sqrt[a*x^3+b*x^2-b*x])	Problem 13 (i)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_13-i…	PMT	(2*x^2+3)/((2*x^2-3)*Sqrt[2*x^3+3*x^2-3*x])	Problem 13 (i) instantiated a=2,b=3	{“failed”, “verification failed”}
laurent_poly_subst-prob_laurent_sub_13-i…	Trager	((a*x^3-2*c)*Sqrt[a*x^3+b*x^2+c])/(a*x^3+c)^2	Problem 13 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_subst-prob_laurent_sub_13-i…	PMT	((a*x^3-2*c)*Sqrt[a*x^3+b*x^2+c])/(a*x^3+c)^2	Problem 13 (ii)	{“failed”, “deep residue sheet undecided”, c + a x^3}
laurent_poly_subst-prob_laurent_sub_13-i…	PMT	((4*x+3)*Sqrt[2*x^4-2*x^2-x])/((2*x+1)*(x^3+2*x+1))	Problem 13 (iv)	{“failed”, “verification failed”}

id	method	integrand	book ref	detail
laurent_poly_substprob_laurent_sub_13-v…	Trager	((2*c-a*x^3)*Sqrt[a*x^3+b*x^2+c])/((a*x^3+(b-3)*x^2+c)*(a*x^3+(b-2)*x^2+c))	Problem 13 (vii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_substprob_laurent_sub_14b-…	Trager	(x^3+4*b)/((x^3+b)*(a*x^4-x^3-b)^(1/4))	Problem 14b (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_substprob_laurent_sub_16-i	PMT	((x^2+1)/(x*(x^2-1)))*Sqrt[(x^2-x-1)/(x^2+x-1)]	Problem 16 (i)	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-I, I}}, {1 + x, {I, -I}}}}
laurent_poly_substprob_laurent_sub_16-i…	Trager	((a*x^2-b)/(x*(a*x^2+b)))*Sqrt[(a*x^2-c*x+b)/(a*x^2+c*x+b)]	Problem 16 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_substprob_laurent_sub_17-i	PMT	((2*x^3-1)/(x*(x^3+1)))*Sqrt[(x^6+2*x^3-x^2+1)/(x^6+2*x^3+x^2+1)]	Problem 17 (i)	{“needs torsion realisation (milestone iii)”, {{x, {-1, 1}}, {1 + x, {I, -I}}, {1 - x + x^2, {I, -I, -I, I}}}}
laurent_poly_substprob_laurent_sub_17-i…	Trager	((2*a*x^3+b)*(a^2*x^6-2*a*b*x^3-a*b*x^2+b^2))/(x*(a*x^3-b)*(a^2*x^6-2*a*b*x^3+x^2+b^2))*Sq…	Problem 17 (ii)	{“failed”, “radicand is not a rational function over Q”}
laurent_poly_substprob_laurent_sub_17-i…	PMT	((2*2*x^3+3)*(2^2*x^6-2*2*3*x^3-2*3*x^2+3^2))/(x*(2*x^3-3)*(2^2*x^6-2*2*3*x^3+x^2+3^2))*Sq…	Problem 17 (ii) instantiated a=2,b=3,c=5	{“needs torsion realisation (milestone iii)”, {{x, {-1, 1}}, {-3 + 2 x^3, {-6*I, 6*I, 6*I, -6*I, 6*I, -6*I}}}}
laurent_poly_substprob_laurent_sub_18-i…	Trager	((2*a*x^3+b)/(x*(a*x^3+x-b)))*((a*x^3-a*x-b)/(a*x^3+b*x-b))^(1/3)	Problem 18 (ii)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_1-v	Trager	1/((x+1)*((a+1)*x^3+3*(1-a)*x^2+3*(a+1)*x-a+1)^(1/3))	Problem 1 (v)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_1-vi	Trager	1/((x-1)*((a-1)*x^3+3*(a+1)*x^2+3*(a-1)*x+a+1)^(1/3))	Problem 1 (vi)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_2-ii	PMT	(x^2+2*x-1)/(x*(x-1)*Sqrt[2*x^4-2*x^3+3*x^2+1])	Problem 2 (ii)	{“needs torsion realisation (milestone iii)”, {{-1 + x, {1, -1}}, {x, {1, -1}}}}
rational_subsprob_rational_sub_2-iii	Trager	(b*x^2-2*a*x-a*b)/((x^2+a)*Sqrt[2*x^4+2*b*x^3+(2*a+b^2)*x^2+a^2])	Problem 2 (iii)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_2-iv	Trager	(b*x^2-2*a*x-a*b)/(x*(x+b)*Sqrt[2*x^4+2*b*x^3+(2*a+b^2)*x^2+a^2])	Problem 2 (iv)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_2-iv-ins…	PMT	(3*x^2-2*2*x-2*3)/(x*(x+3)*Sqrt[2*x^4+2*3*x^3+(2*3^2+3^2)*x^2+2^2])	Problem 2 (iv) instantiated a=2,b=3	{“needs torsion realisation (milestone iii)”, {{x, {-1, 1}}, {3 + x, {-1, 1}}}}
rational_subsprob_rational_sub_3-i	PMT	((2*x^2-2*x-1)*Sqrt[x^4+x])/(2*x-1)^3	Problem 3 (i)	{“failed”, “verification failed”}
rational_subsprob_rational_sub_3-ii	Trager	((a*x^2-2*b*x-b)*Sqrt[a*(x^2+x)*(a*x^2-a*x+2*b)])/(a*(b-a*x)*(a*x-b)^2)	Problem 3 (ii)	{“failed”, “radicand is not a rational function over Q”}
rational_subsprob_rational_sub_3-ii	PMT	((a*x^2-2*b*x-b)*Sqrt[a*(x^2+x)*(a*x^2-a*x+2*b)])/(a*(b-a*x)*(a*x-b)^2)	Problem 3 (ii)	{“failed”, “deep residue sheet undecided”, -b + a x}
rational_subsprob_rational_sub_3-ii-ins…	PMT	((2*x^2-2*3*x-3)*Sqrt[2*(x^2+x)*(2*x^2-2*x+2*3)])/(2*(3-2*x)*(2*x-3)^2)	Problem 3 (ii) instantiated a=2,b=3	{“failed”, “verification failed”}

id	method	integrand	book ref	detail
gunther-prob_gunther_1-i	PMT	x/((2*x^3-1)*Sqrt[x^4-x])	Problem 1 (i)	{“failed”, “verification failed”}
gunther-prob_gunther_1-iii	PMT	(x^2-1)/((x^2+x+1)*Sqrt[x^3+x])	Problem 1 (iii)	{“failed”, “verification failed”}
gunther-prob_gunther_2-i	PMT	(x-1)/((x+1)*Sqrt[x^3+x])	Problem 2 (i)	{“failed”, “verification failed”}
gunther-prob_gunther_2-ii	Trager	(a*x-b)/((a*x+b)*Sqrt[a^2*x^3+b^2*x])	Problem 2 (ii)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_2-ii-inst	PMT	(2*x-3)/((2*x+3)*Sqrt[2^2*x^3+3^2*x])	Problem 2 (ii) instantiated a=2,b=3	{“failed”, “verification failed”}
gunther-prob_gunther_3-i	Trager	x/Sqrt[a*x^4+b*x]	Problem 3 (i)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_3-i	PMT	x/Sqrt[a*x^4+b*x]	Problem 3 (i)	{“failed”, “no solution within bounds”, {0}}
gunther-prob_gunther_3-i-inst	PMT	x/Sqrt[2*x^4+3*x]	Problem 3 (i) instantiated a=2,b=3	{“failed”, “no solution within bounds”, {0}}
gunther-prob_gunther_3-ii	Trager	(2*a*x^6-b)/((a*x^6+b)*Sqrt[a*x^6-c*x^2+b])	Problem 3 (ii)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_3-iii	Trager	(3*a*x^5-2*b)/((a*x^5-c*x^2+b)*Sqrt[a*x^5+b])	Problem 3 (iii)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_3-iii-inst	PMT	(3*2*x^5-2*3)/((2*x^5-5*x^2+3)*Sqrt[2*x^5+3])	Problem 3 (iii) instantiated a=2,b=3,c=5	{“needs torsion realisation (milestone iii)”, {{-1 + x, {2/Sqrt[5], -2/Sqrt[5]}}, {3 + 6 x + 4 x^2 + 2 x^3, {-…
gunther-prob_gunther_3-iv	Trager	(a*x^2+b)/((a*x^2-b)*Sqrt[a*x^3+c*x^2-b*x])	Problem 3 (iv)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_3-iv-inst	PMT	(2*x^2+3)/((2*x^2-3)*Sqrt[2*x^3+5*x^2-3*x])	Problem 3 (iv) instantiated a=2,b=3,c=5	{“failed”, “verification failed”}
gunther-prob_gunther_3-v	Trager	(4*d*x^5+b)/((a*d*x^5-c*x-a*b)*Sqrt[d*x^6-b*x])	Problem 3 (v)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_3-v	PMT	(4*d*x^5+b)/((a*d*x^5-c*x-a*b)*Sqrt[d*x^6-b*x])	Problem 3 (v)	{“failed”, “verification failed”}
gunther-prob_gunther_3-v-inst	PMT	(4*7*x^5+3)/((2*7*x^5-5*x-2*3)*Sqrt[7*x^6-3*x])	Problem 3 (v) instantiated a=2,b=3,c=5,d…	{“failed”, “verification failed”}
gunther-prob_gunther_3-vi	Trager	(x^2-1)/((x^2+1)*Sqrt[a*x^4+b*x^3+c*x^2+b*x+a])	Problem 3 (vi)	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_4	Trager	(2*a*x^4+4*a*x^3-b*x+b-2*c)/((a*x^4+x^2+(b+2)*x+c+1)*Sqrt[a*x^4+b*x+c])	Problem 4	{“failed”, “radicand is not a rational function over Q”}
gunther-prob_gunther_5-iv-mp	PMT	x^3/((x^6-8)*Sqrt[x^6+1])	Problem 5 (iv) x^6-8 with x^6+1, from (i…	{“needs torsion realisation (milestone iii)”, {{-2 + x^2, {1/36, -1/36, 1/36, -1/36}}, {4 + 2 x^2 + x^4, {1/64…
power_expand-power_expand_1-i	PMT	Sqrt[4*x^3-3*x-1]/x	Problem 1 (i)	{“failed”, “verification failed”}
power_expand-power_expand_3	Trager	Sqrt[x]/Sqrt[x^3+x^2-x-1]	Problem 3 (iii) AXIOM example	{“failed”, “more than one radicand (a radical compositum)”}
generalised_gunther-prob_gg_2-ii	Trager	(x-2*a+b)/((x^2+(a*b^2-2*b)*x+b^2-a^2*b^2)*((x-a)*(x-b))^(1/3))	Problem gg2 (ii)	{“failed”, “radicand is not a rational function over Q”}

id	method	integrand	book ref	detail
generalised_gunther-prob_gg_3-main	Trager	(a*x^2+2*b*x+3*c)/((x^3+a*x^2+b*x+c)^(1/3))	Problem gg3 (difficulty ?; elem by u=x/(…	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_6-i	Trager	(x+a)^2/((x+b)*(2*x^4+4*(a+b)*x^3+6*(a^2+b^2)*x^2+4*(a^3+b^3)*x+a^4+b^4))^(3/4))	Problem gg6 (i)	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_6-ii	Trager	(x-a)^2/((x-b)*(2*x^4-4*(a+b)*x^3+6*(a^2+b^2)*x^2-4*(a^3+b^3)*x+a^4+b^4)^(3/4))	Problem gg6 (ii)	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_7-x	PMT	(3*x+1)/((3*x-1)*(x^3-x)^(1/3))	Problem gg7 (x) cube roots of cubics	{“failed”, “verification failed”}
generalised_gunther-prob_gg_7-xi	PMT	(x-1)*(3*x+1)/((3*x-1)*(x^3-x)^(2/3))	Problem gg7 (xi) cube roots of cubics	{“failed”, “verification failed”}
generalised_gunther-prob_gg_7-xii	PMT	(x-1)^2*(3*x+1)/(x*(x+1)*(3*x-1)*(x^3-x)^(1/3))	Problem gg7 (xii) cube roots of cubics	{“failed”, “verification failed”}
generalised_gunther-prob_gg_7-xiii	PMT	x*(x+1)*(3*x+1)/((x-1)*(3*x-1)*(x^3-x)^(2/3))	Problem gg7 (xiii) cube roots of cubics	{“failed”, “verification failed”}
generalised_gunther-prob_gg_7-xiv	PMT	x*(x+1)*(3*x+1)/((x-1)^2*(3*x-1)*(x^3-x)^(1/3))	Problem gg7 (xiv) cube roots of cubics	{“failed”, “verification failed”}
generalised_gunther-prob_gg_9-i	Trager	x/((2*a*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (i)	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_9-i	PMT	x/((2*a*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (i)	{“failed”, “no solution within bounds”, {0}}
generalised_gunther-prob_gg_9-ii	Trager	x^3/((2*a*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (ii)	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_9-ii	PMT	x^3/((2*a*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (ii)	{“failed”, “no solution within bounds”, {0}}
generalised_gunther-prob_gg_9-iii_n1	Trager	(2*a*x^3+b)*x/((a^2*m*x^6+a*b*m*x^3+b^2*(m+1))*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=1	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_9-iii_n2	Trager	(3*a*x^3+2*b)*x^3/((a^2*m*x^9+a*b*m*x^6+b^2*(m+1)*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=2	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_9-iii_n3	Trager	(4*a*x^3+3*b)*x^5/((a^2*m*x^12+a*b*m*x^9+b^2*(m+1)*x^6+b*x^3+b)*(a*x^3+b)^(1/3))	Problem gg9 (iii) general form with n=3	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_10-n2	Trager	1/((2*b*c/a*x^2+c)*Sqrt[a+b*x^2])	Problem gg10 Albert Rich form x^(n-2)/((…	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_10-n3	Trager	x/((2*b*c/a*x^3+c)*(a+b*x^3)^(1/3))	Problem gg10 Albert Rich form x^(n-2)/((…	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_10-n3	PMT	x/((2*b*c/a*x^3+c)*(a+b*x^3)^(1/3))	Problem gg10 Albert Rich form x^(n-2)/((…	{“failed”, “no solution within bounds”, {0}}
generalised_gunther-prob_gg_10-n4	Trager	x^2/((2*b*c/a*x^4+c)*(a+b*x^4)^(1/4))	Problem gg10 Albert Rich form x^(n-2)/((…	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_11-ii_plus	PMT	1/((x+1)*(x^2+2*x^3)^(1/4))	Problem gg11 (ii) signs + +	{“failed”, “partially ramified branch prime”, x}
generalised_gunther-prob_gg_11-iii_plus	Trager	1/((a*x+b)*(a^2*x^3+b^2*x)^(1/4))	Problem gg11 (iii) sign +	{“failed”, “radicand is not a rational function over Q”}

id	method	integrand	book ref	detail
generalised_gunther-prob_gg_11-iii_minus	Trager	1/((a*x-b)*(a^2*x^3+b^2*x)^(1/4))	Problem gg11 (iii) sign -	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_11-iv_plus	Trager	1/((a*x+b)*(b*x^2+2*a*x^3)^(1/4))	Problem gg11 (iv) signs + +	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_11-iv_plus	PMT	1/((a*x+b)*(b*x^2+2*a*x^3)^(1/4))	Problem gg11 (iv) signs + +	{“failed”, “partially ramified branch prime”, x}
generalised_gunther-prob_gg_11-iv_plus-i…	PMT	1/((2*x+3)*(3*x^2+2*2*x^3)^(1/4))	Problem gg11 (iv) signs + + instantiated…	{“failed”, “partially ramified branch prime”, x}
generalised_gunther-prob_gg_11-iv_minus	Trager	1/((a*x-b)*(b*x^2-2*a*x^3)^(1/4))	Problem gg11 (iv) signs - -	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_11-iv_minus	PMT	1/((a*x-b)*(b*x^2-2*a*x^3)^(1/4))	Problem gg11 (iv) signs - -	{“failed”, “partially ramified branch prime”, x}
generalised_gunther-prob_gg_11-iv_minus-…	PMT	1/((2*x-3)*(3*x^2-2*2*x^3)^(1/4))	Problem gg11 (iv) signs - - instantiated…	{“failed”, “partially ramified branch prime”, x}
generalised_gunther-prob_gg_13-i	Trager	(a*c*x^4+2*b*c*x^2+b*d)*Sqrt[(a*x^2+b)/(c*x^2+d)]/(c^2*x^8+… a^2)*x^4-2*a*b*x…	Problem gg13 (i) rational radicand	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_13-i	PMT	(a*c*x^4+2*b*c*x^2+b*d)*Sqrt[(a*x^2+b)/(c*x^2+d)]/(c^2*x^8+… a^2)*x^4-2*a*b*x…	Problem gg13 (i) rational radicand	{“needs extension from QQ (milestone iii)”, {{b + a x^2 - d x^2 - c x^4, {-(2 a^3 + 4 b c Sqrt[4 b c + (a - …
generalised_gunther-prob_gg_13-ii	Trager	(2*a*c*x^4+(a*d+3*b*c)*x^2+2*b*d)*Sqrt[(a*x^2+b)/(c*x^2+d)]/(… a*x^…	Problem gg13 (ii) rational radicand	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_13-iii	Trager	(2*a*c*x^3+a*d*x^2+4*b*c*x+3*b*d)/((b*x+a)^(1/3)*… a*x^2-b)*((a*x^2+b)/(c*x+d))^(1/3)…	Problem gg13 (iii) rational radicand	{“failed”, “radicand is not a rational function over Q”}
generalised_gunther-prob_gg_13-iv	Trager	(a*c*x^6+2*b*c*x^3+b*d)*((a*x^3+b)/(c*x^3+d))^(2/3)/((a*x^3+… a)*x^3-b))	Problem gg13 (iv) rational radicand	{“failed”, “radicand is not a rational function over Q”}
chebychev-intro-1	Trager	x^(-5/6)*Sqrt[x^(1/3)-1]	Chapter intro worked example (type III)	{“failed”, “nested radicals”}
chebychev-chebychev_1-i	Trager	x^(1/3)*(a*Sqrt[x]+b)^3	Problem chebychev_1 (i)	{“failed”, “integrand is not a rational function of x and the radical over Q”}
chebychev-chebychev_1-ii	Trager	x^(3/2)*(a*x^(5/6)+b)^(1/3)	Problem chebychev_1 (ii)	{“failed”, “nested radicals”}
chebychev-chebychev_1-ii	PMT	x^(3/2)*(a*x^(5/6)+b)^(1/3)	Problem chebychev_1 (ii)	{“failed”, “verification failed”}
chebychev-chebychev_1-ii-inst	Trager	x^(3/2)*(2*x^(5/6)+3)^(1/3)	Problem chebychev_1 (ii) instantiated a=…	{“failed”, “nested radicals”}
chebychev-chebychev_1-iii	Trager	x^(4/3)/(a*Sqrt[x]+b)^(2/3)	Problem chebychev_1 (iii)	{“failed”, “nested radicals”}
chebychev-chebychev_1-iii-inst	Trager	x^(4/3)/(2*Sqrt[x]+3)^(2/3)	Problem chebychev_1 (iii) instantiated a…	{“failed”, “nested radicals”}
chebychev-chebychev_1-iv	Trager	(a*x^(3/4)+b)^(2/3)/x^3	Problem chebychev_1 (iv)	{“failed”, “nested radicals”}
chebychev-chebychev_1-iv-inst	Trager	(2*x^(3/4)+3)^(2/3)/x^3	Problem chebychev_1 (iv) instantiated a=…	{“failed”, “nested radicals”}
chebychev-chebychev_3-i	Trager	(a*x^(1/5)+b)^(3/4)	Problem chebychev_3 (i)	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
chebychev-chebychev_3-i-inst	Trager	(2*x^(1/5)+3)^(3/4)	Problem chebychev_3 (i) instantiated a=2…	{“failed”, “nested radicals”}
chebychev-chebychev_3-iii	Trager	x^(1/3)*(a*x^2+b)^(1/3)	Problem chebychev_3 (iii)	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_3-iii	PMT	x^(1/3)*(a*x^2+b)^(1/3)	Problem chebychev_3 (iii)	{“failed”, “internal: solution does not verify”}
chebychev-chebychev_3-iii-inst	Trager	x^(1/3)*(2*x^2+3)^(1/3)	Problem chebychev_3 (iii) instantiated a…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_3-v	Trager	1/(a*x^3+b)^(1/3)	Problem chebychev_3 (v)	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_3-v	PMT	1/(a*x^3+b)^(1/3)	Problem chebychev_3 (v)	{“failed”, “internal: solution does not verify”}
chebychev-chebychev_3-vii	Trager	x^4/(a*x^4+b)^(5/4)	Problem chebychev_3 (vii) (hint: q+(p+1)…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_3-vii	PMT	x^4/(a*x^4+b)^(5/4)	Problem chebychev_3 (vii) (hint: q+(p+1)…	{“failed”, “no solution within bounds”, {4}}
chebychev-chebychev_3-viii	Trager	x^(4/5)*(a/x^(4/5)+b)^(5/4)	Problem chebychev_3 (viii) (expect by bo…	{“failed”, “nested radicals”}
chebychev-chebychev_3-viii-inst	Trager	x^(4/5)*(2/x^(4/5)+3)^(5/4)	Problem chebychev_3 (viii) (expect by bo…	{“failed”, “nested radicals”}
chebychev-chebychev_3-x	Trager	x^(3/2)/Sqrt[a/Sqrt[x]+b]	Problem chebychev_3 (x) (expect by book …	{“failed”, “nested radicals”}
chebychev-chebychev_3-x-inst	Trager	x^(3/2)/Sqrt[2/Sqrt[x]+3]	Problem chebychev_3 (x) (expect by book …	{“failed”, “nested radicals”}
chebychev-chebychev_3b-v	Trager	Sqrt[Tan[x]]	Problem chebychev_3b (v)	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3b-vi	Trager	Sqrt[((Exp[x]-Exp[-x])/(Exp[x]+Exp[-x]))]	Problem chebychev_3b (vi) tanh via Exp	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3b-vii	Trager	Sqrt[Cos[x]/Sin[x]]	Problem chebychev_3b (vii) cot as Cos/Si…	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3b-viii	Trager	Sqrt[((Exp[x]+Exp[-x])/(Exp[x]-Exp[-x]))]	Problem chebychev_3b (viii) coth via Exp	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3c-iii_n2	Trager	1/(Tan[x])^(1/2)	Problem chebychev_3c (iii) 1/tan^(1/n) n…	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3c-vi_n2	Trager	(Tan[x])^(1/2)	Problem chebychev_3c (vi) 1/cot^(1/n)=ta…	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_3c-iii_n3	Trager	1/(Tan[x])^(1/3)	Problem chebychev_3c (iii) 1/tan^(1/n) n…	{“failed”, “transcendental function in the integrand”}

id	method	integrand	book ref	detail
chebychev-chebychev_3c-vi_n3	Trager	(Tan[x])^(1/3)	Problem chebychev_3c (vi) 1/cot^(1/n)=ta…	{“failed”, “transcendental function in the integrand”}
chebychev-chebychev_2-i	Trager	(x^2-2*x-1)*(2*Sqrt[x-1]/Sqrt[x^2+1]-1)^(2/3)/((x^2+1)^2*(Sqrt[x-1]/Sqrt[x^2+1])^(2/3))	Problem chebychev_2 (i) (solved)	{“failed”, “nested radicals”}
chebychev-chebychev_2-i	PMT	(x^2-2*x-1)*(2*Sqrt[x-1]/Sqrt[x^2+1]-1)^(2/3)/((x^2+1)^2*(Sqrt[x-1]/Sqrt[x^2+1])^(2/3))	Problem chebychev_2 (i) (solved)	{“failed”, “tower construction failed”, ((-1 - 2 x + x^2) (-1 + (2 Sqrt[-1 + x]/Sqrt[1 + x^2])^(2/3))/((Sqrt[…
chebychev-chebychev_2-ii	Trager	(x^3-x^2-x-1)*((Sqrt[x-1]/(Sqrt[x+1]*Sqrt[x^2+1]))-1)^(2/3)/(Sqrt[x-1]*(x+1)^(3/2)*(x^2+1)…	Problem chebychev_2 (ii) (hint substitut…	{“failed”, “nested radicals”}
chebychev-chebychev_2-ii	PMT	(x^3-x^2-x-1)*((Sqrt[x-1]/(Sqrt[x+1]*Sqrt[x^2+1]))-1)^(2/3)/(Sqrt[x-1]*(x+1)^(3/2)*(x^2+1)…	Problem chebychev_2 (ii) (hint substitut…	{“failed”, “tower construction failed”, ((-1 - x - x^2 + x^3) (-1 + Sqrt[-1 + x]/(Sqrt[1 + x] Sqrt[1 + x^2]))^…
chebychev-chebychev_5-i	Trager	Sqrt[a*x^2+b]/Sqrt[c/x^2+d]	Problem chebychev_5 (i) type III conject…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-i	PMT	Sqrt[a*x^2+b]/Sqrt[c/x^2+d]	Problem chebychev_5 (i) type III conject…	{“failed”, “tower construction failed”, Sqrt[b + a x^2]/Sqrt[d + c/x^2]}
chebychev-chebychev_5-i-inst	Trager	Sqrt[2*x^2+3]/Sqrt[5/x^2+7]	Problem chebychev_5 (i) type III conject…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-i-inst	PMT	Sqrt[2*x^2+3]/Sqrt[5/x^2+7]	Problem chebychev_5 (i) type III conject…	{“failed”, “tower construction failed”, Sqrt[3 + 2 x^2]/Sqrt[7 + 5/x^2]}
chebychev-chebychev_5-ii	Trager	Sqrt[a/x^2+b]/Sqrt[c*x^2+d]	Problem chebychev_5 (ii) type III conjec…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-ii	PMT	Sqrt[a/x^2+b]/Sqrt[c*x^2+d]	Problem chebychev_5 (ii) type III conjec…	{“failed”, “tower construction failed”, Sqrt[b + a/x^2]/Sqrt[d + c x^2]}
chebychev-chebychev_5-ii-inst	Trager	Sqrt[2/x^2+3]/Sqrt[5*x^2+7]	Problem chebychev_5 (ii) type III conjec…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-ii-inst	PMT	Sqrt[2/x^2+3]/Sqrt[5*x^2+7]	Problem chebychev_5 (ii) type III conjec…	{“failed”, “tower construction failed”, Sqrt[3 + 2/x^2]/Sqrt[7 + 5 x^2]}
chebychev-chebychev_5-iii	Trager	Sqrt[a*x^2+b]*Sqrt[c/x^2+d]	Problem chebychev_5 (iii) type III conje…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-iii	PMT	Sqrt[a*x^2+b]*Sqrt[c/x^2+d]	Problem chebychev_5 (iii) type III conje…	{“failed”, “tower construction failed”, Sqrt[d + c/x^2] Sqrt[b + a x^2]}
chebychev-chebychev_5-iii-inst	Trager	Sqrt[2*x^2+3]*Sqrt[5/x^2+7]	Problem chebychev_5 (iii) type III conje…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-iii-inst	PMT	Sqrt[2*x^2+3]*Sqrt[5/x^2+7]	Problem chebychev_5 (iii) type III conje…	{“failed”, “tower construction failed”, Sqrt[7 + 5/x^2] Sqrt[3 + 2 x^2]}

id	method	integrand	book ref	detail
chebychev-chebychev_5-iv	Trager	1/(Sqrt[a*x^2+b]*Sqrt[c/x^2+d])	Problem chebychev_5 (iv) type III conjec…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-iv	PMT	1/(Sqrt[a*x^2+b]*Sqrt[c/x^2+d])	Problem chebychev_5 (iv) type III conjec…	{“failed”, “tower construction failed”, 1/(Sqrt[d + c/x^2] Sqrt[b + a x^2])}
chebychev-chebychev_5-iv-inst	Trager	1/(Sqrt[2*x^2+3]*Sqrt[5/x^2+7])	Problem chebychev_5 (iv) type III conjec…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-iv-inst	PMT	1/(Sqrt[2*x^2+3]*Sqrt[5/x^2+7])	Problem chebychev_5 (iv) type III conjec…	{“failed”, “tower construction failed”, 1/(Sqrt[7 + 5/x^2] Sqrt[3 + 2 x^2])}
chebychev-chebychev_5-v	Trager	(a*Sqrt[x]+b)^(2/3)/(c*Sqrt[x]+d)^(2/3)	Problem chebychev_5 (v) type IV conjectu…	{“failed”, “nested radicals”}
chebychev-chebychev_5-v-inst	Trager	(2*Sqrt[x]+3)^(2/3)/(5*Sqrt[x]+7)^(2/3)	Problem chebychev_5 (v) type IV conjectu…	{“failed”, “nested radicals”}
chebychev-chebychev_5-vi	Trager	(a/x^(1/3)+b)^(1/4)/(c/x^(1/3)+d)^(1/4)	Problem chebychev_5 (vi) type IV conject…	{“failed”, “nested radicals”}
chebychev-chebychev_5-vi	PMT	(a/x^(1/3)+b)^(1/4)/(c/x^(1/3)+d)^(1/4)	Problem chebychev_5 (vi) type IV conject…	{“failed”, “tower construction failed”, (b + a/x^(1/3))^(1/4)/(d + c/x^(1/3))^(1/4)}
chebychev-chebychev_5-vi-inst	Trager	(2/x^(1/3)+3)^(1/4)/(5/x^(1/3)+7)^(1/4)	Problem chebychev_5 (vi) type IV conject…	{“failed”, “nested radicals”}
chebychev-chebychev_5-vi-inst	PMT	(2/x^(1/3)+3)^(1/4)/(5/x^(1/3)+7)^(1/4)	Problem chebychev_5 (vi) type IV conject…	{“failed”, “tower construction failed”, (3 + 2/x^(1/3))^(1/4)/(7 + 5/x^(1/3))^(1/4)}
chebychev-chebychev_5-vii	Trager	(a/x+b)^(1/3)*(c/x+d)^(2/3)	Problem chebychev_5 (vii) type V conject…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-vii	PMT	(a/x+b)^(1/3)*(c/x+d)^(2/3)	Problem chebychev_5 (vii) type V conject…	{“failed”, “tower construction failed”, (b + a/x)^(1/3) (d + c/x)^(2/3)}
chebychev-chebychev_5-vii-inst	Trager	(2/x+3)^(1/3)*(5/x+7)^(2/3)	Problem chebychev_5 (vii) type V conject…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-vii-inst	PMT	(2/x+3)^(1/3)*(5/x+7)^(2/3)	Problem chebychev_5 (vii) type V conject…	{“failed”, “tower construction failed”, (3 + 2/x)^(1/3) (7 + 5/x)^(2/3)}
chebychev-chebychev_5-viii	Trager	(a*x^(1/3)+b)^(5/3)/(c*x^(1/3)+d)^(2/3)	Problem chebychev_5 (viii) type V conjec…	{“failed”, “nested radicals”}
chebychev-chebychev_5-viii-inst	Trager	(2*x^(1/3)+3)^(5/3)/(5*x^(1/3)+7)^(2/3)	Problem chebychev_5 (viii) type V conjec…	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
chebychev-chebychev_5-final_special_n2	Trager	(a*x^2+b)^(1/2)/(b^2-a^2*x^4)^(1/2)	Problem chebychev_5 closing remark, n=2,…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-final_special_n2	PMT	(a*x^2+b)^(1/2)/(b^2-a^2*x^4)^(1/2)	Problem chebychev_5 closing remark, n=2,…	{“failed”, “tower construction failed”, Sqrt[b + a x^2]/Sqrt[b^2 - a^2 x^4]}
chebychev-chebychev_5-final_special_n2-i…	Trager	(2*x^2+3)^(1/2)/(3^2-2^2*x^4)^(1/2)	Problem chebychev_5 closing remark, n=2,…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-final_special_n2-i…	PMT	(2*x^2+3)^(1/2)/(3^2-2^2*x^4)^(1/2)	Problem chebychev_5 closing remark, n=2,…	{“failed”, “tower construction failed”, Sqrt[3 + 2 x^2]/Sqrt[9 - 4 x^4]}
chebychev-chebychev_5-final_special_n3	Trager	(a*x^3+b)^(1/3)/(b^2-a^2*x^6)^(1/3)	Problem chebychev_5 closing remark, n=3,…	{“failed”, “radicand is not a rational function over Q”}
chebychev-chebychev_5-final_special_n3	PMT	(a*x^3+b)^(1/3)/(b^2-a^2*x^6)^(1/3)	Problem chebychev_5 closing remark, n=3,…	{“failed”, “tower construction failed”, (b + a x^3)^(1/3)/(b^2 - a^2 x^6)^(1/3)}
chebychev-chebychev_5-final_special_n3-i…	Trager	(2*x^3+3)^(1/3)/(3^2-2^2*x^6)^(1/3)	Problem chebychev_5 closing remark, n=3,…	{“failed”, “more than one radicand (a radical compositum)”}
chebychev-chebychev_5-final_special_n3-i…	PMT	(2*x^3+3)^(1/3)/(3^2-2^2*x^6)^(1/3)	Problem chebychev_5 closing remark, n=3,…	{“failed”, “tower construction failed”, (3 + 2 x^3)^(1/3)/(9 - 4 x^6)^(1/3)}
inverse_function-intro-1	Trager	Sqrt[1-3/x+Sqrt[9*x^2-6*x+9]/x]	Chapter intro worked example (solved)	{“failed”, “nested radicals”}
inverse_function-intro-1	PMT	Sqrt[1-3/x+Sqrt[9*x^2-6*x+9]/x]	Chapter intro worked example (solved)	{“failed”, “tower construction failed”, Sqrt[1 - 3/x + Sqrt[9 - 6 x + 9 x^2]/x]}
inverse_function-inverse_function_1-main	Trager	Sqrt[a-b-a/x-b/x+Sqrt[(a+b)^2*x^2-2*(a^2-b^2)*x+(a+b)^2]/x]	Problem inverse_function_1	{“failed”, “nested radicals”}
inverse_function-inverse_function_1-main	PMT	Sqrt[a-b-a/x-b/x+Sqrt[(a+b)^2*x^2-2*(a^2-b^2)*x+(a+b)^2]/x]	Problem inverse_function_1	{“failed”, “tower construction failed”, Sqrt[a - b - a/x - b/x + Sqrt[(a + b)^2 - 2 (a^2 - b^2) x + (a + b)^2 …
inverse_function-inverse_function_1-main…	Trager	Sqrt[2-3-2/x-3/x+Sqrt[(2+3)^2*x^2-2*(2^2-3^2)*x+(2+3)^2]/x]	Problem inverse_function_1 instantiated …	{“failed”, “nested radicals”}
inverse_function-inverse_function_1-main…	PMT	Sqrt[2-3-2/x-3/x+Sqrt[(2+3)^2*x^2-2*(2^2-3^2)*x+(2+3)^2]/x]	Problem inverse_function_1 instantiated …	{“failed”, “tower construction failed”, Sqrt[-1 - 5/x + Sqrt[25 + 10 x + 25 x^2]/x]}

id	method	integrand	book ref	detail
inverse_function-inverse_function_2-i	Trager	Sqrt[(a-b)*x+Sqrt[a+b]*Sqrt[x]*Sqrt[(a+b)*x+4*a*b]]/(x+1)^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (i)	{"failed", "algebraic constant in the integrand (coefficients must be rational)"}
inverse_function-inverse_function_2-i	PMT	Sqrt[(a-b)*x+Sqrt[a+b]*Sqrt[x]*Sqrt[(a+b)*x+4*a*b]]/(x+1)^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (i)	{"failed", "tower construction failed", Sqrt[(a - b) x + Sqrt[a + b] Sqrt[x] Sqrt[4 a b + (a + b) x]]/Sqrt[a +…
inverse_function-inverse_function_2-i-in…	Trager	Sqrt[(2-3)*x+Sqrt[2+3]*Sqrt[x]*Sqrt[(2+3)*x+4*2*3]]/(x+1)^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (i) instantia…	{"failed", "algebraic constant in the integrand (coefficients must be rational)"}
inverse_function-inverse_function_2-i-in…	PMT	Sqrt[(2-3)*x+Sqrt[2+3]*Sqrt[x]*Sqrt[(2+3)*x+4*2*3]]/(x+1)^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (i) instantia…	{"failed", "tower construction failed", Sqrt[-x + Sqrt[5] Sqrt[x] Sqrt[24 + 5 x]]/Sqrt[5 + x]}
inverse_function-inverse_function_2-ii	Trager	((a-b)*x-Sqrt[x]*Sqrt[(a+b)^2*x+4*a*b])^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (ii)	{"failed", "nested radicals"}
inverse_function-inverse_function_2-ii	PMT	((a-b)*x-Sqrt[x]*Sqrt[(a+b)^2*x+4*a*b])^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (ii)	{"failed", "tower construction failed", ((a - b) x - Sqrt[x] Sqrt[4 a b + (a + b)^2 x])^(1/3)/(1 + x)^(1/3)}
inverse_function-inverse_function_2-ii-i…	Trager	((2-3)*x-Sqrt[x]*Sqrt[(2+3)^2*x+4*2*3])^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (ii) instanti…	{"failed", "nested radicals"}
inverse_function-inverse_function_2-ii-i…	PMT	((2-3)*x-Sqrt[x]*Sqrt[(2+3)^2*x+4*2*3])^(1/3)/(x+1)^(1/3)	Problem inverse_function_2 (ii) instanti…	{"failed", "tower construction failed", (-x - Sqrt[x] Sqrt[24 + 25 x])^(1/3)/(1 + x)^(1/3)}
inverse_function-inverse_function_2-iii	Trager	((Sqrt[x^3+81*x^2]+9*x)^(2/3)-x)^(1/3)/(Sqrt[x^3+81*x^2]+9*x)^(1/9)	Problem inverse_function_2 (iii)	{"failed", "nested radicals"}
inverse_function-inverse_function_2-iii	PMT	((Sqrt[x^3+81*x^2]+9*x)^(2/3)-x)^(1/3)/(Sqrt[x^3+81*x^2]+9*x)^(1/9)	Problem inverse_function_2 (iii)	{"failed", "tower construction failed", (-x + (9 x + Sqrt[81 x^2 + x^3])^(2/3))^(1/3)/(9 x + Sqrt[81 x^2 + x^3…
inverse_function-inverse_function_2-iv	Trager	(3^(1/3)*x+(Sqrt[3]*Sqrt[27-x^3]+9)^(2/3))/(Sqrt[3]*Sqrt[27-x^3]+9)^(1/3)	Problem inverse_function_2 (iv)	{"failed", "algebraic constant in the integrand (coefficients must be rational)"}
inverse_function-inverse_function_2-iv	PMT	(3^(1/3)*x+(Sqrt[3]*Sqrt[27-x^3]+9)^(2/3))/(Sqrt[3]*Sqrt[27-x^3]+9)^(1/3)	Problem inverse_function_2 (iv)	{"failed", "tower construction failed", (3^(1/3) x + (9 + Sqrt[3] Sqrt[27 - x^3])^(2/3))/(9 + Sqrt[3] Sqrt[27 …
inverse_function-inverse_function_2-v	Trager	(2^(1/3)*(Sqrt[x^2+4]+x)^(2/3)-2)^(1/3)/(Sqrt[x^2+4]+x)^(1/9)	Problem inverse_function_2 (v)	{"failed", "algebraic constant in the integrand (coefficients must be rational)"}
inverse_function-inverse_function_3-iii	Trager	ArcTan[x^2]	Problem inverse_function_3 (iii) BOOK CL…	{"failed", "transcendental function in the integrand"}

id	method	integrand	book ref	detail
inverse_function-inverse_function_3-vi	Trager	Log[(1+x^2)/(1-x^2)]/2	Problem inverse_function_3 (vi) ArcTanh(…	{“failed”, “transcendental function in the integrand”}
product_rule-product_rule_2-iv	Trager	(a*x^5-5*a*c*x^3+2*b*x^2+2*b*c)/((x^2-c)^2*Sqrt[a*x^3-b])	Problem product_rule_2 (iv) (solved, 2x …	{“failed”, “radicand is not a rational function over Q”}
product_rule-product_rule_5-i	PMT	x^2*(x^4-2)/(x^4-1)*Sqrt[x^2/(x^4-1)]	Problem product_rule_5 (i) (solved)	{“failed”, “verification failed”}
product_rule-product_rule_5-v	Trager	(9*x^9-14*b*x^6-a*x^3-4*a*b)/(x^3*(x^6+a)^2)*((x^6+a)/(x^3-b))^(1/6)	Problem product_rule_5 (v)	{“failed”, “radicand is not a rational function over Q”}
log_term-log_term_prob_7-main	PMT	(29*x^2+18*x-3)/Sqrt[x^6+4*x^5+6*x^4-12*x^3+33*x^2-16*x]	Problem log_term_prob_7 Schultz	{“failed”, “verification failed”}
realis-prob_realis_1-i	PMT	(x^2-1)/((x^2-2)*(x^2+1)*Sqrt[3-x^4])	Problem prob_realis_1 (i)	{“needs torsion realisation (milestone iii)”, {{-2 + x^2, {(1/6*I)/Sqrt[2], (-1/6*I)/Sqrt[2], (-1/6*I)/Sqrt[2]…
realis-prob_realis_1-iii	Trager	(3*b*x^4+a)/((b*c*x^6+a*c*x^2+d)*Sqrt[b*x^4+a])	Problem prob_realis_1 (iii)	{“failed”, “radicand is not a rational function over Q”}
realis-prob_realis_2-ii	PMT	(3*x+1)/((x^2-1)*Sqrt[x^3-x^2-x])	Problem prob_realis_2 (ii) (elem: 2*ArcT…	{“failed”, “verification failed”}
realis-prob_realis_2-iii_upper	PMT	(5*x-1)/((x^2-1)*Sqrt[(x-1)^2*(x+1)^3-1])	Problem prob_realis_2 (iii) upper signs	{“needs torsion realisation (milestone iii)”, {{-1 + Sqrt[x], {-2*I, 2*I}}, {1 + Sqrt[x], {2*I, -2*I}}, {1 + x…
realis-prob_realis_2-iii_lower	PMT	(5*x+1)/((x^2-1)*Sqrt[(x+1)^2*(x-1)^3-1])	Problem prob_realis_2 (iii) lower signs	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-3*I, 3*I}}, {1 + x, {-2*I, 2*I}}}}
realis-prob_realis_2-v_lower_n1	PMT	(3*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^1-1])	Problem prob_realis_2 (v) n=1, lower sig…	{“failed”, “verification failed”}
realis-prob_realis_2-v_upper_n2	PMT	(5*x-1)/((x^2-1)*Sqrt[(x+1)*(x^2-1)^2-1])	Problem prob_realis_2 (v) n=2, numerator…	{“needs torsion realisation (milestone iii)”, {{-1 + Sqrt[x], {-2*I, 2*I}}, {1 + Sqrt[x], {2*I, -2*I}}, {1 + x…
realis-prob_realis_2-v_lower_n2	PMT	(5*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^2-1])	Problem prob_realis_2 (v) n=2, lower sig…	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-3*I, 3*I}}, {1 + x, {-2*I, 2*I}}}}
realis-prob_realis_2-v_upper_n3	PMT	(7*x-1)/((x^2-1)*Sqrt[(x+1)*(x^2-1)^3-1])	Problem prob_realis_2 (v) n=3, numerator…	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-3*I, 3*I}}, {1 + x, {-4*I, 4*I}}}}
realis-prob_realis_2-v_lower_n3	PMT	(7*x+1)/((x^2-1)*Sqrt[(x-1)*(x^2-1)^3-1])	Problem prob_realis_2 (v) n=3, lower sig…	{“needs torsion realisation (milestone iii)”, {{-1 + Sqrt[x], {-4*I, 4*I}}, {1 + Sqrt[x], {4*I, -4*I}}, {1 + x…
realis-prob_realis_3-ii	Trager	(3*a*x^4+2*b*x^2+c)/((a*e*x^6+b*e*x^4+c*e*x^2+d)*Sqrt[a*x^4+b*x^2+c])	Problem prob_realis_3 (ii)	{“failed”, “radicand is not a rational function over Q”}

id	method	integrand	book ref	detail
realis-prob_realis_4-i	PMT	1/((1-x^2)*Sqrt[(2-x^2)*(x^4-2*x^2+2)])	Problem prob_realis_4 (i) (has a concise…	{“needs torsion realisation (milestone iii)”, {{-1 + x, {-1/2, 1/2}}, {1 + x, {1/2, -1/2}}}}
realis-prob_realis_4-ii	Trager	(a-2*x^2)/((2*x^4-2*a*x^2+a)*Sqrt[(a-x^2)*(x^4-a*x^2+a)])	Problem prob_realis_4 (ii) (a=2 gives (i…	{“failed”, “radicand is not a rational function over Q”}
nested-intro-1	Trager	Sqrt[a*x+b*Sqrt[x]]	Section intro, nested radicals (closed f…	{“failed”, “nested radicals”}
nested-intro-1-inst	Trager	Sqrt[2*x+3*Sqrt[x]]	Section intro, nested radicals (closed f…	{“failed”, “nested radicals”}
nested-nested_1-i	Trager	Sqrt[a/x^2+b*x^2*Sqrt[x]]	Problem nested_1 (i), n=-2	{“failed”, “nested radicals”}
nested-nested_1-i	PMT	Sqrt[a/x^2+b*x^2*Sqrt[x]]	Problem nested_1 (i), n=-2	{“needs torsion realisation (milestone iii)”, {{Sqrt[x], {2 Sqrt[a], -2 Sqrt[a]}}}}
nested-nested_1-i-inst	Trager	Sqrt[2/x^2+3*x^2*Sqrt[x]]	Problem nested_1 (i), n=-2; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_1-i-inst	PMT	Sqrt[2/x^2+3*x^2*Sqrt[x]]	Problem nested_1 (i), n=-2; inst a=2, b=…	{“needs torsion realisation (milestone iii)”, {{Sqrt[x], {2 Sqrt[2], -2 Sqrt[2]}}}}
nested-nested_1-ii	Trager	Sqrt[a*x+b*x^2*Sqrt[x]]	Problem nested_1 (ii), n=1	{“failed”, “nested radicals”}
nested-nested_1-ii	PMT	Sqrt[a*x+b*x^2*Sqrt[x]]	Problem nested_1 (ii), n=1	{“failed”, “verification failed”}
nested-nested_1-ii-inst	Trager	Sqrt[2*x+3*x^2*Sqrt[x]]	Problem nested_1 (ii), n=1; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_1-iii	Trager	Sqrt[a*x^2+b*x^2*Sqrt[x]]	Problem nested_1 (iii), n=2	{“failed”, “nested radicals”}
nested-nested_1-iii	PMT	Sqrt[a*x^2+b*x^2*Sqrt[x]]	Problem nested_1 (iii), n=2	{“failed”, “verification failed”}
nested-nested_1-iii-inst	Trager	Sqrt[2*x^2+3*x^2*Sqrt[x]]	Problem nested_1 (iii), n=2; inst a=2, b…	{“failed”, “nested radicals”}
nested-nested_1-iii-inst	PMT	Sqrt[2*x^2+3*x^2*Sqrt[x]]	Problem nested_1 (iii), n=2; inst a=2, b…	{“failed”, “verification failed”}
nested-nested_1-iv	Trager	Sqrt[a*x^3+b*x^2*Sqrt[x]]	Problem nested_1 (iv), n=3	{“failed”, “nested radicals”}
nested-nested_1-iv-inst	Trager	Sqrt[2*x^3+3*x^2*Sqrt[x]]	Problem nested_1 (iv), n=3; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_1-v	Trager	Sqrt[a*x^4+b*x^2*Sqrt[x]]	Problem nested_1 (v), n=4	{“failed”, “nested radicals”}
nested-nested_1-v	PMT	Sqrt[a*x^4+b*x^2*Sqrt[x]]	Problem nested_1 (v), n=4	{“failed”, “no solution within bounds”, {12}}
nested-nested_1-v-inst	Trager	Sqrt[2*x^4+3*x^2*Sqrt[x]]	Problem nested_1 (v), n=4; inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_1-v-inst	PMT	Sqrt[2*x^4+3*x^2*Sqrt[x]]	Problem nested_1 (v), n=4; inst a=2, b=3	{“failed”, “no solution within bounds”, {12}}

id	method	integrand	book ref	detail
nested-nested_1-vi	Trager	Sqrt[a*x^7+b*x^2*Sqrt[x]]	Problem nested_1 (vi), n=7	{“failed”, “nested radicals”}
nested-nested_1-vi	PMT	Sqrt[a*x^7+b*x^2*Sqrt[x]]	Problem nested_1 (vi), n=7	{“failed”, “no solution within bounds”, {18}}
nested-nested_1-vi-inst	Trager	Sqrt[2*x^7+3*x^2*Sqrt[x]]	Problem nested_1 (vi), n=7; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_1-vi-inst	PMT	Sqrt[2*x^7+3*x^2*Sqrt[x]]	Problem nested_1 (vi), n=7; inst a=2, b=…	{“failed”, “no solution within bounds”, {18}}
nested-nested_2-n1	Trager	Sqrt[a*x+b*x^2/Sqrt[x]]	Problem nested_2, n=1 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n1	PMT	Sqrt[a*x+b*x^2/Sqrt[x]]	Problem nested_2, n=1 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_2-n1-inst	Trager	Sqrt[2*x+3*x^2/Sqrt[x]]	Problem nested_2, n=1 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n1-inst	PMT	Sqrt[2*x+3*x^2/Sqrt[x]]	Problem nested_2, n=1 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_2-n2	Trager	Sqrt[a*x^2+b*x^3/Sqrt[x]]	Problem nested_2, n=2 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n2	PMT	Sqrt[a*x^2+b*x^3/Sqrt[x]]	Problem nested_2, n=2 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_2-n2-inst	Trager	Sqrt[2*x^2+3*x^3/Sqrt[x]]	Problem nested_2, n=2 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n2-inst	PMT	Sqrt[2*x^2+3*x^3/Sqrt[x]]	Problem nested_2, n=2 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_2-n3	Trager	Sqrt[a*x^3+b*x^4/Sqrt[x]]	Problem nested_2, n=3 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n3	PMT	Sqrt[a*x^3+b*x^4/Sqrt[x]]	Problem nested_2, n=3 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_2-n3-inst	Trager	Sqrt[2*x^3+3*x^4/Sqrt[x]]	Problem nested_2, n=3 (algebraic for all…	{“failed”, “nested radicals”}
nested-nested_2-n3-inst	PMT	Sqrt[2*x^3+3*x^4/Sqrt[x]]	Problem nested_2, n=3 (algebraic for all…	{“failed”, “verification failed”}
nested-nested_3-i	Trager	(a/x^3+b*x*Sqrt[x])^(1/3)	Problem nested_3 (i), n=-3	{“failed”, “nested radicals”}
nested-nested_3-i-inst	Trager	(2/x^3+3*x*Sqrt[x])^(1/3)	Problem nested_3 (i), n=-3; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_3-ii	Trager	(a+b*x*Sqrt[x])^(1/3)	Problem nested_3 (ii), n=0	{“failed”, “nested radicals”}
nested-nested_3-ii	PMT	(a+b*x*Sqrt[x])^(1/3)	Problem nested_3 (ii), n=0	{“failed”, “internal: solution does not verify”}

id	method	integrand	book ref	detail
nested-nested_3-ii-inst	Trager	(2+3*x*Sqrt[x])^(1/3)	Problem nested_3 (ii), n=0; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_3-iii	Trager	(a*x+b*x*Sqrt[x])^(1/3)	Problem nested_3 (iii), n=1	{“failed”, “nested radicals”}
nested-nested_3-iii-inst	Trager	(2*x+3*x*Sqrt[x])^(1/3)	Problem nested_3 (iii), n=1; inst a=2, b…	{“failed”, “nested radicals”}
nested-nested_3-iv	Trager	(a*x^2+b*x*Sqrt[x])^(1/3)	Problem nested_3 (iv), n=2	{“failed”, “nested radicals”}
nested-nested_3-iv	PMT	(a*x^2+b*x*Sqrt[x])^(1/3)	Problem nested_3 (iv), n=2	{“failed”, “verification failed”}
nested-nested_3-iv-inst	Trager	(2*x^2+3*x*Sqrt[x])^(1/3)	Problem nested_3 (iv), n=2; inst a=2, b=…	{“failed”, “nested radicals”}
nested-nested_3-iv-inst	PMT	(2*x^2+3*x*Sqrt[x])^(1/3)	Problem nested_3 (iv), n=2; inst a=2, b=…	{“failed”, “verification failed”}
nested-nested_3-v	Trager	(a*x^3+b*x*Sqrt[x])^(1/3)	Problem nested_3 (v), n=3	{“failed”, “nested radicals”}
nested-nested_3-v	PMT	(a*x^3+b*x*Sqrt[x])^(1/3)	Problem nested_3 (v), n=3	{“failed”, “internal: solution does not verify”}
nested-nested_3-v-inst	Trager	(2*x^3+3*x*Sqrt[x])^(1/3)	Problem nested_3 (v), n=3; inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_4-ex1	Trager	(a/x^4+b*x^(4/3))^(1/4)	Problem nested_4, solution example 1 (n=…	{“failed”, “nested radicals”}
nested-nested_4-ex1	PMT	(a/x^4+b*x^(4/3))^(1/4)	Problem nested_4, solution example 1 (n=…	{“needs torsion realisation (milestone iii)”, {{x^(1/3), {-3 a^(1/4)}}, {x^(1/3), {-(3*I) a^(1/4)}}, {x^(1/3),…
nested-nested_4-ex1-inst	Trager	(2/x^4+3*x^(4/3))^(1/4)	Problem nested_4, solution example 1 (n=…	{“failed”, “nested radicals”}
nested-nested_4-ex2	Trager	(a*x^3+b*x^(16/3))^(1/4)	Problem nested_4, solution example 2 (n=…	{“failed”, “nested radicals”}
nested-nested_4-ex2	PMT	(a*x^3+b*x^(16/3))^(1/4)	Problem nested_4, solution example 2 (n=…	{“failed”, “no solution within bounds”, {28}}
nested-nested_4-ex2-inst	Trager	(2*x^3+3*x^(16/3))^(1/4)	Problem nested_4, solution example 2 (n=…	{“failed”, “nested radicals”}
nested-nested_4-ex2-inst	PMT	(2*x^3+3*x^(16/3))^(1/4)	Problem nested_4, solution example 2 (n=…	{“failed”, “no solution within bounds”, {28}}
nested-nested_5-i	Trager	Sqrt[d*x+c*x^2*Sqrt[a/x+b]]/Sqrt[a/x+b]	Problem nested_5 (i)	{“failed”, “nested radicals”}
nested-nested_5-i	PMT	Sqrt[d*x+c*x^2*Sqrt[a/x+b]]/Sqrt[a/x+b]	Problem nested_5 (i)	{“failed”, “tower construction failed”, Sqrt[d x + c Sqrt[b + a/x] x^2]/Sqrt[b + a/x]}
nested-nested_5-i-inst	Trager	Sqrt[7*x+5*x^2*Sqrt[2/x+3]]/Sqrt[2/x+3]	Problem nested_5 (i); inst a=2, b=3, c=5…	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_5-i-inst	PMT	Sqrt[7*x+5*x^2*Sqrt[2/x+3]]/Sqrt[2/x+3]	P3, problem nested_5 (i); inst a=2, b=3, c=5…	{“failed”, “tower construction failed”, Sqrt[7 x + 5 Sqrt[3 + 2/x] x^2]/Sqrt[3 + 2/x]}
nested-nested_5-ii	Trager	Sqrt[a/x+b]/Sqrt[d*x+c*x^2*Sqrt[a/x+b]]	P3, problem nested_5 (ii)	{“failed”, “nested radicals”}
nested-nested_5-ii	PMT	Sqrt[a/x+b]/Sqrt[d*x+c*x^2*Sqrt[a/x+b]]	P3, problem nested_5 (ii)	{“failed”, “tower construction failed”, Sqrt[b + a/x]/Sqrt[d x + c Sqrt[b + a/x] x^2]}
nested-nested_5-ii-inst	Trager	Sqrt[2/x+3]/Sqrt[7*x+5*x^2*Sqrt[2/x+3]]	P3, problem nested_5 (ii); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_5-ii-inst	PMT	Sqrt[2/x+3]/Sqrt[7*x+5*x^2*Sqrt[2/x+3]]	P3, problem nested_5 (ii); inst a=2, b=3, c=…	{“failed”, “tower construction failed”, Sqrt[3 + 2/x]/Sqrt[7 x + 5 Sqrt[3 + 2/x] x^2]}
nested-nested_5-iii	Trager	Sqrt[d*x+c*x^2*Sqrt[a/x+b]]/((c*x^2+d*x)*Sqrt[a/x+b])	P3, problem nested_5 (iii)	{“failed”, “nested radicals”}
nested-nested_5-iii	PMT	Sqrt[d*x+c*x^2*Sqrt[a/x+b]]/((c*x^2+d*x)*Sqrt[a/x+b])	P3, problem nested_5 (iii)	{“failed”, “tower construction failed”, Sqrt[d x + c Sqrt[b + a/x] x^2]/(Sqrt[b + a/x] (d x + c x^2))}
nested-nested_5-iii-inst	Trager	Sqrt[7*x+5*x^2*Sqrt[2/x+3]]/((5*x^2+7*x)*Sqrt[2/x+3])	P3, problem nested_5 (iii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_5-iii-inst	PMT	Sqrt[7*x+5*x^2*Sqrt[2/x+3]]/((5*x^2+7*x)*Sqrt[2/x+3])	P3, problem nested_5 (iii); inst a=2, b=3, c…	{“failed”, “tower construction failed”, Sqrt[7 x + 5 Sqrt[3 + 2/x] x^2]/(Sqrt[3 + 2/x] (7 x + 5 x^2))}
nested-nested_5-iv	Trager	Sqrt[a/x+b]/((c*x^2+d*x)*Sqrt[d*x+c*x^2*Sqrt[a/x+b]])	P3, problem nested_5 (iv)	{“failed”, “nested radicals”}
nested-nested_5-iv	PMT	Sqrt[a/x+b]/((c*x^2+d*x)*Sqrt[d*x+c*x^2*Sqrt[a/x+b]])	P3, problem nested_5 (iv)	{“failed”, “tower construction failed”, Sqrt[b + a/x]/((d x + c x^2) Sqrt[d x + c Sqrt[b + a/x] x^2])}
nested-nested_5-iv-inst	Trager	Sqrt[2/x+3]/((5*x^2+7*x)*Sqrt[7*x+5*x^2*Sqrt[2/x+3]])	P3, problem nested_5 (iv); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_5-iv-inst	PMT	Sqrt[2/x+3]/((5*x^2+7*x)*Sqrt[7*x+5*x^2*Sqrt[2/x+3]])	P3, problem nested_5 (iv); inst a=2, b=3, c=…	{“failed”, “tower construction failed”, Sqrt[3 + 2/x]/((7 x + 5 x^2) Sqrt[7 x + 5 Sqrt[3 + 2/x] x^2])}
nested-nested_6-i	Trager	Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]/Sqrt[a/x+b]	P3, problem nested_6 (i)	{“failed”, “nested radicals”}
nested-nested_6-i-inst	Trager	Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]/Sqrt[2/x+3]	P3, problem nested_6 (i); inst a=2, b=3, c=5…	{“failed”, “nested radicals”}
nested-nested_6-i-inst	PMT	Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]/Sqrt[2/x+3]	P3, problem nested_6 (i); inst a=2, b=3, c=5…	{“failed”, “verification failed”}
nested-nested_6-ii	Trager	Sqrt[a/x+b]/Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]	P3, problem nested_6 (ii)	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_6-ii-inst	Trager	Sqrt[2/x+3]/Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]	Problem nested_6 (ii); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_6-ii-inst	PMT	Sqrt[2/x+3]/Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]	Problem nested_6 (ii); inst a=2, b=3, c=…	{“failed”, “verification failed”}
nested-nested_6-iii	Trager	Sqrt[a/x+b]*Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]]	Problem nested_6 (iii)	{“failed”, “nested radicals”}
nested-nested_6-iii-inst	Trager	Sqrt[2/x+3]*Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]	Problem nested_6 (iii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_6-iii-inst	PMT	Sqrt[2/x+3]*Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]]	Problem nested_6 (iii); inst a=2, b=3, c…	{“failed”, “verification failed”}
nested-nested_6-iv	Trager	1/(Sqrt[a/x+b]*Sqrt[d*x^2+c*x^2/Sqrt[a/x+b]])	Problem nested_6 (iv)	{“failed”, “nested radicals”}
nested-nested_6-iv-inst	Trager	1/(Sqrt[2/x+3]*Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]])	Problem nested_6 (iv); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_6-iv-inst	PMT	1/(Sqrt[2/x+3]*Sqrt[7*x^2+5*x^2/Sqrt[2/x+3]])	Problem nested_6 (iv); inst a=2, b=3, c=…	{“failed”, “verification failed”}
nested-nested_7-i	Trager	Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]/((b*x+c)*Sqrt[a^2*x^2+b*x+c])	Problem nested_7 (i)	{“failed”, “nested radicals”}
nested-nested_7-i	PMT	Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]/((b*x+c)*Sqrt[a^2*x^2+b*x+c])	Problem nested_7 (i)	{“failed”, “tower construction failed”, Sqrt[a x + Sqrt[c + b x + a^2 x^2]]/((c + b x) Sqrt[c + b x + a^2 x^2]…
nested-nested_7-i-inst	Trager	Sqrt[2*x+Sqrt[2^2*x^2+3*x+5]]/((3*x+b)*Sqrt[2^2*x^2+3*x+5])	Problem nested_7 (i); inst a=2, b=3, c=5	{“failed”, “nested radicals”}
nested-nested_7-ii	Trager	(a^2*x^2+b*x+c)/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (ii)	{“failed”, “nested radicals”}
nested-nested_7-ii	PMT	(a^2*x^2+b*x+c)/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (ii)	{“failed”, “tower construction failed”, (c + b x + a^2 x^2)/Sqrt[a x + Sqrt[c + b x + a^2 x^2]]}
nested-nested_7-ii-inst	Trager	(2^2*x^2+3*x+5)/Sqrt[2*x+Sqrt[2^2*x^2+3*x+5]]	Problem nested_7 (ii); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_7-iii	Trager	Sqrt[a^2*x^2+b*x+c]/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (iii)	{“failed”, “nested radicals”}
nested-nested_7-iii	PMT	Sqrt[a^2*x^2+b*x+c]/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (iii)	{“failed”, “tower construction failed”, Sqrt[c + b x + a^2 x^2]/Sqrt[a x + Sqrt[c + b x + a^2 x^2]]}
nested-nested_7-iii-inst	Trager	Sqrt[2^2*x^2+3*x+5]/Sqrt[2*x+Sqrt[2^2*x^2+3*x+5]]	Problem nested_7 (iii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_7-iv	Trager	(a*x^2-b*x-c)/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (iv)	{“failed”, “nested radicals”}
nested-nested_7-iv	PMT	(a*x^2-b*x-c)/Sqrt[a*x+Sqrt[a^2*x^2+b*x+c]]	Problem nested_7 (iv)	{“failed”, “tower construction failed”, (-c - b x + a x^2)/Sqrt[a x + Sqrt[c + b x + a^2 x^2]]}

id	method	integrand	book ref	detail
nested-nested_7-iv-inst	Trager	(2*x^2-3*x-5)/Sqrt[2*x+Sqrt[2^2*x^2+3*x+5]]	Problem nested_7 (iv); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_7-v	Trager	Sqrt[a^2*x^2+b*x+c]/(x^2*Sqrt[-a*x+Sqrt[a^2*x^2+b*x+c]])	Problem nested_7 (v)	{“failed”, “nested radicals”}
nested-nested_7-v	PMT	Sqrt[a^2*x^2+b*x+c]/(x^2*Sqrt[-a*x+Sqrt[a^2*x^2+b*x+c]])	Problem nested_7 (v)	{“failed”, “tower construction failed”, Sqrt[c + b x + a^2 x^2]/(x^2 Sqrt[-a x + Sqrt[c + b x + a^2 x^2]])}
nested-nested_7-v-inst	Trager	Sqrt[2^2*x^2+3*x+5]/(x^2*Sqrt[-2*x+Sqrt[2^2*x^2+3*x+5]])	Problem nested_7 (v); inst a=2, b=3, c=5	{“failed”, “nested radicals”}
nested-nested_8-i	Trager	Sqrt[d+b*Sqrt[x]+Sqrt[a*Sqrt[x]+b^2*x+c]]/x	Problem nested_8 (i)	{“failed”, “nested radicals”}
nested-nested_8-i	PMT	Sqrt[d+b*Sqrt[x]+Sqrt[a*Sqrt[x]+b^2*x+c]]/x	Problem nested_8 (i)	{“failed”, “tower construction failed”, Sqrt[d + b Sqrt[x] + Sqrt[c + a Sqrt[x] + b^2 x]]/x}
nested-nested_8-i-inst	Trager	Sqrt[7+3*Sqrt[x]+Sqrt[2*Sqrt[x]+3^2*x+5]]/x	Problem nested_8 (i); inst a=2, b=3, c=5…	{“failed”, “nested radicals”}
nested-nested_8-ii	Trager	Sqrt[d+b*Sqrt[x]+Sqrt[a*Sqrt[x]+b^2*x+c]]	Problem nested_8 (ii)	{“failed”, “nested radicals”}
nested-nested_8-ii	PMT	Sqrt[d+b*Sqrt[x]+Sqrt[a*Sqrt[x]+b^2*x+c]]	Problem nested_8 (ii)	{“failed”, “tower construction failed”, Sqrt[d + b Sqrt[x] + Sqrt[c + a Sqrt[x] + b^2 x]]}
nested-nested_8-ii-inst	Trager	Sqrt[7+3*Sqrt[x]+Sqrt[2*Sqrt[x]+3^2*x+5]]	Problem nested_8 (ii); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_9-i	Trager	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/4)	Problem nested_9 (i)	{“failed”, “nested radicals”}
nested-nested_9-i	PMT	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/4)	Problem nested_9 (i)	{“failed”, “tower construction failed”, 1/(-a x + Sqrt[c + b x + a^2 x^2])^(1/4)}
nested-nested_9-i-inst	Trager	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/4)	Problem nested_9 (i); inst a=2, b=3, c=5	{“failed”, “nested radicals”}
nested-nested_9-ii	Trager	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(3/4)	Problem nested_9 (ii)	{“failed”, “nested radicals”}
nested-nested_9-ii	PMT	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(3/4)	Problem nested_9 (ii)	{“failed”, “tower construction failed”, 1/(-a x + Sqrt[c + b x + a^2 x^2])^(3/4)}
nested-nested_9-ii-inst	Trager	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(3/4)	Problem nested_9 (ii); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_9-iii	Trager	(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/4)/(b*x+c)	Problem nested_9 (iii)	{“failed”, “nested radicals”}
nested-nested_9-iii	PMT	(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/4)/(b*x+c)	Problem nested_9 (iii)	{“failed”, “tower construction failed”, (-a x + Sqrt[c + b x + a^2 x^2])^(1/4)/(c + b x)}

id	method	integrand	book ref	detail
nested-nested_9-iii-inst	Trager	(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/4)/(3*x+5)	Problem nested_9 (iii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_9-iv	Trager	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(5/6)	Problem nested_9 (iv)	{“failed”, “nested radicals”}
nested-nested_9-iv	PMT	1/(-a*x+Sqrt[a^2*x^2+b*x+c])^(5/6)	Problem nested_9 (iv)	{“failed”, “tower construction failed”, 1/(-a x + Sqrt[c + b x + a^2 x^2])^(5/6)}
nested-nested_9-iv-inst	Trager	1/(-2*x+Sqrt[2^2*x^2+3*x+5])^(5/6)	Problem nested_9 (iv); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_9-v	Trager	x/(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/8)	Problem nested_9 (v)	{“failed”, “nested radicals”}
nested-nested_9-v	PMT	x/(-a*x+Sqrt[a^2*x^2+b*x+c])^(1/8)	Problem nested_9 (v)	{“failed”, “tower construction failed”, x/(-a x + Sqrt[c + b x + a^2 x^2])^(1/8)}
nested-nested_9-v-inst	Trager	x/(-2*x+Sqrt[2^2*x^2+3*x+5])^(1/8)	Problem nested_9 (v); inst a=2, b=3, c=5	{“failed”, “nested radicals”}
nested-nested_10-i	Trager	Sqrt[a^2*x^2+b]/Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]]	Problem nested_10 (i)	{“failed”, “nested radicals”}
nested-nested_10-i	PMT	Sqrt[a^2*x^2+b]/Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]]	Problem nested_10 (i)	{“failed”, “tower construction failed”, Sqrt[b + a^2 x^2]/Sqrt[a x^2 + x Sqrt[b + a^2 x^2]]}
nested-nested_10-i-inst	Trager	Sqrt[2^2*x^2+3]/Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]]	Problem nested_10 (i); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_10-i-inst	PMT	Sqrt[2^2*x^2+3]/Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]]	Problem nested_10 (i); inst a=2, b=3	{“failed”, “tower construction failed”, Sqrt[3 + 4 x^2]/Sqrt[2 x^2 + x Sqrt[3 + 4 x^2]]}
nested-nested_10-ii	Trager	Sqrt[a^2*x^2+b]*Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]]/x	Problem nested_10 (ii)	{“failed”, “nested radicals”}
nested-nested_10-ii	PMT	Sqrt[a^2*x^2+b]*Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]]/x	Problem nested_10 (ii)	{“failed”, “tower construction failed”, (Sqrt[b + a^2 x^2] Sqrt[a x^2 + x Sqrt[b + a^2 x^2]])/x}
nested-nested_10-ii-inst	Trager	Sqrt[2^2*x^2+3]*Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]]/x	Problem nested_10 (ii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_10-ii-inst	PMT	Sqrt[2^2*x^2+3]*Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]]/x	Problem nested_10 (ii); inst a=2, b=3	{“failed”, “tower construction failed”, (Sqrt[3 + 4 x^2] Sqrt[2 x^2 + x Sqrt[3 + 4 x^2]])/x}
nested-nested_10-iii	Trager	x^2/(Sqrt[a^2*x^2+b]*Sqrt[-a*x^2+x*Sqrt[a^2*x^2+b]])	Problem nested_10 (iii)	{“failed”, “nested radicals”}
nested-nested_10-iii	PMT	x^2/(Sqrt[a^2*x^2+b]*Sqrt[-a*x^2+x*Sqrt[a^2*x^2+b]])	Problem nested_10 (iii)	{“failed”, “tower construction failed”, x^2/(Sqrt[b + a^2 x^2] Sqrt[-a x^2 + x Sqrt[b + a^2 x^2]])}
nested-nested_10-iii-inst	Trager	x^2/(Sqrt[2^2*x^2+3]*Sqrt[-2*x^2+x*Sqrt[2^2*x^2+3]])	Problem nested_10 (iii); inst a=2, b=3	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_10-iii-inst	PMT	x^2/(Sqrt[2^2*x^2+3]*Sqrt[-2*x^2+x*Sqrt[2^2*x^2+3]])	Problem nested_10 (iii); inst a=2, b=3	{“failed”, “tower construction failed”, x^2/(Sqrt[3 + 4 x^2] Sqrt[-2 x^2 + x Sqrt[3 + 4 x^2]])}
nested-nested_10-iv	Trager	Sqrt[-a*x^2+x*Sqrt[a^2*x^2+b]]/(a^2*x^2+b*x)	Problem nested_10 (iv)	{“failed”, “nested radicals”}
nested-nested_10-iv	PMT	Sqrt[-a*x^2+x*Sqrt[a^2*x^2+b]]/(a^2*x^2+b*x)	Problem nested_10 (iv)	{“failed”, “tower construction failed”, Sqrt[-a x^2 + x Sqrt[b + a^2 x^2]]/(b x + a^2 x^2)}
nested-nested_10-iv-inst	Trager	Sqrt[-2*x^2+x*Sqrt[2^2*x^2+3]]/(2^2*x^2+3*x)	Problem nested_10 (iv); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_10-iv-inst	PMT	Sqrt[-2*x^2+x*Sqrt[2^2*x^2+3]]/(2^2*x^2+3*x)	Problem nested_10 (iv); inst a=2, b=3	{“failed”, “tower construction failed”, Sqrt[-2 x^2 + x Sqrt[3 + 4 x^2]]/(3 x + 4 x^2)}
nested-nested_10-v	Trager	(a^2*x^2-b)^2/((a^2*x^2+b)^2*Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]])	Problem nested_10 (v)	{“failed”, “nested radicals”}
nested-nested_10-v	PMT	(a^2*x^2-b)^2/((a^2*x^2+b)^2*Sqrt[a*x^2+x*Sqrt[a^2*x^2+b]])	Problem nested_10 (v)	{“failed”, “tower construction failed”, (-b + a^2 x^2)^2/((b + a^2 x^2)^2 Sqrt[a x^2 + x Sqrt[b + a^2 x^2]])}
nested-nested_10-v-inst	Trager	(2^2*x^2-3)^2/((2^2*x^2+3)^2*Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]])	Problem nested_10 (v); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_10-v-inst	PMT	(2^2*x^2-3)^2/((2^2*x^2+3)^2*Sqrt[2*x^2+x*Sqrt[2^2*x^2+3]])	Problem nested_10 (v); inst a=2, b=3	{“failed”, “tower construction failed”, (-3 + 4 x^2)^2/((3 + 4 x^2)^2 Sqrt[2 x^2 + x Sqrt[3 + 4 x^2]])}
nested-nested_11-i	Trager	Sqrt[x^2+Sqrt[x^4+1]]/((x+1)^2*Sqrt[Sqrt[x^4+1]])	Problem nested_11 (i)	{“failed”, “nested radicals”}
nested-nested_11-i	PMT	Sqrt[x^2+Sqrt[x^4+1]]/((x+1)^2*Sqrt[Sqrt[x^4+1]])	Problem nested_11 (i)	{“failed”, “tower construction failed”, Sqrt[x^2 + Sqrt[1 + x^4]]/((1 + x)^2 Sqrt[1 + x^4])}
nested-nested_11-ii	Trager	Sqrt[a*x^2+Sqrt[a^2*x^4+b]]/((c*x+d)^2*Sqrt[a^2*x^4+b])	Problem nested_11 (ii)	{“failed”, “nested radicals”}
nested-nested_11-ii	PMT	Sqrt[a*x^2+Sqrt[a^2*x^4+b]]/((c*x+d)^2*Sqrt[a^2*x^4+b])	Problem nested_11 (ii)	{“failed”, “tower construction failed”, Sqrt[a x^2 + Sqrt[b + a^2 x^4]]/((d + c x)^2 Sqrt[b + a^2 x^4])}
nested-nested_11-ii-inst	Trager	Sqrt[2*x^2+Sqrt[2^2*x^4+3]]/((5*x+7)^2*Sqrt[2^2*x^4+3])	Problem nested_11 (ii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_11-ii-inst	PMT	Sqrt[2*x^2+Sqrt[2^2*x^4+3]]/((5*x+7)^2*Sqrt[2^2*x^4+3])	Problem nested_11 (ii); inst a=2, b=3, c…	{“failed”, “tower construction failed”, Sqrt[2 x^2 + Sqrt[3 + 4 x^4]]/((7 + 5 x)^2 Sqrt[3 + 4 x^4])}
nested-nested_11-iii	Trager	Sqrt[a*x^2+Sqrt[a^2*x^4+b]]	Problem nested_11 (iii)	{“failed”, “nested radicals”}
nested-nested_11-iii	PMT	Sqrt[a*x^2+Sqrt[a^2*x^4+b]]	Problem nested_11 (iii)	{“failed”, “tower construction failed”, Sqrt[a x^2 + Sqrt[b + a^2 x^4]]}

id	method	integrand	book ref	detail
nested-nested_11-iii-inst	Trager	Sqrt[2*x^2+Sqrt[2^2*x^4+3]]	Problem nested_11 (iii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_11-iii-inst	PMT	Sqrt[2*x^2+Sqrt[2^2*x^4+3]]	Problem nested_11 (iii); inst a=2, b=3	{“failed”, “tower construction failed”, Sqrt[2 x^2 + Sqrt[3 + 4 x^4]]}
nested-nested_11-iv	Trager	(a^2*x^4+b)^2/((a^2*x^4-b)^2*Sqrt[a*x^2+Sqrt[a^2*x^4+b]])	Problem nested_11 (iv)	{“failed”, “nested radicals”}
nested-nested_11-iv	PMT	(a^2*x^4+b)^2/((a^2*x^4-b)^2*Sqrt[a*x^2+Sqrt[a^2*x^4+b]])	Problem nested_11 (iv)	{“failed”, “tower construction failed”, (b + a^2 x^4)^2/((-b + a^2 x^4)^2 Sqrt[a x^2 + Sqrt[b + a^2 x^4]])}
nested-nested_11-iv-inst	Trager	(2^2*x^4+3)^2/((2^2*x^4-3)^2*Sqrt[2*x^2+Sqrt[2^2*x^4+3]])	Problem nested_11 (iv); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_11-iv-inst	PMT	(2^2*x^4+3)^2/((2^2*x^4-3)^2*Sqrt[2*x^2+Sqrt[2^2*x^4+3]])	Problem nested_11 (iv); inst a=2, b=3	{“failed”, “tower construction failed”, (3 + 4 x^4)^2/((-3 + 4 x^4)^2 Sqrt[2 x^2 + Sqrt[3 + 4 x^4]])}
nested-nested_11-v	Trager	(x^4-b)/((x^4+b)*Sqrt[a^2*x^4-b]*Sqrt[a*x^2+Sqrt[a^2*x^4-b]])	Problem nested_11 (v)	{“failed”, “nested radicals”}
nested-nested_11-v	PMT	(x^4-b)/((x^4+b)*Sqrt[a^2*x^4-b]*Sqrt[a*x^2+Sqrt[a^2*x^4-b]])	Problem nested_11 (v)	{“failed”, “tower construction failed”, (-b + x^4)/((b + x^4) Sqrt[-b + a^2 x^4] Sqrt[a x^2 + Sqrt[-b + a^2 x^…
nested-nested_11-v-inst	Trager	(x^4-3)/((x^4+3)*Sqrt[2^2*x^4-3]*Sqrt[2*x^2+Sqrt[2^2*x^4-3]])	Problem nested_11 (v); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_11-v-inst	PMT	(x^4-3)/((x^4+3)*Sqrt[2^2*x^4-3]*Sqrt[2*x^2+Sqrt[2^2*x^4-3]])	Problem nested_11 (v); inst a=2, b=3	{“failed”, “tower construction failed”, (-3 + x^4)/((3 + x^4) Sqrt[-3 + 4 x^4] Sqrt[2 x^2 + Sqrt[-3 + 4 x^4]])…
nested-nested_12-i	Trager	Sqrt[a^2*x^2+b]/(a*x-Sqrt[a^2*x^2+b])^(1/3)	Problem nested_12 (i)	{“failed”, “nested radicals”}
nested-nested_12-i	PMT	Sqrt[a^2*x^2+b]/(a*x-Sqrt[a^2*x^2+b])^(1/3)	Problem nested_12 (i)	{“failed”, “tower construction failed”, Sqrt[b + a^2 x^2]/(a x - Sqrt[b + a^2 x^2])^(1/3)}
nested-nested_12-i-inst	Trager	Sqrt[2^2*x^2+3]/(2*x-Sqrt[2^2*x^2+3])^(1/3)	Problem nested_12 (i); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_12-ii	Trager	(a*x-Sqrt[a^2*x^2+b])^(1/3)/(x*Sqrt[a^2*x^2+b])	Problem nested_12 (ii)	{“failed”, “nested radicals”}
nested-nested_12-ii	PMT	(a*x-Sqrt[a^2*x^2+b])^(1/3)/(x*Sqrt[a^2*x^2+b])	Problem nested_12 (ii)	{“failed”, “tower construction failed”, (a x - Sqrt[b + a^2 x^2])^(1/3)/(x Sqrt[b + a^2 x^2])}
nested-nested_12-ii-inst	Trager	(2*x-Sqrt[2^2*x^2+3])^(1/3)/(x*Sqrt[2^2*x^2+3])	Problem nested_12 (ii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_12-iii	Trager	(a*x-Sqrt[a^2*x^2+b])^(1/3)/((a^2*x^2+b)^2*Sqrt[a^2*x^2+b])	Problem nested_12 (iii)	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_12-iii	PMT	(a*x-Sqrt[a^2*x^2+b])^(1/3)/((a^2*x^2+b)^2*Sqrt[a^2*x^2+b])	Problem nested_12 (iii)	{“failed”, “tower construction failed”, (a x - Sqrt[b + a^2 x^2])^(1/3)/(b + a^2 x^2)^(5/2)}
nested-nested_12-iii-inst	Trager	(2*x-Sqrt[2^2*x^2+3])^(1/3)/((2^2*x^2+3)^2*Sqrt[2^2*x^2+3])	Problem nested_12 (iii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_12-iv	Trager	(a^2*x^2-b)/((a^2*x^2+b)*(a*x-Sqrt[a^2*x^2+b])^(1/3))	Problem nested_12 (iv)	{“failed”, “nested radicals”}
nested-nested_12-iv	PMT	(a^2*x^2-b)/((a^2*x^2+b)*(a*x-Sqrt[a^2*x^2+b])^(1/3))	Problem nested_12 (iv)	{“failed”, “tower construction failed”, (-b + a^2 x^2)/((b + a^2 x^2) (a x - Sqrt[b + a^2 x^2])^(1/3))}
nested-nested_12-iv-inst	Trager	(2^2*x^2-3)/((2^2*x^2+3)*(2*x-Sqrt[2^2*x^2+3])^(1/3))	Problem nested_12 (iv); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_13-i	Trager	x*Sqrt[a*x^2-b]/Sqrt[(c*x*Sqrt[a*x^2-b]-a*x^2+b)]	Problem nested_13 (i)	{“failed”, “nested radicals”}
nested-nested_13-i	PMT	x*Sqrt[a*x^2-b]/Sqrt[(c*x*Sqrt[a*x^2-b]-a*x^2+b)]	Problem nested_13 (i)	{“failed”, “tower construction failed”, (x Sqrt[-b + a x^2])/Sqrt[b - a x^2 + c x Sqrt[-b + a x^2]]}
nested-nested_13-i-inst	Trager	x*Sqrt[2*x^2-3]/Sqrt[(5*x*Sqrt[2*x^2-3]-2*x^2+3)]	Problem nested_13 (i); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_13-i-inst	PMT	x*Sqrt[2*x^2-3]/Sqrt[(5*x*Sqrt[2*x^2-3]-2*x^2+3)]	Problem nested_13 (i); inst a=2, b=3, c=…	{“failed”, “tower construction failed”, (x Sqrt[-3 + 2 x^2])/Sqrt[3 - 2 x^2 + 5 x Sqrt[-3 + 2 x^2]]}
nested-nested_13-ii	Trager	(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(3/2)/(x*Sqrt[a*x^2-b])	Problem nested_13 (ii)	{“failed”, “nested radicals”}
nested-nested_13-ii	PMT	(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(3/2)/(x*Sqrt[a*x^2-b])	Problem nested_13 (ii)	{“failed”, “tower construction failed”, (b - a x^2 + c x Sqrt[-b + a x^2])^(3/2)/(x Sqrt[-b + a x^2])}
nested-nested_13-ii-inst	Trager	(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(3/2)/(x*Sqrt[2*x^2-3])	Problem nested_13 (ii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_13-ii-inst	PMT	(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(3/2)/(x*Sqrt[2*x^2-3])	Problem nested_13 (ii); inst a=2, b=3, c…	{“failed”, “tower construction failed”, (3 - 2 x^2 + 5 x Sqrt[-3 + 2 x^2])^(3/2)/(x Sqrt[-3 + 2 x^2])}
nested-nested_13-iii	Trager	Sqrt[(c*x*Sqrt[a*x^2-b]-a*x^2+b)]/(c*x*Sqrt[a*x^2-b]-a*x^2)	Problem nested_13 (iii)	{“failed”, “nested radicals”}
nested-nested_13-iii	PMT	Sqrt[(c*x*Sqrt[a*x^2-b]-a*x^2+b)]/(c*x*Sqrt[a*x^2-b]-a*x^2)	Problem nested_13 (iii)	{“failed”, “tower construction failed”, Sqrt[b - a x^2 + c x Sqrt[-b + a x^2]]/(-a x^2 + c x Sqrt[-b + a x^2])…
nested-nested_13-iii-inst	Trager	Sqrt[(5*x*Sqrt[2*x^2-3]-2*x^2+3)]/(5*x*Sqrt[2*x^2-3]-2*x^2)	Problem nested_13 (iii); inst a=2, b=3, …	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_13-iii-inst	PMT	Sqrt[(5*x*Sqrt[2*x^2-3]-2*x^2+3)]/(5*x*Sqrt[2*x^2-3]-2*x^2)	Problem nested_13 (iii); inst a=2, b=3, …	{“failed”, “tower construction failed”, Sqrt[3 - 2 x^2 + 5 x Sqrt[-3 + 2 x^2]]/(-2 x^2 + 5 x Sqrt[-3 + 2 x^2])…
nested-nested_13-iv	Trager	(c*x*Sqrt[a*x^2-b]+b)/(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(5/2)	Problem nested_13 (iv)	{“failed”, “nested radicals”}
nested-nested_13-iv	PMT	(c*x*Sqrt[a*x^2-b]+b)/(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(5/2)	Problem nested_13 (iv)	{“failed”, “tower construction failed”, (b + c x Sqrt[-b + a x^2])/(b - a x^2 + c x Sqrt[-b + a x^2])^(5/2)}
nested-nested_13-iv-inst	Trager	(5*x*Sqrt[2*x^2-3]+3)/(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(5/2)	Problem nested_13 (iv); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_13-iv-inst	PMT	(5*x*Sqrt[2*x^2-3]+3)/(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(5/2)	Problem nested_13 (iv); inst a=2, b=3, c…	{“failed”, “tower construction failed”, (3 + 5 x Sqrt[-3 + 2 x^2])/(3 - 2 x^2 + 5 x Sqrt[-3 + 2 x^2])^(5/2)}
nested-nested_13-v	Trager	(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(3/2)	Problem nested_13 (v)	{“failed”, “nested radicals”}
nested-nested_13-v	PMT	(c*x*Sqrt[a*x^2-b]-a*x^2+b)^(3/2)	Problem nested_13 (v)	{“failed”, “tower construction failed”, (b - a x^2 + c x Sqrt[-b + a x^2])^(3/2)}
nested-nested_13-v-inst	Trager	(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(3/2)	Problem nested_13 (v); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_13-v-inst	PMT	(5*x*Sqrt[2*x^2-3]-2*x^2+3)^(3/2)	Problem nested_13 (v); inst a=2, b=3, c=…	{“failed”, “tower construction failed”, (3 - 2 x^2 + 5 x Sqrt[-3 + 2 x^2])^(3/2)}
nested-nested_14-i	Trager	Sqrt[Sqrt[a*x^2+b^2]+b]	Problem nested_14 (i)	{“failed”, “nested radicals”}
nested-nested_14-i	PMT	Sqrt[Sqrt[a*x^2+b^2]+b]	Problem nested_14 (i)	{“failed”, “tower construction failed”, Sqrt[b + Sqrt[b^2 + a x^2]]}
nested-nested_14-i-inst	Trager	Sqrt[Sqrt[2*x^2+3^2]+3]	Problem nested_14 (i); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_14-ii	Trager	Sqrt[Sqrt[a*x^2+b^2]+b]/Sqrt[a*x^2+b^2]	Problem nested_14 (ii)	{“failed”, “nested radicals”}
nested-nested_14-ii	PMT	Sqrt[Sqrt[a*x^2+b^2]+b]/Sqrt[a*x^2+b^2]	Problem nested_14 (ii)	{“failed”, “tower construction failed”, Sqrt[b + Sqrt[b^2 + a x^2]]/Sqrt[b^2 + a x^2]}
nested-nested_14-ii-inst	Trager	Sqrt[Sqrt[2*x^2+3^2]+3]/Sqrt[2*x^2+3^2]	Problem nested_14 (ii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_14-iii	Trager	Sqrt[Sqrt[a*x^2+b^2]+b]/(x^2*Sqrt[a*x^2+b^2])	Problem nested_14 (iii)	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_14-iii	PMT	Sqrt[Sqrt[a*x^2+b^2]+b]/(x^2*Sqrt[a*x^2+b^2])	Problem nested_14 (iii)	{“failed”, “tower construction failed”, Sqrt[b + Sqrt[b^2 + a x^2]]/(x^2 Sqrt[b^2 + a x^2])}
nested-nested_14-iii-inst	Trager	Sqrt[Sqrt[2*x^2+3^2]+3]/(x^2*Sqrt[2*x^2+3^2])	Problem nested_14 (iii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_14-iii-inst	PMT	Sqrt[Sqrt[2*x^2+3^2]+3]/(x^2*Sqrt[2*x^2+3^2])	Problem nested_14 (iii); inst a=2, b=3	{“failed”, “no solution within bounds”, {1}}
nested-nested_14-iv	Trager	Sqrt[Sqrt[a*x^2+b^2]+b]/((a*x^2-b)*Sqrt[a*x^2+b^2])	Problem nested_14 (iv)	{“failed”, “nested radicals”}
nested-nested_14-iv	PMT	Sqrt[Sqrt[a*x^2+b^2]+b]/((a*x^2-b)*Sqrt[a*x^2+b^2])	Problem nested_14 (iv)	{“failed”, “tower construction failed”, Sqrt[b + Sqrt[b^2 + a x^2]]/((-b + a x^2) Sqrt[b^2 + a x^2])}
nested-nested_14-iv-inst	Trager	Sqrt[Sqrt[2*x^2+3^2]+3]/((2*x^2-3)*Sqrt[2*x^2+3^2])	Problem nested_14 (iv); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_14-v	Trager	(a*x^2+b)^2*Sqrt[a*x^2+b^2]/((a*x^2-b)^2*Sqrt[Sqrt[a*x^2+b^2]+b])	Problem nested_14 (v)	{“failed”, “nested radicals”}
nested-nested_14-v	PMT	(a*x^2+b)^2*Sqrt[a*x^2+b^2]/((a*x^2-b)^2*Sqrt[Sqrt[a*x^2+b^2]+b])	Problem nested_14 (v)	{“failed”, “tower construction failed”, ((b + a x^2)^2 Sqrt[b^2 + a x^2])/((-b + a x^2)^2 Sqrt[b + Sqrt[b^2 + …
nested-nested_14-v-inst	Trager	(2*x^2+3)^2*Sqrt[2*x^2+3^2]/((2*x^2-3)^2*Sqrt[Sqrt[2*x^2+3^2]+3])	Problem nested_14 (v); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_15-i	Trager	Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_15 (i)	{“failed”, “nested radicals”}
nested-nested_15-i	PMT	Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_15 (i)	{“failed”, “tower construction failed”, Sqrt[a x + Sqrt[b + a^2 x^2]]}
nested-nested_15-i-inst	Trager	Sqrt[Sqrt[2^2*x^2+3]+2*x]	Problem nested_15 (i); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_15-ii	Trager	x^2*Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_15 (ii)	{“failed”, “nested radicals”}
nested-nested_15-ii	PMT	x^2*Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_15 (ii)	{“failed”, “tower construction failed”, x^2 Sqrt[a x + Sqrt[b + a^2 x^2]]}
nested-nested_15-ii-inst	Trager	x^2*Sqrt[Sqrt[2^2*x^2+3]+2*x]	Problem nested_15 (ii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_15-iii	Trager	1/(x^2*Sqrt[Sqrt[a^2*x^2+b]+a*x])	Problem nested_15 (iii)	{“failed”, “nested radicals”}
nested-nested_15-iii	PMT	1/(x^2*Sqrt[Sqrt[a^2*x^2+b]+a*x])	Problem nested_15 (iii)	{“failed”, “tower construction failed”, 1/(x^2 Sqrt[a x + Sqrt[b + a^2 x^2]])}
nested-nested_15-iii-inst	Trager	1/(x^2*Sqrt[Sqrt[2^2*x^2+3]+2*x])	Problem nested_15 (iii); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_15-iv	Trager	(a^2*x^2+b)^(3/2)/(x*(Sqrt[a^2*x^2+b]+a*x)^(3/2))	Problem nested_15 (iv)	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_15-iv	PMT	(a^2*x^2+b)^(3/2)/(x*(Sqrt[a^2*x^2+b]+a*x)^(3/2))	Problem nested_15 (iv)	{“failed”, “tower construction failed”, (b + a^2 x^2)^(3/2)/(x (a x + Sqrt[b + a^2 x^2])^(3/2))}
nested-nested_15-iv-inst	Trager	(2^2*x^2+3)^(3/2)/(x*(Sqrt[2^2*x^2+3]+2*x)^(3/2))	Problem nested_15 (iv); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_15-v	Trager	x^2/((a^2*x^2+b)^(3/2)*(Sqrt[a^2*x^2+b]+a*x)^(3/2))	Problem nested_15 (v)	{“failed”, “nested radicals”}
nested-nested_15-v	PMT	x^2/((a^2*x^2+b)^(3/2)*(Sqrt[a^2*x^2+b]+a*x)^(3/2))	Problem nested_15 (v)	{“failed”, “tower construction failed”, x^2/((b + a^2 x^2)^(3/2) (a x + Sqrt[b + a^2 x^2])^(3/2))}
nested-nested_15-v-inst	Trager	x^2/((2^2*x^2+3)^(3/2)*(Sqrt[2^2*x^2+3]+2*x)^(3/2))	Problem nested_15 (v); inst a=2, b=3	{“failed”, “nested radicals”}
nested-nested_16-i	Trager	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]	Problem nested_16 (i)	{“failed”, “nested radicals”}
nested-nested_16-i	PMT	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]	Problem nested_16 (i)	{“failed”, “tower construction failed”, Sqrt[c + Sqrt[a x + Sqrt[b + a^2 x^2]]]}
nested-nested_16-i-inst	Trager	Sqrt[Sqrt[Sqrt[2^2*x^2+3]+2*x]+5]	Problem nested_16 (i); inst a=2, b=3, c=…	{“failed”, “nested radicals”}
nested-nested_16-ii	Trager	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]/(Sqrt[a^2*x^2+b]*Sqrt[Sqrt[a^2*x^2+b]+a*x])	Problem nested_16 (ii)	{“failed”, “nested radicals”}
nested-nested_16-ii	PMT	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]/(Sqrt[a^2*x^2+b]*Sqrt[Sqrt[a^2*x^2+b]+a*x])	Problem nested_16 (ii)	{“failed”, “tower construction failed”, Sqrt[c + Sqrt[a x + Sqrt[b + a^2 x^2]]]/(Sqrt[b + a^2 x^2] Sqrt[a x + …
nested-nested_16-ii-inst	Trager	Sqrt[Sqrt[Sqrt[2^2*x^2+3]+2*x]+5]/(Sqrt[2^2*x^2+3]*Sqrt[Sqrt[2^2*x^2+3]+2*x])	Problem nested_16 (ii); inst a=2, b=3, c…	{“failed”, “nested radicals”}
nested-nested_16-iii	Trager	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]/Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_16 (iii)	{“failed”, “nested radicals”}
nested-nested_16-iii	PMT	Sqrt[Sqrt[Sqrt[a^2*x^2+b]+a*x]+c]/Sqrt[Sqrt[a^2*x^2+b]+a*x]	Problem nested_16 (iii)	{“failed”, “tower construction failed”, Sqrt[c + Sqrt[a x + Sqrt[b + a^2 x^2]]]/Sqrt[a x + Sqrt[b + a^2 x^2]]}
nested-nested_16-iii-inst	Trager	Sqrt[Sqrt[Sqrt[2^2*x^2+3]+2*x]+5]/Sqrt[Sqrt[2^2*x^2+3]+2*x]	Problem nested_16 (iii); inst a=2, b=3, …	{“failed”, “nested radicals”}
nested-nested_17-i	Trager	((x^2+1)^(3/2)+x^3)^(1/3)/(x^2*(x^2+1))	Problem nested_17 (i)	{“failed”, “nested radicals”}
nested-nested_17-ii	Trager	(a*(c*x^2+d)^(3/2)+b*x^3)^(1/3)/(x^2*(c*x^2+d))	Problem nested_17 (ii)	{“failed”, “nested radicals”}
nested-nested_17-ii	PMT	(a*(c*x^2+d)^(3/2)+b*x^3)^(1/3)/(x^2*(c*x^2+d))	Problem nested_17 (ii)	{“failed”, “tower construction failed”, (b x^3 + a (d + c x^2)^(3/2))^(1/3)/(x^2 (d + c x^2))}
nested-nested_17-ii-inst	Trager	(2*(5*x^2+7)^(3/2)+3*x^3)^(1/3)/(x^2*(5*x^2+7))	Problem nested_17 (ii); inst a=2, b=3, c…	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
nested-nested_17-ii-inst	PMT	(2*(5*x^2+7)^(3/2)+3*x^3)^(1/3)/(x^2*(5*x^2+7))	Problem nested_17 (ii); inst a=2, b=3, c…	{"failed", "tower construction failed", (3 x^3 + 2 (7 + 5 x^2)^(3/2))^(1/3)/(x^2 (7 + 5 x^2))}
nested-nested_17-iii	Trager	(2*x^2+1)*(x^3*(x^2+1)^(3/2)+1)^(1/3)/(x^4*(x^2+1))	Problem nested_17 (iii)	{"failed", "nested radicals"}
nested-nested_17-iv	Trager	(2*c*x^2+d)*(a*x^3*(c*x^2+d)^(3/2)+b)^(1/3)/(x*(c*x^2+d))	Problem nested_17 (iv)	{"failed", "nested radicals"}
nested-nested_17-iv	PMT	(2*c*x^2+d)*(a*x^3*(c*x^2+d)^(3/2)+b)^(1/3)/(x*(c*x^2+d))	Problem nested_17 (iv)	{"failed", "tower construction failed", ((d + 2 c x^2) (b + a x^3 (d + c x^2)^(3/2))^(1/3))/(x (d + c x^2))}
nested-nested_17-iv-inst	Trager	(2*5*x^2+7)*(2*x^3*(5*x^2+7)^(3/2)+3)^(1/3)/(x*(5*x^2+7))	Problem nested_17 (iv); inst a=2, b=3, c…	{"failed", "nested radicals"}
nested-nested_17-iv-inst	PMT	(2*5*x^2+7)*(2*x^3*(5*x^2+7)^(3/2)+3)^(1/3)/(x*(5*x^2+7))	Problem nested_17 (iv); inst a=2, b=3, c…	{"failed", "tower construction failed", ((7 + 10 x^2) (3 + 2 x^3 (7 + 5 x^2)^(3/2))^(1/3))/(x (7 + 5 x^2))}
nested-nested_17-v	Trager	x^2*(5*x^3+2)*Sqrt[x^3+1]/(x^3*(x^3+1)^(3/2)+1)^(1/3)	Problem nested_17 (v)	{"failed", "nested radicals"}
nested-nested_17-v	PMT	x^2*(5*x^3+2)*Sqrt[x^3+1]/(x^3*(x^3+1)^(3/2)+1)^(1/3)	Problem nested_17 (v)	{"failed", "tower construction failed", (x^2 Sqrt[1 + x^3] (2 + 5 x^3))/(1 + x^3 (1 + x^3)^(3/2))^(1/3)}
nested-nested_17-vi	Trager	x^2*(5*c*x^3+2*d)*Sqrt[c*x^3+d]/(a*x^3*(c*x^3+d)^(3/2)+b)^(1/3)	Problem nested_17 (vi)	{"failed", "nested radicals"}
nested-nested_17-vi	PMT	x^2*(5*c*x^3+2*d)*Sqrt[c*x^3+d]/(a*x^3*(c*x^3+d)^(3/2)+b)^(1/3)	Problem nested_17 (vi)	{"failed", "tower construction failed", (x^2 Sqrt[d + c x^3] (2 d + 5 c x^3))/(b + a x^3 (d + c x^3)^(3/2))^(1…
nested-nested_17-vi-inst	Trager	x^2*(5*5*x^3+2*7)*Sqrt[5*x^3+7]/(2*x^3*(5*x^3+7)^(3/2)+3)^(1/3)	Problem nested_17 (vi); inst a=2, b=3, c…	{"failed", "nested radicals"}
nested-nested_17-vi-inst	PMT	x^2*(5*5*x^3+2*7)*Sqrt[5*x^3+7]/(2*x^3*(5*x^3+7)^(3/2)+3)^(1/3)	Problem nested_17 (vi); inst a=2, b=3, c…	{"failed", "tower construction failed", (x^2 Sqrt[7 + 5 x^3] (14 + 25 x^3))/(3 + 2 x^3 (7 + 5 x^3)^(3/2))^(1/3…
nested-nested_19-i	Trager	1/Sqrt[x^4+x^2*Sqrt[x^4+4]]	Problem nested_19 (i)	{"failed", "nested radicals"}
nested-nested_19-i	PMT	1/Sqrt[x^4+x^2*Sqrt[x^4+4]]	Problem nested_19 (i)	{"failed", "tower construction failed", 1/Sqrt[x^4 + x^2 Sqrt[4 + x^4]]}
nested-nested_19-ii	Trager	Sqrt[1/x^4-Sqrt[4*x^4+1]/x^4]	Problem nested_19 (ii)	{"failed", "nested radicals"}
nested-nested_19-ii	PMT	Sqrt[1/x^4-Sqrt[4*x^4+1]/x^4]	Problem nested_19 (ii)	{"failed", "tower construction failed", Sqrt[1/x^4 - Sqrt[1 + 4 x^4]/x^4]}

id	method	integrand	book ref	detail
nested-nested_19-iii	Trager	(Sqrt[2]*x^3-Sqrt[2*x^6-1])^(1/3)	Problem nested_19 (iii)	{“failed”, “algebraic constant in the integrand (coefficients must be rational)”}
nested-nested_19-iii	PMT	(Sqrt[2]*x^3-Sqrt[2*x^6-1])^(1/3)	Problem nested_19 (iii)	{“failed”, “tower construction failed”, (Sqrt[2] x^3 - Sqrt[-1 + 2 x^6])^(1/3)}
nested-nested_19-iv	Trager	(Sqrt[2]/x^3-Sqrt[2-x^6]/x^3)^(1/3)	Problem nested_19 (iv)	{“failed”, “algebraic constant in the integrand (coefficients must be rational)”}
nested-nested_19-iv	PMT	(Sqrt[2]/x^3-Sqrt[2-x^6]/x^3)^(1/3)	Problem nested_19 (iv)	{“failed”, “tower construction failed”, (Sqrt[2]/x^3 - Sqrt[2 - x^6]/x^3)^(1/3)}
nested-nested_19-v	Trager	(x^3+Sqrt[x^6-1])^(1/3)	Problem nested_19 (v)	{“failed”, “nested radicals”}
nested-nested_19-v	PMT	(x^3+Sqrt[x^6-1])^(1/3)	Problem nested_19 (v)	{“failed”, “tower construction failed”, (x^3 + Sqrt[-1 + x^6])^(1/3)}
nested-nested_19-vi	Trager	(1/x^3+Sqrt[1/x^6-4])^(1/3)	Problem nested_19 (vi)	{“failed”, “nested radicals”}
nested-nested_19-vi	PMT	(1/x^3+Sqrt[1/x^6-4])^(1/3)	Problem nested_19 (vi)	{“failed”, “tower construction failed”, (Sqrt[-4 + 1/x^6] + 1/x^3)^(1/3)}
nested-nested_19-vii	Trager	(1+x^3+Sqrt[x^6+2*x^3])^(1/3)	Problem nested_19 (vii)	{“failed”, “nested radicals”}
nested-nested_19-vii	PMT	(1+x^3+Sqrt[x^6+2*x^3])^(1/3)	Problem nested_19 (vii)	{“failed”, “tower construction failed”, (1 + x^3 + Sqrt[2 x^3 + x^6])^(1/3)}
nested-nested_19-viii	Trager	(1+1/x^3+Sqrt[2*x^3+1]/x^3)^(1/3)	Problem nested_19 (viii)	{“failed”, “nested radicals”}
nested-nested_19-viii	PMT	(1+1/x^3+Sqrt[2*x^3+1]/x^3)^(1/3)	Problem nested_19 (viii)	{“failed”, “tower construction failed”, (1 + 1/x^3 + Sqrt[1 + 2 x^3]/x^3)^(1/3)}
nested-nested_19-ix	Trager	(x^3-Sqrt[x^6+4])^(1/3)	Problem nested_19 (ix)	{“failed”, “nested radicals”}
nested-nested_19-ix	PMT	(x^3-Sqrt[x^6+4])^(1/3)	Problem nested_19 (ix)	{“failed”, “tower construction failed”, (x^3 - Sqrt[4 + x^6])^(1/3)}
nested-nested_19-x	Trager	(1/x^3-Sqrt[4*x^6+1]/x^3)^(1/3)	Problem nested_19 (x)	{“failed”, “nested radicals”}
nested-nested_19-x	PMT	(1/x^3-Sqrt[4*x^6+1]/x^3)^(1/3)	Problem nested_19 (x)	{“failed”, “tower construction failed”, (1/x^3 - Sqrt[1 + 4 x^6]/x^3)^(1/3)}
nested-nested_19-xi	Trager	(2+1/x^2-Sqrt[8*x^2+1]/x^2)^(1/4)	Problem nested_19 (xi)	{“failed”, “nested radicals”}
nested-nested_19-xii	Trager	(1-Sqrt[x^8+8*x^4]/x^4)^(1/4)	Problem nested_19 (xii)	{“failed”, “nested radicals”}
nested-nested_19-xii	PMT	(1-Sqrt[x^8+8*x^4]/x^4)^(1/4)	Problem nested_19 (xii)	{“failed”, “tower construction failed”, (1 - Sqrt[8 x^4 + x^8]/x^4)^(1/4)}

id	method	integrand	book ref	detail
nested-nested_19-xiii	Trager	(1+Sqrt[x^8+1]/x^4)^(1/4)	Problem nested_19 (xiii)	{“failed”, “nested radicals”}
nested-nested_19-xiii	PMT	(1+Sqrt[x^8+1]/x^4)^(1/4)	Problem nested_19 (xiii)	{“failed”, “tower construction failed”, (1 + Sqrt[1 + x^8]/x^4)^(1/4)}
multrad-intro-1	Trager	(x^4-12)/((x^4-8*x^2+12)*Sqrt[x^2-4]*Sqrt[x^2-3])	Section intro example (solved, -ArcTanh)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-i	Trager	(x^4-a*b)/((d*x^4-(a*d+b*d-c)*x^2+a*b*d)*Sqrt[x^2-a]*Sqrt[x^2-b])	Problem multiple_radicals_0 (i)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-i	PMT	(x^4-a*b)/((d*x^4-(a*d+b*d-c)*x^2+a*b*d)*Sqrt[x^2-a]*Sqrt[x^2-b])	Problem multiple_radicals_0 (i)	{“failed”, “verification failed”}
multrad-multiple_radicals_0-i-inst	Trager	(x^4-2*3)/((7*x^4-(2*7+3*7-5)*x^2+2*3*7)*Sqrt[x^2-2]*Sqrt[x^2-3])	Problem multiple_radicals_0 (i); inst a=…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-i-inst	PMT	(x^4-2*3)/((7*x^4-(2*7+3*7-5)*x^2+2*3*7)*Sqrt[x^2-2]*Sqrt[x^2-3])	Problem multiple_radicals_0 (i); inst a=…	{“failed”, “verification failed”}
multrad-multiple_radicals_0-ii	Trager	(x^4-a*b)/((x^4+a*b)*Sqrt[x^2-a]*Sqrt[x^2-b])	Problem multiple_radicals_0 (ii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-ii	PMT	(x^4-a*b)/((x^4+a*b)*Sqrt[x^2-a]*Sqrt[x^2-b])	Problem multiple_radicals_0 (ii)	{“failed”, “verification failed”}
multrad-multiple_radicals_0-ii-inst	Trager	(x^4-2*3)/((x^4+2*3)*Sqrt[x^2-2]*Sqrt[x^2-3])	Problem multiple_radicals_0 (ii); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-ii-inst	PMT	(x^4-2*3)/((x^4+2*3)*Sqrt[x^2-2]*Sqrt[x^2-3])	Problem multiple_radicals_0 (ii); inst a…	{“failed”, “verification failed”}
multrad-multiple_radicals_0-iii	Trager	(a*x^4+b)^(1/4)/(b^2-a^2*x^8)^(1/4)	Problem multiple_radicals_0 (iii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-iii	PMT	(a*x^4+b)^(1/4)/(b^2-a^2*x^8)^(1/4)	Problem multiple_radicals_0 (iii)	{“failed”, “tower construction failed”, (b + a x^4)^(1/4)/(b^2 - a^2 x^8)^(1/4)}
multrad-multiple_radicals_0-iii-inst	Trager	(2*x^4+3)^(1/4)/(3^2-2^2*x^8)^(1/4)	Problem multiple_radicals_0 (iii); inst …	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-iii-inst	PMT	(2*x^4+3)^(1/4)/(3^2-2^2*x^8)^(1/4)	Problem multiple_radicals_0 (iii); inst …	{“failed”, “tower construction failed”, (3 + 2 x^4)^(1/4)/(9 - 4 x^8)^(1/4)}
multrad-multiple_radicals_0-iv	Trager	(x^4-a*b)*Sqrt[x^2-a]*Sqrt[x^2-b]/(x^2*(d*x^4+(c-a*d-b*d)*x^2+a*b*d))	Problem multiple_radicals_0 (iv)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-iv	PMT	(x^4-a*b)*Sqrt[x^2-a]*Sqrt[x^2-b]/(x^2*(d*x^4+(c-a*d-b*d)*x^2+a*b*d))	Problem multiple_radicals_0 (iv)	{“failed”, “verification failed”}
multrad-multiple_radicals_0-iv-inst	Trager	(x^4-2*3)*Sqrt[x^2-2]*Sqrt[x^2-3]/(x^2*(7*x^4+(5-2*7-3*7)*x^2+2*3*7))	Problem multiple_radicals_0 (iv); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-iv-inst	PMT	(x^4-2*3)*Sqrt[x^2-2]*Sqrt[x^2-3]/(x^2*(7*x^4+(5-2*7-3*7)*x^2+2*3*7))	Problem multiple_radicals_0 (iv); inst a…	{“failed”, “verification failed”}

id	method	integrand	book ref	detail
multrad-multiple_radicals_0-v	Trager	x^2*(x^4-2*b*x^2+a*b)*Sqrt[x^2-a]/((d*x^8-2*a*d*x^6+(a^2*d+c)*x^4-2*b*c*x^2+b^2*c)*Sqrt[x^…	Problem multiple_radicals_0 (v)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-v-inst	Trager	x^2*(x^4-2*3*x^2+2*3)*Sqrt[x^2-2]/((7*x^8-2*2*7*x^6+(2^2*7+5)*x^4-2*3*5*x^2+3^2*5)*Sqrt[x^…	Problem multiple_radicals_0 (v); inst a=…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_0-v-inst	PMT	x^2*(x^4-2*3*x^2+2*3)*Sqrt[x^2-2]/((7*x^8-2*2*7*x^6+(2^2*7+5)*x^4-2*3*5*x^2+3^2*5)*Sqrt[x^…	Problem multiple_radicals_0 (v); inst a=…	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, 45927 + 40824 …
multrad-multiple_radicals_0-vi	Trager	(x^4+2*a*x^2+a^2)*(x^6+3*a*x^4-a*(2*a+3*b)*x^2+a^2*b)/((c*x^12-2*c*x^10+(a^2*c+4*a*b*c+b^2…	Problem multiple_radicals_0 (vi)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_0-vi-inst	Trager	(x^4+2*2*x^2+2^2)*(x^6+3*2*x^4-2*(2*2+3*3)*x^2+2^2*3)/((5*x^12-2*5*x^10+(2^2*5+4*2*3*5+3^2…	Problem multiple_radicals_0 (vi); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_1-i	Trager	Sqrt[x^2+1]*Sqrt[1-x^4]	Problem multiple_radicals_1 (i)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_1-ii	Trager	Sqrt[a^2*x^2+b^2]*Sqrt[b^4-a^4*x^4]	Problem multiple_radicals_1 (ii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_1-ii	PMT	Sqrt[a^2*x^2+b^2]*Sqrt[b^4-a^4*x^4]	Problem multiple_radicals_1 (ii)	{“failed”, “tower construction failed”, Sqrt[b^2 + a^2 x^2] Sqrt[b^4 - a^4 x^4]}
multrad-multiple_radicals_1-ii-inst	Trager	Sqrt[2^2*x^2+3^2]*Sqrt[3^4-2^4*x^4]	Problem multiple_radicals_1 (ii); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_1-iii	Trager	1/(Sqrt[x^2+1]*Sqrt[1-x^4])	Problem multiple_radicals_1 (iii)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_1-iv	Trager	1/(Sqrt[a^2*x^2+b^2]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_1 (iv)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_1-iv	PMT	1/(Sqrt[a^2*x^2+b^2]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_1 (iv)	{“failed”, “tower construction failed”, 1/(Sqrt[b^2 + a^2 x^2] Sqrt[b^4 - a^4 x^4])}
multrad-multiple_radicals_1-iv-inst	Trager	1/(Sqrt[2^2*x^2+3^2]*Sqrt[3^4-2^4*x^4])	Problem multiple_radicals_1 (iv); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_2-i	Trager	(x^4+2)/((x^4-2)*Sqrt[x^2-1]*Sqrt[x^2+2])	Problem multiple_radicals_2 (i)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_2-ii	Trager	(x^4-2)*Sqrt[x^2+1]*Sqrt[x^2+2]/(x^2*(x^4+2*x^2+2))	Problem multiple_radicals_2 (ii)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_2-iii	Trager	(x^4+2*a*x^2-a*b)*Sqrt[x^2+a]/((x^4-(b+1)*x^2-a)*(x^4+(1-b)*x^2+a)*Sqrt[x^2-b])	Problem multiple_radicals_2 (iii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_2-iii-inst	Trager	(x^4+2*2*x^2-2*3)*Sqrt[x^2+2]/((x^4-(3+1)*x^2-2)*(x^4+(1-3)*x^2+2)*Sqrt[x^2-3])	Problem multiple_radicals_2 (iii); inst …	{“failed”, “more than one radicand (a radical compositum)”}

id	method	integrand	book ref	detail
multrad-multiple_radicals_2-iv	Trager	(3*x^8+(a-b)*x^4+a*b)*Sqrt[x^4+a]*Sqrt[x^4-b]/((x^8+(a-b)*x^4+x^2-a*b)*(x^8+(a-b)*x^4-x^2-…	Problem multiple_radicals_2 (iv)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_2-iv	PMT	(3*x^8+(a-b)*x^4+a*b)*Sqrt[x^4+a]*Sqrt[x^4-b]/((x^8+(a-b)*x^4+x^2-a*b)*(x^8+(a-b)*x^4-x^2-…	Problem multiple_radicals_2 (iv)	{“failed”, “tower construction failed”, (Sqrt[a + x^4] Sqrt[-b + x^4] (a b + (a - b) x^4 + 3 x^8))/((-a b - x^…
multrad-multiple_radicals_2-iv-inst	Trager	(3*x^8+(2-3)*x^4+2*3)*Sqrt[x^4+2]*Sqrt[x^4-3]/((x^8+(2-3)*x^4+x^2-2*3)*(x^8+(2-3)*x^4-x^2-…	Problem multiple_radicals_2 (iv); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_2-iv-inst	PMT	(3*x^8+(2-3)*x^4+2*3)*Sqrt[x^4+2]*Sqrt[x^4-3]/((x^8+(2-3)*x^4+x^2-2*3)*(x^8+(2-3)*x^4-x^2-…	Problem multiple_radicals_2 (iv); inst a…	{“failed”, “tower construction failed”, (Sqrt[-3 + x^4] Sqrt[2 + x^4] (6 - x^4 + 3 x^8))/((-6 - x^2 - x^4 + x^…
multrad-multiple_radicals_2-v-inst	Trager	(2*x^5-3*3*x^4+4*5*x^3-2*2*x+2*3)*Sqrt[x^4+2]/((x^4-x^2+3*x+2-5)*(x^2-3*x+5)^(3/2))	Problem multiple_radicals_2 (v); inst a=…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_2-v-inst	PMT	(2*x^5-3*3*x^4+4*5*x^3-2*2*x+2*3)*Sqrt[x^4+2]/((x^4-x^2+3*x+2-5)*(x^2-3*x+5)^(3/2))	Problem multiple_radicals_2 (v); inst a=…	{“failed”, “tower construction failed”, (Sqrt[2 + x^4] (6 - 4 x + 20 x^3 - 9 x^4 + 2 x^5))/((5 - 3 x + x^2)^(3…
multrad-multiple_radicals_3-i-inst	Trager	(x^2-2*2*x+2*3+2*5-3*5)*Sqrt[x-3]*Sqrt[x-5]/((x^2-(3+5-1)*x+3*5-2)*(x-2)^(3/2))	Problem multiple_radicals_3 (i); inst a=…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_3-i-inst	PMT	(x^2-2*2*x+2*3+2*5-3*5)*Sqrt[x-3]*Sqrt[x-5]/((x^2-(3+5-1)*x+3*5-2)*(x-2)^(3/2))	Problem multiple_radicals_3 (i); inst a=…	{“failed”, “verification failed”}
multrad-multiple_radicals_3-ii	Trager	(x^2-2*a*x+a*b+a*c-b*c)/((x^2-(b+c+1)*x+a+b*c)*Sqrt[x-a]*Sqrt[x-b]*Sqrt[x-c])	Problem multiple_radicals_3 (ii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_3-ii	PMT	(x^2-2*a*x+a*b+a*c-b*c)/((x^2-(b+c+1)*x+a+b*c)*Sqrt[x-a]*Sqrt[x-b]*Sqrt[x-c])	Problem multiple_radicals_3 (ii)	{“failed”, “verification failed”}
multrad-multiple_radicals_3-ii-inst	Trager	(x^2-2*2*x+2*3+2*5-3*5)/((x^2-(3+5+1)*x+2+3*5)*Sqrt[x-2]*Sqrt[x-3]*Sqrt[x-5])	Problem multiple_radicals_3 (ii); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_3-iii	Trager	(x^2-2*b*x+a*b-a*c+b*c)*Sqrt[x-b]/((x^2-(a+c-1)*x-b+a*c)*(x^2-(a+c+1)*x+b+a*c)*Sqrt[x-a]*S…	Problem multiple_radicals_3 (iii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_3-iii-inst	Trager	(x^2-2*3*x+2*3-2*5+3*5)*Sqrt[x-3]/((x^2-(2+5-1)*x-3+2*5)*(x^2-(2+5+1)*x+3+2*5)*Sqrt[x-2]*S…	Problem multiple_radicals_3 (iii); inst …	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-i	Trager	Sqrt[1-x]*Sqrt[x+1]*Sqrt[1-x^4]	Problem multiple_radicals_4 (i)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-i	PMT	Sqrt[1-x]*Sqrt[x+1]*Sqrt[1-x^4]	Problem multiple_radicals_4 (i)	{“needs torsion realisation (milestone iii)”, {{1 + (-1 + Sqrt[1 - x])^2/(-1 + Sqrt[1 + x])^2, {5/4, -5/4, -5/…
multrad-multiple_radicals_4-ii	Trager	Sqrt[b-a*x]*Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4]	Problem multiple_radicals_4 (ii)	{“failed”, “radicand is not a rational function over Q”}

id	method	integrand	book ref	detail
multrad-multiple_radicals_4-ii	PMT	Sqrt[b-a*x]*Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4]	Problem multiple_radicals_4 (ii)	{“failed”, “tower construction failed”, Sqrt[b - a x] Sqrt[b + a x] Sqrt[b^4 - a^4 x^4]}
multrad-multiple_radicals_4-ii-inst	Trager	Sqrt[3-2*x]*Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4]	Problem multiple_radicals_4 (ii); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-ii-inst	PMT	Sqrt[3-2*x]*Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4]	Problem multiple_radicals_4 (ii); inst a…	{“failed”, “tower construction failed”, Sqrt[3 - 2 x] Sqrt[3 + 2 x] Sqrt[81 - 16 x^4]}
multrad-multiple_radicals_4-iii	Trager	Sqrt[x+1]*Sqrt[1-x^4]/Sqrt[1-x]	Problem multiple_radicals_4 (iii)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-iii	PMT	Sqrt[x+1]*Sqrt[1-x^4]/Sqrt[1-x]	Problem multiple_radicals_4 (iii)	{“needs torsion realisation (milestone iii)”, {{1 + (-1 + Sqrt[1 - x])^2/(-1 + Sqrt[1 + x])^2, {1, -1, -1, 1}}…
multrad-multiple_radicals_4-iv	Trager	Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4]/Sqrt[b-a*x]	Problem multiple_radicals_4 (iv)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_4-iv	PMT	Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4]/Sqrt[b-a*x]	Problem multiple_radicals_4 (iv)	{“failed”, “tower construction failed”, (Sqrt[b + a x] Sqrt[b^4 - a^4 x^4])/Sqrt[b - a x]}
multrad-multiple_radicals_4-iv-inst	Trager	Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4]/Sqrt[3-2*x]	Problem multiple_radicals_4 (iv); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-iv-inst	PMT	Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4]/Sqrt[3-2*x]	Problem multiple_radicals_4 (iv); inst a…	{“failed”, “tower construction failed”, (Sqrt[3 + 2 x] Sqrt[81 - 16 x^4])/Sqrt[3 - 2 x]}
multrad-multiple_radicals_4-v	Trager	Sqrt[x+1]/(Sqrt[1-x]*Sqrt[1-x^4])	Problem multiple_radicals_4 (v)	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-v	PMT	Sqrt[x+1]/(Sqrt[1-x]*Sqrt[1-x^4])	Problem multiple_radicals_4 (v)	{“needs torsion realisation (milestone iii)”, {{-1 + (-1 + Sqrt[1 - x])^2/(-1 + Sqrt[1 + x])^2 + (2 (-1 + Sqrt…
multrad-multiple_radicals_4-vi	Trager	Sqrt[a*x+b]/(Sqrt[b-a*x]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_4 (vi)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_4-vi	PMT	Sqrt[a*x+b]/(Sqrt[b-a*x]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_4 (vi)	{“failed”, “tower construction failed”, Sqrt[b + a x]/(Sqrt[b - a x] Sqrt[b^4 - a^4 x^4])}
multrad-multiple_radicals_4-vi-inst	Trager	Sqrt[2*x+3]/(Sqrt[3-2*x]*Sqrt[3^4-2^4*x^4])	Problem multiple_radicals_4 (vi); inst a…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-vi-inst	PMT	Sqrt[2*x+3]/(Sqrt[3-2*x]*Sqrt[3^4-2^4*x^4])	Problem multiple_radicals_4 (vi); inst a…	{“failed”, “tower construction failed”, Sqrt[3 + 2 x]/(Sqrt[3 - 2 x] Sqrt[81 - 16 x^4])}
multrad-multiple_radicals_4-vii	Trager	1/(Sqrt[1-x]*Sqrt[x+1]*Sqrt[1-x^4])	Problem multiple_radicals_4 (vii)	{“failed”, “more than one radicand (a radical compositum)”}

id	method	integrand	book ref	detail
multrad-multiple_radicals_4-vii	PMT	1/(Sqrt[1-x]*Sqrt[x+1]*Sqrt[1-x^4])	Problem multiple_radicals_4 (vii)	{“needs torsion realisation (milestone iii)”, {{-1 + (-1 + Sqrt[1 - x])^2/(-1 + Sqrt[1 + x])^2 - (2 (-1 + Sqrt…
multrad-multiple_radicals_4-viii	Trager	1/(Sqrt[b-a*x]*Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_4 (viii)	{“failed”, “radicand is not a rational function over Q”}
multrad-multiple_radicals_4-viii	PMT	1/(Sqrt[b-a*x]*Sqrt[a*x+b]*Sqrt[b^4-a^4*x^4])	Problem multiple_radicals_4 (viii)	{“failed”, “tower construction failed”, 1/(Sqrt[b - a x] Sqrt[b + a x] Sqrt[b^4 - a^4 x^4])}
multrad-multiple_radicals_4-viii-inst	Trager	1/(Sqrt[3-2*x]*Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4])	Problem multiple_radicals_4 (viii); inst…	{“failed”, “more than one radicand (a radical compositum)”}
multrad-multiple_radicals_4-viii-inst	PMT	1/(Sqrt[3-2*x]*Sqrt[2*x+3]*Sqrt[3^4-2^4*x^4])	Problem multiple_radicals_4 (viii); inst…	{“failed”, “tower construction failed”, 1/(Sqrt[3 - 2 x] Sqrt[3 + 2 x] Sqrt[81 - 16 x^4])}
multrad-multiple_radicals_7-ii	Trager	ArcTan[Sqrt[x+1]-Sqrt[x]]	Problem multiple_radicals_7 (ii), Charlw…	{“failed”, “transcendental function in the integrand”}
multrad-multiple_radicals_7-iv	Trager	ArcTan[d*Sqrt[a*x+b]-d*Sqrt[a*x+c]]	Problem multiple_radicals_7 (iv)	{“failed”, “transcendental function in the integrand”}
multrad-multiple_radicals_7-iv-inst	Trager	ArcTan[7*Sqrt[2*x+3]-7*Sqrt[2*x+5]]	Problem multiple_radicals_7 (iv); inst a…	{“failed”, “transcendental function in the integrand”}
euler-prob_euler_outrageous_1-ii	Trager	(a^2*x^2-2*a*b*x+b^2)*(a^2*x^2+2*a*b*x+b^2)/(6*a^2*x^2+b^2)*(a^4*x^4+6*a^2*b^2*x^2+b^4)^(3/…	Problem prob_euler_outrageous_1 (ii)	{“failed”, “radicand is not a rational function over Q”}
euler-prob_euler_outrageous_2-p	PMT	1/((3+x^2)*(1+3*x^2)^(1/3))	Problem prob_euler_outrageous_2 (+ sign)…	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, 3 + x^2, (-1/4…
euler-prob_euler_outrageous_3-ii	Trager	(a*x^2-x-2*b)/((x^4-a*x^2+b)*(a*x^2-b)^(1/4))	Problem prob_euler_outrageous_3 (ii)	{“failed”, “radicand is not a rational function over Q”}
euler-prob_euler_outrageous_3-ii	PMT	(a*x^2-x-2*b)/((x^4-a*x^2+b)*(a*x^2-b)^(1/4))	Problem prob_euler_outrageous_3 (ii)	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, -b + a x^2 - x…
euler-prob_euler_outrageous_3-ii-inst	PMT	(2*x^2-x-2*3)/((x^4-2*x^2+3)*(2*x^2-3)^(1/4))	Problem prob_euler_outrageous_3 (ii); in…	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, 3 - 2 x^2 + x^…
euler-prob_euler_outrageous_4-iii	Trager	Sqrt[a*x^4+c*x^2+b]/(b-a*x^4)	Problem prob_euler_outrageous_4 (iii)	{“failed”, “radicand is not a rational function over Q”}
euler-prob_euler_outrageous_4-iii-inst	PMT	Sqrt[2*x^4+5*x^2+3]/(3-2*x^4)	Problem prob_euler_outrageous_4 (iii); i…	{“needs torsion realisation (milestone iii)”, {{-3 + 2 x^4, {Root[1 - 960 #1^2 + 9216 #1^4 &, 2], Root[1 - 960…

id	method	integrand	book ref	detail
euler-prob_euler_outrageous_4-iv	Trager	x^2/((b-a*x^4)*Sqrt[a*x^4+b])	Problem prob_euler_outrageous_4 (iv)	{“failed”, “radicand is not a rational function over Q”}
euler-prob_euler_outrageous_5	Trager	(a*x^2-b)/((a*x^2+b)*Sqrt[a^2*x^4+b^2])	Problem prob_euler_outrageous_5	{“failed”, “radicand is not a rational function over Q”}
euler-prob_euler_outrageous_6-i	Trager	1/((x^4+1)*Sqrt[Sqrt[x^4+1]-x^2])	Problem prob_euler_outrageous_6 (i)	{“failed”, “nested radicals”}
euler-prob_euler_outrageous_6-i	PMT	1/((x^4+1)*Sqrt[Sqrt[x^4+1]-x^2])	Problem prob_euler_outrageous_6 (i)	{“failed”, “tower construction failed”, 1/((1 + x^4) Sqrt[-x^2 + Sqrt[1 + x^4]])}
euler-prob_euler_outrageous_6-ii	Trager	1/((a^2*x^4+b)*Sqrt[Sqrt[a^2*x^4+b]-a*x^2])	Problem prob_euler_outrageous_6 (ii)	{“failed”, “nested radicals”}
euler-prob_euler_outrageous_6-ii	PMT	1/((a^2*x^4+b)*Sqrt[Sqrt[a^2*x^4+b]-a*x^2])	Problem prob_euler_outrageous_6 (ii)	{“failed”, “tower construction failed”, 1/((b + a^2 x^4) Sqrt[-a x^2 + Sqrt[b + a^2 x^4]])}
euler-prob_euler_outrageous_6-ii-inst	Trager	1/((2^2*x^4+3)*Sqrt[Sqrt[2^2*x^4+3]-2*x^2])	Problem prob_euler_outrageous_6 (ii); in…	{“failed”, “nested radicals”}
euler-prob_euler_outrageous_6-ii-inst	PMT	1/((2^2*x^4+3)*Sqrt[Sqrt[2^2*x^4+3]-2*x^2])	Problem prob_euler_outrageous_6 (ii); in…	{“failed”, “tower construction failed”, 1/((3 + 4 x^4) Sqrt[-2 x^2 + Sqrt[3 + 4 x^4]])}
euler-prob_euler_outrageous_6-iii-n2	Trager	1/((x^4+1)*Sqrt[(x^4+1)^(1/2)-x^2])	Problem prob_euler_outrageous_6 (iii), n…	{“failed”, “nested radicals”}
euler-prob_euler_outrageous_6-iii-n2	PMT	1/((x^4+1)*Sqrt[(x^4+1)^(1/2)-x^2])	Problem prob_euler_outrageous_6 (iii), n…	{“failed”, “tower construction failed”, 1/((1 + x^4) Sqrt[-x^2 + Sqrt[1 + x^4]])}
euler-prob_euler_outrageous_6-iii-n3	Trager	1/((x^6+1)*Sqrt[(x^6+1)^(1/3)-x^2])	Problem prob_euler_outrageous_6 (iii), n…	{“failed”, “nested radicals”}
euler-prob_euler_outrageous_6-iii-n3	PMT	1/((x^6+1)*Sqrt[(x^6+1)^(1/3)-x^2])	Problem prob_euler_outrageous_6 (iii), n…	{“failed”, “tower construction failed”, 1/((1 + x^6) Sqrt[-x^2 + (1 + x^6)^(1/3)])}
timofeev-timofeev_1-vii	PMT	1/Sqrt[(x^3-5*x^2+3*x+9)]	Problem timofeev_1 (vii)	{“failed”, “verification failed”}
timofeev-timofeev_1-viii	PMT	1/Sqrt[(x^3-5*x^2+3*x+9)^3]	Problem timofeev_1 (viii)	{“failed”, “verification failed”}
timofeev-timofeev_1-xi	PMT	1/((x^3-5*x^2+3*x+9)^4)^(1/3)	Problem timofeev_1 (xi)	{“failed”, “verification failed”}
timofeev-timofeev_3-i	Trager	Sqrt[x-5]*Sqrt[x+3]/((x-1)*(x^2-25))	Problem timofeev_3 (i)	{“failed”, “more than one radicand (a radical compositum)”}
timofeev-timofeev_3-ii	Trager	x^2*Sqrt[x+1]*(1-x^2)^(1/4)/(Sqrt[1-x]*(Sqrt[1-x]-Sqrt[x+1]))	Problem timofeev_3 (ii)	{“failed”, “more than one radicand (a radical compositum)”}
timofeev-timofeev_3-ii	PMT	x^2*Sqrt[x+1]*(1-x^2)^(1/4)/(Sqrt[1-x]*(Sqrt[1-x]-Sqrt[x+1]))	Problem timofeev_3 (ii)	{“failed”, “partially ramified branch prime”, -1 + (-1 + Sqrt[1 - x])^2/(-1 + Sqrt[1 + x])^2 - (2 (-1 + Sqrt[1…

id	method	integrand	book ref	detail
timofeev-timofeev_3-iii	Trager	x*Sqrt[1-x]*(x+1)^(2/3)/((1-x)^(2/3)*Sqrt[x+1]-(1-x)^(5/6)*(x+1)^(1/3))	Problem timofeev_3 (iii)	{“failed”, “more than one radicand (a radical compositum)”}
timofeev-timofeev_3-iii	PMT	x*Sqrt[1-x]*(x+1)^(2/3)/((1-x)^(2/3)*Sqrt[x+1]-(1-x)^(5/6)*(x+1)^(1/3))	Problem timofeev_3 (iii)	{“failed”, “no solution within bounds”, {12}}
timofeev-timofeev_3-v	Trager	((x^2+2)/x^2)^(7/9)/(x^2+2)^(3/2)	Problem timofeev_3 (v)	{“failed”, “more than one radicand (a radical compositum)”}
potpourri-p1	PMT	(x^2+1)*(x^2-x-1)/((x^2-1)^2*Sqrt[x^4-x^3-x^2+x+1])	Potpourri problem 1	{“needs torsion realisation (milestone iii)”, {{-1 + x, {1/2, -1/2}}, {1 + x, {-1/2, 1/2}}}}
potpourri-p2	PMT	1/((x-1)*Sqrt[3*x^4-12*x^3+18*x^2-12*x-1])	Potpourri problem 2	{“failed”, “no solution within bounds”, {0}}
potpourri-p6-i	PMT	Sqrt[x^4-x]	Potpourri problem 6 (i)	{“failed”, “verification failed”}
potpourri-p6-vii	PMT	1/(x^8+2*x^4+1)^(1/8)	Potpourri problem 6 (vii)	{“failed”, “partially ramified branch prime”, 1 + x^4}
potpourri-p6-viii	PMT	(x^3-1)*Sqrt[x^6-1]/(x^4*(x^3+1))	Potpourri problem 6 (viii)	{“needs torsion realisation (milestone iii)”, {{x, {2*I, -2*I}}}}
potpourri-p6-ix	PMT	(2*x^4-1)/((x^8-1)*(x^4-1)^(1/4))	Potpourri problem 6 (ix)	{“failed”, “no solution within bounds”, {4}}
potpourri-p7-ii	PMT	1/((x+1)*(x^4-x^2-2*x+2)^(3/2))	Potpourri problem 7 (ii)	{“failed”, “verification failed”}
potpourri-p7-iv	PMT	1/(81*x^4-135*x^3+54*x^2+12*x-8)^(1/3)	Potpourri problem 7 (iv)	{“failed”, “verification failed”}
potpourri-p8-iv	Trager	x/((x^2-1)*Sqrt[-x+Sqrt[x^2+1]])	Potpourri problem 8 (nested radicals) (i…	{“failed”, “nested radicals”}
potpourri-p8-vi	Trager	Sqrt[1+Sqrt[x+Sqrt[x^2+1]]]/Sqrt[x^2+1]	Potpourri problem 8 (nested radicals) (v…	{“failed”, “nested radicals”}
potpourri-p8-viii	Trager	(1+(x+Sqrt[x^2-1])^(1/4))^(1/6)/Sqrt[x^2-1]	Potpourri problem 8 (nested radicals) (v…	{“failed”, “nested radicals”}
potpourri-p9-i	Trager	(a+Sqrt[x]+Sqrt[x+2*a*Sqrt[x]])^(1/3)	Potpourri problem 9 (parametric nested r…	{“failed”, “nested radicals”}
potpourri-p9-i-inst	Trager	(2+Sqrt[x]+Sqrt[x+2*2*Sqrt[x]])^(1/3)	Potpourri problem 9 (parametric nested r…	{“failed”, “nested radicals”}
potpourri-p9-ii	Trager	Sqrt[c*x^2-x*Sqrt[a*x^2-b*x]]/(x^4*Sqrt[a*x^2-b*x])	Potpourri problem 9 (parametric nested r…	{“failed”, “nested radicals”}
potpourri-p9-ii	PMT	Sqrt[c*x^2-x*Sqrt[a*x^2-b*x]]/(x^4*Sqrt[a*x^2-b*x])	Potpourri problem 9 (parametric nested r…	{“failed”, “tower construction failed”, Sqrt[c x^2 - x Sqrt[-b x + a x^2]]/(x^4 Sqrt[-b x + a x^2])}
potpourri-p9-ii-inst	Trager	Sqrt[5*x^2-x*Sqrt[2*x^2-3*x]]/(x^4*Sqrt[2*x^2-3*x])	Potpourri problem 9 (parametric nested r…	{“failed”, “nested radicals”}

id	method	integrand	book ref	detail
potpourri-p9-ii-inst	PMT	Sqrt[5*x^2-x*Sqrt[2*x^2-3*x]]/(x^4*Sqrt[2*x^2-3*x])	Potpourri problem 9 (parametric nested r…	{“failed”, “tower construction failed”, Sqrt[5 x^2 - x Sqrt[-3 x + 2 x^2]]/(x^4 Sqrt[-3 x + 2 x^2])}
danger-d3-i	PMT	1/(x^4-4*x^2+2)^(1/4)	Danger zone problem 3; todo says not tha…	{“failed”, “no solution within bounds”, {0}}
danger-d3-ii	PMT	1/(x^4+4*x^2+2)^(1/4)	Danger zone problem 3; todo says not tha…	{“failed”, “no solution within bounds”, {0}}
danger-d3-iii	PMT	1/(x^4-8*x^2+8)^(1/4)	Danger zone problem 3; todo says not tha…	{“failed”, “no solution within bounds”, {0}}
danger-d3-iv	PMT	1/(x^4+8*x^2+8)^(1/4)	Danger zone problem 3; todo says not tha…	{“failed”, “no solution within bounds”, {0}}
danger-d4-i	PMT	1/(x^4-4*x^3+2*x^2)^(1/4)	Danger zone problem 4; todo says not tha…	{“failed”, “partially ramified branch prime”, x}
danger-d4-ii	PMT	1/(x^4+4*x^3+2*x^2)^(1/4)	Danger zone problem 4; todo says not tha…	{“failed”, “partially ramified branch prime”, x}
danger-d4-iii	PMT	1/(x^4-8*x^3+8*x^2)^(1/4)	Danger zone problem 4; todo says not tha…	{“failed”, “partially ramified branch prime”, x}
danger-d4-iv	PMT	1/(x^4+8*x^3+8*x^2)^(1/4)	Danger zone problem 4; todo says not tha…	{“failed”, “partially ramified branch prime”, x}
intro-can_cannot-1	Trager	Sqrt[Tan[x]]	Introduction, Integrals that Can and Can…	{“failed”, “transcendental function in the integrand”}
intro-can_cannot-2	Trager	ArcTan[Sqrt[x]]	Introduction, Integrals that Can and Can…	{“failed”, “transcendental function in the integrand”}
intro-can_cannot-3	Trager	Tan[Sqrt[x]]/Sqrt[x]	Introduction, Integrals that Can and Can…	{“failed”, “transcendental function in the integrand”}
intro-can_cannot-4	Trager	ArcTan[x]/Sqrt[x]	Introduction, Integrals that Can and Can…	{“failed”, “transcendental function in the integrand”}
intro-validity-2	Trager	(x^8-1)/((x^8-a*x^4+1)*Sqrt[x^4-1])	Introduction, Validity of Solutions, par…	{“failed”, “integrand is not a rational function of x and the radical over Q”}
intro-validity-2	PMT	(x^8-1)/((x^8-a*x^4+1)*Sqrt[x^4-1])	Introduction, Validity of Solutions, par…	{“failed”, “residue class lost: no places over the prime for a root of the residue polynomial”, -1 + a x^4 - x…
intro-multiple_approaches-1	Trager	1/((x-1)*Sqrt[x^2+b*x+c])	Introduction, Multiple Approaches to Int…	{“failed”, “radicand is not a rational function over Q”}
intro-further-schultz	PMT	(29*x^2+18*x-3)/Sqrt[x^6+4*x^5+6*x^4-12*x^3+33*x^2-16*x]	Introduction, Further Reading: Schultz p…	{“failed”, “verification failed”}

FriCAS answers that fail their own derivative check (1)

- potpourri-p7-iii: `1/(x^6+2*x^5-x^4-4*x^3-x^2+2*x+1)^(1/6)`